COMP2711: Discrete Mathematics

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Contents

1 Propositional Logic **6**
1.1 Propositions . . . 6
1.2 Compound Propositions . . . 6
1.2.1 Negation . . . 7
1.2.2 Conjunction . . . 7
1.2.3 Disjunction . . . 7
1.2.4 Exclusive Or . . . 7
1.2.5 Conditional Statement . . . 8
1.2.6 Converse, Contrapositive, Inverse . . . 9
1.2.7 Biconditional Statement . . . 9
1.2.8 Precedence of Logical Operators . . . 9
1.3 Propositional Equivalences . . . 9
1.3.1 Tautology, Contradiction, Contingency . . . 9
1.3.2 Logical Equivalence . . . 10
1.3.3 De Morgan's Laws . . . 10
1.3.4 Expressing Other Connectives . . . 11
1.3.5 Table of Propositional Equivalences . . . 11
1.3.6 More Examples of Equivalence Proofs . . . 12
1.3.7 Application: Knight and Knave Puzzle . . . 13

2 Predicate Logic **13**
2.1 Predicates . . . 13
2.2 Quantifiers . . . 14
2.2.1 Universal Quantification . . . 14
2.2.2 Existential Quantification . . . 15
2.3 Quantifiers with Restricted Domains . . . 15
2.3.1 Precedence of Quantifiers and Binding Variables . . . 16
2.4 Logical Equivalences Involving Quantifiers . . . 16
2.5 Negating Quantified Expressions . . . 17
2.6 Nested Quantifiers . . . 18
2.6.1 Order of Quantifiers . . . 18
2.6.2 Translating Between English and Logic . . . 19
2.6.3 Negating Nested Quantifiers . . . 20
2.6.4 Null Quantifications . . . 20

3 Rules of Inference **21**
3.1 Rules of Inference for Propositional Logic . . . 21
3.1.1 Table of Rules of Inference . . . 22
3.1.2 Examples . . . 22
3.1.3 Invalid Arguments . . . 23
3.2 Rules of Inference for Predicate Logic . . . 24
3.2.1 Examples . . . 24
3.3 Basic Proof Techniques . . . 25
3.3.1 Some Terminology . . . 25
3.3.2 The Axiomatic Method . . . 25
3.3.3 Direct Proof . . . 26
3.3.4 Proof by Contraposition . . . 27
3.3.5 Proof by Contradiction . . . 28
3.3.6 Proof by Contradiction for Conditional Statements . . . 28
3.3.7 Proving Biconditional Statements . . . 29
3.3.8 Nonconstructive Existence Proof . . . 29
3.4 Some Comments on Proofs . . . 29
3.4.1 Theorems and Proofs (Hilbert, Gödel) . . . 30

4 Sets and Functions **30**
4.1 Sets . . . 30
4.2 Power Set . . . 31
4.3 Ordered Tuples and Cartesian Products . . . 31
4.4 Set Operations . . . 32
4.5 Functions . . . 33
4.6 Injective, Surjective, Bijective Functions . . . 33
4.7 Inverse Functions and Composition . . . 34
4.8 Cardinality of Infinite Sets . . . 35
4.9 Comparing Infinite Sets . . . 36
4.10 Uncomputable Functions . . . 37

5 Mathematical Induction **38**
5.1 Mathematical Induction: An Axiom . . . 38
5.2 Variants of Induction . . . 39
5.3 Proving Summations . . . 39
5.4 Proving Inequalities . . . 40
5.5 Proving Divisibility Results . . . 40
5.6 Number of Subsets of a Finite Set . . . 41
5.7 Odd Pie Fight Problem . . . 41
5.8 Tiling Checkerboards . . . 41
5.9 Strong Induction . . . 42
5.10 Induction with Two Variables . . . 43

6 Recursion **43**
6.1 Recursively Defined Functions . . . 43
6.2 Solving First-Order Linear Recurrences . . . 44
6.3 Fibonacci Numbers . . . 45
6.4 Recursively Defined Sets . . . 46
6.5 Structural Induction . . . 46

6.6 Well-Formed Formulae . . . 47
6.7 Strings . . . 48
6.8 String Concatenation . . . 48
6.9 Length of a String . . . 49

7 Modular Arithmetic **49**
7.1 Divisibility . . . 50
7.2 Euclid's Division Theorem . . . 50
7.3 Modular Arithmetic . . . 52
7.4 Modular Arithmetic on $\mathbb{Z}_m$ . . . 53
7.4.1 Additive and Multiplicative Inverses . . . 53
7.5 Congruences . . . 54
7.5.1 Modular Arithmetic and Congruences . . . 54
7.6 Applications of Modular Arithmetic . . . 55
7.6.1 Parity Bits . . . 55
7.6.2 Hash Functions . . . 55
7.6.3 Hash Table . . . 55
7.6.4 Hashing Strings . . . 56
7.6.5 Pseudorandom Numbers . . . 56

8 Greatest Common Divisor **57**
8.1 Primes and the Fundamental Theorem of Arithmetic . . . 57
8.2 Greatest Common Divisor . . . 57
8.3 The Euclidean Algorithm . . . 57
8.3.1 Analysis: Lamé's Theorem . . . 58
8.4 Bézout's Theorem and the Extended Euclidean Algorithm . . . 59
8.5 Multiplicative Inverses . . . 59
8.5.1 Finding Inverses . . . 60
8.6 Solving Linear Congruences . . . 61
8.7 Solving Systems of Congruences . . . 62
8.8 Revisiting the String Hash Function . . . 62
8.9 Checksums: the HKID Check Digit . . . 62
8.9.1 Single Error Detection . . . 63
8.9.2 Transposition Error Detection . . . 63
8.10 The Chinese Remainder Theorem . . . 63

9 Cryptography **64**
9.1 Secret Key Cryptography . . . 65
9.1.1 Caesar Cipher (Shift Cipher) . . . 65
9.1.2 Affine Ciphers . . . 65
9.1.3 Block Ciphers . . . 65
9.1.4 Advanced Encryption Standard (AES) . . . 66
9.2 Key Exchange . . . 66
9.2.1 Modular Exponentiation . . . 66
9.2.2 Fast Modular Exponentiation: Repeated Squaring . . . 67
9.2.3 A Hard Problem: Discrete Logarithm . . . 67
9.2.4 Diffie–Hellman Key Exchange . . . 67
9.3 Public Key Cryptography and RSA . . . 68

9.3.1 The RSA Cryptosystem 68
9.4 Fermat's Little Theorem 69
9.5 RSA Correctness 70
9.6 RSA in Use 70

10 Analysis of Algorithms **71**
10.1 Measuring Running Time 71
10.2 Growth of Functions and Big-Theta 72
10.3 Comparing Algorithms via Limits 72
10.4 Important Examples of Growth Rates 73
10.5 Limitations of Big-Theta 74
10.6 Big-Oh and Big-Omega 74
10.7 A Warning: Big-O Inside Induction 75
10.8 Insertion Sort 75
10.9 Best-Case, Worst-Case, and Average-Case Analysis 76
10.10 Searching Algorithms 76
10.10.1 Linear Search 76
10.10.2 Binary Search 77
10.11 Recursive Algorithms 77
10.11.1 Euclid's GCD Algorithm 78
10.11.2 Fibonacci Numbers 78
10.11.3 The Tower of Hanoi 78
10.12 Summary of Recursive Algorithms 79

11 Basics of Counting **79**
11.1 The Product Rule 79
11.2 The Sum Rule 81
11.3 Using Both Product and Sum Rules 81
11.4 Tree Diagrams 82
11.5 The Pigeonhole Principle 82
11.6 The Generalized Pigeonhole Principle 83

12 Permutations and Combinations **84**
12.1 Permutations 84
12.2 Combinations 86
12.3 Combinatorial Proofs and the Bijection Principle 87
12.4 Permutations with Indistinguishable Objects 88
12.5 Throwing Balls into Bins 88
12.6 Indistinguishable Balls into Distinguishable Bins 89
12.7 Combinations with Repetition 89
12.8 Summary of Balls and Bins 90

13 Binomial Coefficients **90**
13.1 The Binomial Theorem 90
13.2 Corollaries of the Binomial Theorem 91
13.3 Pascal's Identity and Pascal's Triangle 92
13.4 Vandermonde's Identity 94
13.5 Counting Bit Strings 94

14 Inclusion-Exclusion Principle **95**
14.1 Two Finite Sets . . . 95
14.2 Three Finite Sets . . . 96
14.3 The General Inclusion-Exclusion Principle . . . 97
14.4 Counting Onto Functions . . . 98
14.5 Distributing Objects into Boxes . . . 99
14.6 Derangements . . . 99

15 Introduction to Probability **101**
15.1 The Hatcheck Problem Revisited . . . 101
15.2 Experiments, Sample Spaces, and Events . . . 101
15.3 Probability by Counting . . . 101
15.4 Poker and Lottery . . . 102
15.5 Sampling With and Without Replacement . . . 102
15.6 The Complement of an Event . . . 103
15.7 The Union of Events . . . 103
15.8 Probability Distributions . . . 104
15.9 Independence . . . 105
15.10 Pairwise and Mutual Independence . . . 106

16 Conditional Probability and Bayes' Theorem **107**
16.1 Conditional Probability . . . 107
16.2 The Law of Total Probability . . . 108
16.3 The Birthday Problem . . . 108
16.4 Independence Revisited . . . 109
16.5 Bayes' Theorem . . . 109
16.6 Generalized Bayes' Theorem . . . 112
16.7 The Monty Hall Problem . . . 113

17 Random Variables **115**
17.1 Random Variables and Their Distributions . . . 115
17.2 Bernoulli Trials and the Binomial Distribution . . . 115
17.3 Expected Value . . . 117
17.4 Linearity of Expectations . . . 118
17.5 Average-Case Analysis of Algorithms . . . 120
17.6 The Geometric Distribution and the Coupon Collector Problem . . . 121
17.7 Conditional Expectation and the Two Envelopes Paradox . . . 122
17.8 Independent Random Variables . . . 123
17.9 Variance and Standard Deviation . . . 124

18 Graph Theory **127**
18.1 Motivating Problems . . . 127
18.2 Graph Terminology . . . 127
18.3 Representing Graphs . . . 129
18.4 Connectivity . . . 130
18.5 Euler Paths and Circuits . . . 132
18.6 The Chinese Postman Problem . . . 133
18.7 Hamilton Paths and Circuits . . . 134

1 Propositional Logic

Logic is the study of the rules of mathematical reasoning. It has many practical applications:

- **Puzzles and law-school admission (LSAT).** Example puzzle: Riders A, B, C, D, E, F, G take buses 1, 2, 3 to work. Neither E nor G takes bus 1 on a day when B does. G does not take bus 2 on a day when D does. When A and F take the same bus, it is always bus 3. C always takes bus 3. Who takes which bus?
- **Automated reasoning and artificial intelligence.**
- **Automated theorem proving**, e.g., the Four Color Theorem.
- **Proof verification.**

Reading: Kenneth Rosen, Sections 1.1–1.3.

1.1 Propositions

Definition (Proposition)

A **proposition** is a declarative statement (i.e., a sentence that declares a fact) that is either true or false, but not both.
The **truth value** of a proposition is true, denoted by T, if it is a true proposition, and false, denoted by F, if it is a false proposition.

Remark. Propositions are the basic building blocks of logic. The area of logic that deals with propositions is called **propositional logic**.

Example. Each of the following declarative statements is a proposition:

(a) Beijing is the capital of China. (true)
(b) COMP 2711 is an elective course for the COMP program. (false)
(c) $2 + 2 = 2^2$. (true)
(d) $1 + 1 = 3$. (false)

Propositions (a) and (c) are true, while (b) and (d) are false.

Example. The following statements are not propositions:

(a) No parking. (not declarative)
(b) Who has an iPad? (a question, not declarative)
(c) $y = \log(x + 1)$. (truth value depends on x and y)
(d) $x^2 - 3x + 1 = 0$. (truth value depends on x)

Note that (c) and (d) can become propositions if x and y are assigned values.

Logical paradoxes are also not propositions:

(a) "This sentence is false." (Liar paradox)
(b) A male barber shaves all and only those men who do not shave themselves. (Barber paradox)

1.2 Compound Propositions

We use lowercase letters $p, q, r, \ldots$ as **propositional variables** to represent propositions. **Logical operators** (or **logical connectives**) can be used to turn existing propositions into new propositions. The definition of a logical operator can be given in the form of a **truth table** by enumerating all possible truth values of the proposition(s) involved.

1.2.1 Negation

Definition (Negation)

Let p be a proposition. The **negation** of p, denoted by $\neg p$ or $\bar{p}$ and read as "not p", is the statement "it is not the case that p". The truth value of $\neg p$ is the opposite of the truth value of p.

p	$\neg p$
T	F
F	T

Example. Let p: "Tom's iPhone has at least 32GB of memory." Then $\neg p$: "It is not the case that Tom's iPhone has at least 32GB of memory", or more naturally, $\neg p$: "Tom's iPhone has less than 32GB of memory."

1.2.2 Conjunction

Definition (Conjunction)

Let p and q be propositions. The **conjunction** of p and q, denoted by $p \wedge q$, is the proposition "p and q". The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

1.2.3 Disjunction

Definition (Disjunction)

The **disjunction** of p and q, denoted by $p \vee q$, is the proposition "p or q". The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

1.2.4 Exclusive Or

Definition (Exclusive Or)

The **exclusive or** of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

1.2.5 Conditional Statement

Definition (Conditional Statement / Implication)

The **conditional statement** $p \to q$ is the proposition "if p, then q". Here p is called the **hypothesis** (or **premise**) and q is called the **conclusion** (or **consequence**). It is also called an **implication**. The conditional statement $p \to q$ is false when p is true and q is false, and true otherwise.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Note. There does not need to be any connection between the hypothesis and the conclusion. The truth value of $p \to q$ depends only on the truth values of p and q.

Example. Consider the following conditional statements:

- "If Beijing is the capital of China, then Beijing is a city of China." — True (both p and q true).
- "If Beijing is a city of China, then Beijing is the capital of China." — True (p true, q true in fact; but even in general we only evaluate by truth values).
- "If the moon is made of green cheese, then I have more money than Bill Gates." — True (hypothesis false, so the implication is vacuously true).
- "If the moon is not made of green cheese, then I have more money than Bill Gates." — False (hypothesis true, conclusion false).

Necessary and sufficient conditions. If $p \to q$ is true, then p is a **sufficient condition** for q, and q is a **necessary condition** for p.

Example. Consider "If I am elected, then taxes will be lowered."

- "I am elected" is a sufficient condition for "lower taxes" (other candidates may also lower taxes).
- "Lower taxes" is a necessary condition for "I am elected": if taxes are not lowered, then I am definitely not elected.

Other ways to express $p \to q$ in English.

- if p, then q
- q if p
- q when p
- p implies q
- q follows from p
- p is a sufficient condition for q
- q is a necessary condition for p
- p only if q (falsity of q implies falsity of p)

Example of "only if": "The company can succeed only if it has sufficient funding."

1.2.6 Converse, Contrapositive, Inverse

Definition (Converse, Contrapositive, Inverse)

Given the conditional statement $p \to q$:
- The **converse** of $p \to q$ is $q \to p$.
- The **contrapositive** of $p \to q$ is $\neg q \to \neg p$.
- The **inverse** of $p \to q$ is $\neg p \to \neg q$.

A conditional statement $p \to q$ and its contrapositive $\neg q \to \neg p$ are logically equivalent. The converse and the inverse are equivalent to each other, but not to the original statement.

1.2.7 Biconditional Statement

Definition (Biconditional Statement)

The **biconditional statement** $p \leftrightarrow q$ is the proposition "p if and only if q". The statement is true when p and q have the same truth value and is false otherwise.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Equivalent ways of expressing $p \leftrightarrow q$:
- p if and only if q
- p iff q
- p is necessary and sufficient for q
- if p then q, and conversely

Ambiguity in natural language. Sometimes "if ... then ..." in natural language actually means "if and only if". For example: "If you finish your meal, then you can have dessert." What if you don't finish your meal? In natural language: then you can't have dessert. In logic: unspecified. In this course, we should avoid such ambiguities.

1.2.8 Precedence of Logical Operators

Multiple logical operators can be used to construct compound propositions. Parentheses can be used for clarity. For example, $p \vee \neg q \to p \wedge q$ may be written more clearly as $(p \vee \neg q) \to (p \wedge q)$.

Remark. Boolean operations in programming languages (e.g., `&&`, `||`, `!`, `==` in C++) correspond to $\wedge$, $\vee$, $\neg$, $\leftrightarrow$. Note that "if ... then ..." statements in programming languages do not correspond to $\to$.

1.3 Propositional Equivalences

1.3.1 Tautology, Contradiction, Contingency

Definition (Tautology, Contradiction, Contingency)

- A **tautology** is a compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it.
- A **contradiction** is a compound proposition that is always false.

- A **contingency** is a compound proposition that is neither a tautology nor a contradiction.

Example.
- $p \vee \neg p$ is a tautology.
- $p \wedge \neg p$ is a contradiction.
- $p \to \neg p$ is a contingency.

This can be verified with a truth table:

p	$p \vee \neg p$	$p \wedge \neg p$	$p \to \neg p$
T	T	F	F
F	T	F	T

1.3.2 Logical Equivalence

Definition (Logical Equivalence)

Compound propositions that always have the same truth values are called **logically equivalent**. Equivalently, the compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology. The notation is $p \equiv q$ (or $p \Leftrightarrow q$).

Remark. The symbol $\equiv$ is not a logical operator and $p \equiv q$ is not a compound proposition but rather is the statement that $p \leftrightarrow q$ is a tautology. Sometimes $\Leftrightarrow$ is used instead of $\equiv$ to denote logical equivalence.

The use of $\equiv$, $\leftrightarrow$, $=$.
- $\equiv$ ($\Leftrightarrow$) denotes logical equivalence regardless of truth values of the propositional variables.
- $\leftrightarrow$ is a binary logical operator.
- $=$ is not used on propositions. It is only used between two mathematical objects.

Example.
- $(p \to q) \leftrightarrow (\neg p \vee q) \equiv T$
- $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
- $T \leftrightarrow F \equiv F$
- $(2 = 3) \wedge (5 \neq 4) \equiv F$
- Let p: $2 + 2 = 4$; then p is true.

1.3.3 De Morgan's Laws

Example. Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

Solution. We construct the truth table:

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Since the columns for $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are identical, the two compound propositions are logically equivalent.

Example. Use De Morgan's laws to express the negation of "today is Saturday and today is a holiday."

Solution. Let p: "today is Saturday" and q: "today is a holiday". By De Morgan's laws, $\neg(p \wedge q) \equiv \neg p \vee \neg q$. So the negation is: "today is not Saturday or today is not a holiday."

Example. Use De Morgan's laws to express the negation of "Alice will send a secret message or Bob will send a secret message."

Solution. By De Morgan's laws, the negation of a disjunction is the conjunction of the negations: "Alice will not send a secret message and Bob will not send a secret message."

1.3.4 Expressing Other Connectives

Example. Can we express $p \to q$ using negations and disjunctions?

Solution. Yes:

$$p \to q \equiv \neg p \vee q.$$

This can be proved using a truth table: $p \to q$ is false only when $p = T$ and $q = F$; likewise $\neg p \vee q$ is false only when $\neg p = F$ (i.e., $p = T$) and $q = F$.

Example. Can we express $p \oplus q$ using negations, conjunctions and disjunctions?

Solution. Yes:

$$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q).$$

This holds because $p \oplus q$ is true exactly when exactly one of p, q is true: either p is true and q is false (first disjunct), or p is false and q is true (second disjunct).

Example. Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Solution.

$$\begin{aligned} \neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{(since } p \to q \equiv \neg p \vee q) \\ &\equiv \neg(\neg p) \wedge \neg q && \text{(De Morgan's law)} \\ &\equiv p \wedge \neg q && \text{(double negation law)} \end{aligned}$$

1.3.5 Table of Propositional Equivalences

Major Propositional Equivalences

Equivalence	Name
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	*Doublenegationlaw*
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation laws

Equivalences Involving Conditional and Biconditional Statements

Involving conditional statements:

$$p \to q \equiv \neg p \vee q$$
$$p \to q \equiv \neg q \to \neg p$$
$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$
$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$
$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$
$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

Involving biconditional statements:

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$
$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$
$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

1.3.6 More Examples of Equivalence Proofs

Example. Show that $\neg\big(p \vee (\neg p \wedge q)\big)$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Solution.

$$\begin{aligned}
\neg\big(p \vee (\neg p \wedge q)\big) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(De Morgan's law)}\\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(De Morgan's law)}\\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(double negation law)}\\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{(distributive law)}\\
&\equiv F \vee (\neg p \wedge \neg q) && \text{(negation law)}\\
&\equiv \neg p \wedge \neg q && \text{(identity law)}
\end{aligned}$$

Example. Show that $(p \wedge q) \to (p \vee q)$ is a tautology by developing a series of logical equivalences.

Solution.

$$\begin{aligned}
(p \wedge q) \to (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{(implication law)} \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{(De Morgan's law)} \\
&\equiv (p \vee \neg p) \vee (q \vee \neg q) && \text{(associative and commutative laws)} \\
&\equiv T \vee T && \text{(negation laws)} \\
&\equiv T && \text{(domination law)}
\end{aligned}$$

Since the proposition is equivalent to T, it is a tautology.

1.3.7 Application: Knight and Knave Puzzle

Example. Recall the knight and knave puzzle: knights always tell the truth, and knaves always lie. A says "We are both knaves." B says nothing. What are A and B?

Solution. Let p be "A is a knight" and q be "B is a knight". A's statement "We are both knaves" is $\neg p \wedge \neg q$. We consider two possibilities:

Case 1: A's claim is true. Then $\neg p \wedge \neg q$ is true. If A is telling the truth, then he is a knight and p is true. However, $\neg p \wedge \neg q$ and p cannot both be true (since $\neg p \wedge p \equiv F$). Contradiction, so this case is impossible.

Case 2: A's claim is false. Then $\neg p \wedge \neg q$ is false, i.e., $\neg(\neg p \wedge \neg q)$ is true. Using De Morgan's law,

$$\neg(\neg p \wedge \neg q) \equiv p \vee q.$$

Since A's claim is false, he is a knave, so p is false. Therefore q must be true.

Conclusion: A is a knave and B is a knight.

2 Predicate Logic

Predicate logic (also called first-order logic) is more powerful than propositional logic. Suppose we know that "every COMP student is required to take either COMP 2711 or COMP 2711H". No rules of propositional logic allow us to conclude the truth of the statement "Chan Tai Man, a COMP student, is required to take either COMP 2711 or COMP 2711H". Predicate logic provides the tools needed for such reasoning.

Reading: Kenneth Rosen, Sections 1.4–1.5.

2.1 Predicates

Definition (Predicate)

Informally, a **predicate** is a statement that may be true or false depending on the choice of values of its variables. Each choice of values produces a proposition.
More formally, a statement involving n variables $x_1, x_2, \ldots, x_n$, denoted by $P(x_1, x_2, \ldots, x_n)$, is the value of the **propositional function** P at the n-tuple $(x_1, x_2, \ldots, x_n)$ and P is called an n-ary predicate.

Example. $P(x)$ denotes the statement "x is greater than 3". Here x is the variable, and P is the predicate "is greater than 3".

Example. Let $P(x)$ denote the statement "$x > 3$". What are the truth values of $P(4)$ and $P(2)$?

Solution. $P(4)$ is the proposition "$4 > 3$", which is true. $P(2)$ is the proposition "$2 > 3$", which is false.

Example. Let $A(x)$ denote the statement "student x is required to take either COMP 2711 or COMP 2711H". Suppose Alice is a COMP student and Bob is a CHEM student. What are the truth values of $A(\text{Alice})$ and $A(\text{Bob})$?

Solution. $A(\text{Alice})$ is true because Alice is a COMP student. $A(\text{Bob})$ is false because Bob is not a COMP student.

Example. Let $Q(x, y)$ denote the statement "$x = y + 3$". What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution. $Q(1, 2)$: "$1 = 2 + 3$" is false. $Q(3, 0)$: "$3 = 0 + 3$" is true.

2.2 Quantifiers

We saw that when the variables in a propositional function are assigned values, the resulting proposition has a certain truth value. Sometimes we may want to say that a predicate is true over a set of values. This is done using **quantifiers**.

2.2.1 Universal Quantification

Definition (Universal Quantification)

The **universal quantification** of $P(x)$ is the statement "for all elements x in the domain such that $P(x)$". It is denoted as $\forall x\, P(x)$. We read it as "for all x, $P(x)$" or "for every x, $P(x)$". Here $\forall$ is called the **universal quantifier**.
An element for which $P(x)$ is false is called a **counterexample** of $\forall x\, P(x)$.

Definition (Domain / Universe)

The **domain** or **universe** is the set of all possible values of a variable. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not well defined.
Generally, an implicit assumption is made that the domain is nonempty. Otherwise $\forall x\, P(x)$ is true for any propositional function $P(x)$ because there are no elements x in the domain for which $P(x)$ is false.

Example. Let $P(x)$ be the statement "$x + 1 > x$". What is the truth value of $\forall x\, P(x)$, where the domain consists of all real numbers?

Solution. The statement is true, because for every real number x, adding 1 to it yields a strictly greater number.

Example. Let $Q(x)$ be the statement "$x < 2$". What is the truth value of $\forall x\, Q(x)$, where the domain consists of all real numbers?

Solution. The statement is false. For example, $x = 3$ is a counterexample.

Example. Let $P(x)$ be the statement "$x^2 > 0$". What is the truth value of $\forall x\, P(x)$, where the domain consists of all real numbers?

Solution. The statement is false, because $x = 0$ is a counterexample: $0^2 = 0 \not> 0$.

Example. What is the truth value of "$\forall x\, (x^2 \geq x)$" if the domain consists of all real numbers? What is its truth value if the domain consists of all integers?

Solution. Over the real numbers, the statement is false because, for example, $x = 0.5$ is a counterexample: $0.5^2 = 0.25 < 0.5$. Over the integers, the statement is true because if $x \geq 1$ then $x^2 \geq x$, if $x = 0$ then $0^2 = 0 \geq 0$, and if $x \leq -1$ then $x^2 \geq 1 > x$.

2.2.2 Existential Quantification

Definition (Existential Quantification)

The **existential quantification** of $P(x)$ is the statement "there exists an element x in the domain such that $P(x)$". The notation $\exists x\, P(x)$ denotes the existential quantification of $P(x)$. Here $\exists$ is called the **existential quantifier**.
Note that $\exists x$ means "there exists at least one x in the domain" but not "there exists one and only one x in the domain" or "there exists a unique x in the domain".

Connection to finite domains. When the domain has n elements $x_1, x_2, \ldots, x_n$:

$$\forall x\, P(x) \text{ is the same as } P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n),$$
$$\exists x\, P(x) \text{ is the same as } P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n).$$

Example. Let $P(x)$ be the statement "$x > 3$". What is the truth value of $\exists x\, P(x)$, where the domain consists of all real numbers?

Solution. The statement is true, because for example $x = 4$ satisfies $4 > 3$.

Example. Let $Q(x)$ be the statement "$x = x + 1$". What is the truth value of $\exists x\, Q(x)$, where the domain consists of all real numbers?

Solution. The statement is false, because $x = x + 1$ simplifies to $0 = 1$, which has no solution over the real numbers.

2.3 Quantifiers with Restricted Domains

Sometimes quantifiers are written with restricted domains, e.g., $\forall x < 0$, $\exists z > 0$. These can be rewritten using conditional or conjunctive forms.

Example. Rewrite the following statements into equivalent statements, where the domain consists of all real numbers.

(a) $\forall x < 0\ (x^2 > 0)$ — "The square of a negative real number is positive." This is equivalent to $\forall x\, (x < 0 \to x^2 > 0)$, which is true.

(b) $\forall y \neq 0\ (y^3 \neq 0)$ — "The cube of every nonzero real number is nonzero." This is equivalent to $\forall y\, (y \neq 0 \to y^3 \neq 0)$, which is true.

(c) $\exists z > 0\ (z^2 = 2)$ — "There is a positive square root of 2." This is equivalent to $\exists z\, (z > 0 \wedge z^2 = 2)$, which is true (namely $z = \sqrt{2}$).

General Rule for Restricted Domains

Let U_1 and U_2 be two domains with $U_1 \subseteq U_2$. Define $Q(x)$ such that $U_1 = \{x \in U_2 \mid Q(x) \text{ is true}\}$. Then:
(a) $\forall x \in U_1\, (P(x)) \equiv \forall x \in U_2\, (Q(x) \to P(x))$
(b) $\exists x \in U_1\, (P(x)) \equiv \exists x \in U_2\, (Q(x) \wedge P(x))$

Example. Let U_1 be all CSE students and U_2 be all HKUST undergraduates. Let $Q(x)$: "x is a CSE student" (so $U_1 = \{x \in U_2 \mid Q(x)\}$) and $P(x)$: "x is required to take COMP2711". Then

$$\forall x \in U_1 \, (P(x)) \equiv \forall x \in U_2 \, (Q(x) \to P(x)).$$

Example. Consider the following argument:

- Premise 1: "All lions are fierce."
- Premise 2: "Some lions do not drink coffee."
- Conclusion: "Some fierce creatures do not drink coffee."

Let $P(x)$: "x is a lion", $Q(x)$: "x is fierce", $R(x)$: "x drinks coffee", with the domain of all creatures. Then:

$$\forall x \, (P(x) \to Q(x))$$
$$\exists x \, (P(x) \wedge \neg R(x))$$
$$\therefore \exists x \, (Q(x) \wedge \neg R(x))$$

We will show in L03 that this argument is valid.

Important note: "Some lions do not drink coffee" cannot be written as $\exists x \, (P(x) \to \neg R(x))$. The reason is that $P(x) \to \neg R(x)$ is true whenever x is not a lion, so $\exists x \, (P(x) \to \neg R(x))$ is true as long as there is at least one creature that is not a lion, even if every lion drinks coffee.

2.3.1 Precedence of Quantifiers and Binding Variables

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators.

Example. $\forall x \, P(x) \vee Q(x)$ means $(\forall x \, P(x)) \vee Q(x)$, which is not a valid proposition if x in $Q(x)$ is not bound.

A variable can be:

- **bound** by a quantifier (the part of a logical expression bound by a quantifier is called its **scope**);
- set to a particular value;
- otherwise, **free**.

All variables in a propositional function must be bound or set to a particular value to turn it into a proposition.

Example. In the expression $\exists x \, P(x) \wedge Q(x) \vee \forall x \, R(x) \vee S(x)$, the quantifiers bind the occurrences of x within their respective scopes, and the final $S(x)$ would need its own quantifier or value assignment.

2.4 Logical Equivalences Involving Quantifiers

Definition (Logical Equivalence with Quantifiers)

Statements involving predicates and quantifiers are **logically equivalent** if and only if they have the same truth value under every interpretation, i.e., no matter which predicates are substituted into these statements and which (common) domain is used for the variables in these propositional functions. We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Example. Show that $\forall x\,(P(x) \wedge Q(x))$ and $\forall x\,P(x) \wedge \forall x\,Q(x)$ are logically equivalent.

Proof. We prove this bi-implication in two directions.

($\Rightarrow$) Suppose $\forall x\,(P(x) \wedge Q(x))$ is true. That is, if a is in the domain, then $P(a)$ and $Q(a)$ are both true. Thus, $\forall x\,P(x)$ is true and so is $\forall x\,Q(x)$. Therefore $\forall x\,P(x) \wedge \forall x\,Q(x)$ is true.

($\Leftarrow$) Suppose $\forall x\,P(x) \wedge \forall x\,Q(x)$ is true. Then $\forall x\,P(x)$ is true and $\forall x\,Q(x)$ is true. Thus, if a is in the domain, then $P(a)$ is true and $Q(a)$ is true. Therefore $P(a) \wedge Q(a)$ is true for all a. Thus $\forall x\,(P(x) \wedge Q(x))$ is true.

■

This equivalence shows that we can distribute a universal quantifier over a conjunction:

$$\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x).$$

We can also distribute an existential quantifier over a disjunction:

$$\exists x\,(P(x) \vee Q(x)) \equiv \exists x\,P(x) \vee \exists x\,Q(x).$$

However, we cannot distribute a universal quantifier over a disjunction, nor can we distribute an existential quantifier over a conjunction:

$$\forall x\,(P(x) \vee Q(x)) \not\equiv \forall x\,P(x) \vee \forall x\,Q(x),$$
$$\exists x\,(P(x) \wedge Q(x)) \not\equiv \exists x\,P(x) \wedge \exists x\,Q(x).$$

Counterexample for $\forall x\,(P(x) \vee Q(x)) \Rightarrow \forall x\,P(x) \vee \forall x\,Q(x)$. Let the domain be $\{a, b, c\}$ with truth values:

	a	b	c
P	T	F	T
Q	F	T	T

Then $P(a) \vee Q(a) = T$, $P(b) \vee Q(b) = T$, $P(c) \vee Q(c) = T$, so $\forall x\,(P(x) \vee Q(x))$ is true. However, $\forall x\,P(x)$ is false (since $P(b) = F$) and $\forall x\,Q(x)$ is false (since $Q(a) = F$), so $\forall x\,P(x) \vee \forall x\,Q(x)$ is false. Therefore the implication does not hold in reverse.

Counterexample for $\exists x\,(P(x) \wedge Q(x)) \Leftarrow \exists x\,P(x) \wedge \exists x\,Q(x)$. Let the domain be $\{a, b\}$ with $P(a) = T, P(b) = F$ and $Q(a) = F, Q(b) = T$. Then $\exists x\,P(x)$ is true and $\exists x\,Q(x)$ is true, so $\exists x\,P(x) \wedge \exists x\,Q(x)$ is true. However, there is no element in the domain for which both $P(x)$ and $Q(x)$ are true, so $\exists x\,(P(x) \wedge Q(x))$ is false.

Counterexample with integer domain. $P(x)$: $2x + 1 = 5$ (true when $x = 2$); $Q(x)$: $x^2 = 9$ (true when $x = 3$). Then $\exists x\,P(x) \wedge \exists x\,Q(x)$ is true, but $\exists x\,(P(x) \wedge Q(x))$ is false because there is no single integer satisfying both equations.

2.5 Negating Quantified Expressions

Motivation. Let $P(x)$ denote "student x has an iPhone." The domain is all students in this class.

- "Every student in the class has an iPhone": $\forall x\,P(x)$. Its negation: "It is not the case that every student has an iPhone", i.e., "there is at least one student who does not have an iPhone": $\exists x\,\neg P(x)$.
- "There is a student in the class who has an iPhone": $\exists x\,P(x)$. Its negation: "It is not the case that some student has an iPhone", i.e., "every student does not have an iPhone": $\forall x\,\neg P(x)$.

De Morgan's Laws for Quantifiers

$$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$$
$$\neg\exists x\, Q(x) \equiv \forall x\, \neg Q(x)$$

When the domain has n elements $x_1, x_2, \ldots, x_n$:

- $\neg\forall x\, P(x)$ is the same as $\neg(P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n))$, which by De Morgan's laws is $\neg P(x_1) \vee \neg P(x_2) \vee \cdots \vee \neg P(x_n)$, i.e., $\exists x\, \neg P(x)$.
- $\neg\exists x\, P(x)$ is the same as $\neg(P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n))$, which by De Morgan's laws is $\neg P(x_1) \wedge \neg P(x_2) \wedge \cdots \wedge \neg P(x_n)$, i.e., $\forall x\, \neg P(x)$.

Negation	Equivalent statement	When is it true?	When is it false?
$\neg\forall x\, P(x)$	$\exists x\, \neg P(x)$	There exists an x for which $P(x)$ is false.	$P(x)$ is true for every x.
$\neg\exists x\, Q(x)$	$\forall x\, \neg Q(x)$	$Q(x)$ is false for every x.	There exists an x for which $Q(x)$ is true.

Example. What is the negation of $\forall x\, (x^2 > x)$?

Solution. $\neg\forall x\, (x^2 > x) \equiv \exists x\, \neg(x^2 > x) \equiv \exists x\, (x^2 \leq x)$.

Example. What is the negation of $\exists x\, (x^2 = 2)$?

Solution. $\neg\exists x\, (x^2 = 2) \equiv \forall x\, \neg(x^2 = 2) \equiv \forall x\, (x^2 \neq 2)$.

Example. Show that $\neg\forall x\, (P(x) \to Q(x))$ and $\exists x\, (P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution.

$$\begin{aligned}
\neg\forall x\, (P(x) \to Q(x)) &\equiv \exists x\, \neg(P(x) \to Q(x)) && \text{(De Morgan's law for quantifiers)}\\
&\equiv \exists x\, \neg(\neg P(x) \vee Q(x)) && \text{(implication law)}\\
&\equiv \exists x\, (P(x) \wedge \neg Q(x)) && \text{(De Morgan's law)}
\end{aligned}$$

2.6 Nested Quantifiers

Example. Assume that the domain for the variables x and y consists of all real numbers. The statement

$$\forall x\, \forall y\, (x + y = y + x)$$

says that $x + y = y + x$ for all real numbers x and y. This is the commutative law for the addition of real numbers.

2.6.1 Order of Quantifiers

The order in which quantifiers appear can make a difference when they are of different types.

Example. The statement

$$\forall x\, \exists y\, (x + y = 0)$$

says that for every real number x, there is a real number y such that $x + y = 0$. This states that every real number has an additive inverse, which is true.

What about

$$\exists y\, \forall x\, (x + y = 0)?$$

It is false, since there is no single value of y that satisfies $x + y = 0$ for all values of x.

Quantifications of Two Variables

Statement	When is it true?	When is it false?
$\forall x \forall y\, P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x, y for which $P(x,y)$ is false.
$\forall x \exists y\, P(x,y)$	For every x there is a y for which $P(x,y)$ is true.	There is an x such that $P(x,y)$ is false for every y.
$\exists x \forall y\, P(x,y)$	There is an x such that $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
$\exists x \exists y\, P(x,y)$	There is a pair x, y for which $P(x,y)$ is true.	$P(x,y)$ is false for every pair x, y.

Example. Let $Q(x,y,z)$ be the statement "$x + y = z$". What are the truth values of $\forall x \forall y \exists z\, Q(x,y,z)$ and $\exists z \forall x \forall y\, Q(x,y,z)$, where the domain is all real numbers?

Solution.

- For the first statement, suppose x and y are assigned values. Then there exists a real number z (namely $z = x + y$) such that $x + y = z$. Thus the first statement is true.
- For the second statement, there is no single value of z that satisfies $x + y = z$ for all values of x and y. Thus the second statement is false.

2.6.2 Translating Between English and Logic

Example. Translate "the sum of two positive integers is always positive" into a logical expression.

Solution. Domain: all integers.

$$\forall x \forall y\, (x > 0 \wedge y > 0 \rightarrow x + y > 0).$$

Example. Translate "every nonzero real number has a multiplicative inverse" into a logical expression.

Solution. Domain: all real numbers.

$$\forall x\, (x \neq 0 \rightarrow \exists y\, (xy = 1)).$$

Example. Translate $\forall x\, (C(x) \vee \exists y\, (C(y) \wedge F(x,y)))$ into English, where $C(x)$ is "x has a computer", $F(x,y)$ is "x and y are friends", and the domain for both x and y consists of all students in the school.

Solution. For every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends. In other words: "Every student either has a computer or has a friend who has a computer."

Example. Translate $\exists x \forall y \forall z\, (F(x,y) \wedge F(x,z) \wedge (y \neq z) \rightarrow \neg F(y,z))$ into English, where $F(a,b)$ means a and b are friends and the domain consists of all students in the school.

Solution. The statement says that there is a student x such that for all students y and all students z other than y, if x and y are friends and x and z are friends, then y and z are not friends. In other words: "There is a student none of whose friends are also friends with each other."

Example. Express "if a person is female and is a parent, then this person is someone's mother" as a logical expression using the predicates $F(x)$: "x is female", $P(x)$: "x is a parent", $M(x,y)$: "x is y's mother", with domain of all people.

Solution.

$$\forall x\,(F(x) \wedge P(x) \rightarrow \exists y\, M(x,y)).$$

An equivalent form is $\forall x\,\exists y\,(F(x) \wedge P(x) \rightarrow M(x,y))$.

2.6.3 Negating Nested Quantifiers

Example. Express the negation of $\forall x\,\exists y\,(xy = 1)$ so that no negation precedes a quantifier.

Solution.

$$\begin{aligned}\neg\forall x\,\exists y\,(xy=1) &\equiv \exists x\,\neg\exists y\,(xy=1)\\ &\equiv \exists x\,\forall y\,\neg(xy=1)\\ &\equiv \exists x\,\forall y\,(xy\neq 1).\end{aligned}$$

Example. Use quantifiers to express "there is a woman who has taken a flight on every airline in the world".

Solution. Let $P(w,f)$ be "woman w has taken flight f" and $Q(f,a)$ be "f is a flight on airline a".

$$\exists w\,\forall a\,\exists f\,(P(w,f) \wedge Q(f,a)).$$

The negation, with no negation preceding a quantifier, is:

$$\forall w\,\exists a\,\forall f\,(\neg P(w,f) \vee \neg Q(f,a)).$$

Example. Express "everyone has exactly one best friend" as a logical expression using $B(x,y)$: "y is x's best friend", domain = all people.

Solution. "x has exactly one best friend" can be represented as:

$$\exists y\,(B(x,y) \wedge \forall z\,((z \neq y) \rightarrow \neg B(x,z))).$$

There is a person y who is the best friend of x, and for every person z, if z is not person y, then z is not the best friend of x. Thus the original statement is:

$$\forall x\,\exists y\,(B(x,y) \wedge \forall z\,((z \neq y) \rightarrow \neg B(x,z))).$$

2.6.4 Null Quantifications

If x does not occur as a free variable in A, then:

$$\begin{aligned}\forall x\,A \wedge P(x) &\equiv A \wedge \forall x\,P(x)\\ \forall x\,A \vee P(x) &\equiv A \vee \forall x\,P(x)\\ \exists x\,A \wedge P(x) &\equiv A \wedge \exists x\,P(x)\\ \exists x\,A \vee P(x) &\equiv A \vee \exists x\,P(x)\end{aligned}$$

Proof sketch (for the first rule). Let the domain of x be $\{x_1,\ldots,x_n\}$.

$$\begin{aligned}\forall x\,(A \wedge P(x)) &\equiv (A \wedge P(x_1)) \wedge \cdots \wedge (A \wedge P(x_n))\\ &\equiv A \wedge P(x_1) \wedge \cdots \wedge P(x_n)\\ &\equiv A \wedge \forall x\,P(x).\end{aligned}$$

Example. $\forall x\,P(x) \vee \forall x\,Q(x) \equiv \forall x\,P(x) \vee \forall y\,Q(y) \equiv \forall x\,\forall y\,(P(x) \vee Q(y))$. Similarly, $\exists x\,P(x) \vee \forall y\,Q(y) \equiv \exists x\,\forall y\,(P(x) \vee Q(y)) \equiv \forall y\,\exists x\,(P(x) \vee Q(y))$. In this particular case, $\exists x$ and $\forall y$ can be swapped (in general they cannot).

Example (caution). One might guess that $\forall x\, P(x) \to \exists y\, Q(y) \equiv \forall x\, \exists y\, (P(x) \to Q(y))$, but this is **not** correct. The actual equivalences are:

$$\begin{aligned}\forall x\, P(x) \to \exists y\, Q(y) &\equiv \neg\forall x\, P(x) \vee \exists y\, Q(y)\\ &\equiv \exists x\, \neg P(x) \vee \exists y\, Q(y)\\ &\equiv \exists x\, \exists y\, (\neg P(x) \vee Q(y))\\ &\equiv \exists x\, \exists y\, (P(x) \to Q(y)).\end{aligned}$$

On the other hand:

$$\begin{aligned}\forall x\, \exists y\, (P(x) \to Q(y)) &\equiv \forall x\, \exists y\, (\neg P(x) \vee Q(y))\\ &\equiv \forall x\, (\neg P(x) \vee \exists y\, Q(y))\\ &\equiv \neg\exists x\, P(x) \vee \exists y\, Q(y)\\ &\equiv \exists x\, P(x) \to \exists y\, Q(y).\end{aligned}$$

The first statement is equivalent to $\exists x\, \exists y\, (P(x) \to Q(y))$, while the second is equivalent to $\exists x\, P(x) \to \exists y\, Q(y)$. These are different in general.

Counterexample. Consider the domain $\{a, b\}$ with $P(a) = T$, $P(b) = F$, and $Q(a) = Q(b) = F$. Then:

- $\forall x\, P(x) \to \exists y\, Q(y)$: Since $P(b) = F$, $\forall x\, P(x)$ is false, so the implication $F \to F$ is **true**.
- $\forall x\, \exists y\, (P(x) \to Q(y))$: At $x = a$, $P(a) = T$, so we need $\exists y\, Q(y)$, but Q is false everywhere, so the inner statement is false at $x = a$. Hence the whole $\forall x$ statement is **false**.

Therefore the two statements are not logically equivalent.

3 Rules of Inference

Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an **argument**, we mean a sequence of statements that end with a conclusion. By **valid**, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or **premises**, of the argument. That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false.

To deduce new statements from statements we already have, we use **rules of inference**, which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements.

Reading: Kenneth Rosen, Sections 1.6, 1.8.

3.1 Rules of Inference for Propositional Logic

Definition (Argument, Valid Argument)

An **argument** in propositional logic is a sequence of propositions. All but the final proposition in the argument are called **premises** or **hypotheses** and the final proposition is called the **conclusion**.
An argument is **valid** if the truth of all its premises implies that the conclusion is true.

Remark. An argument with premises $p_1, p_2, \ldots, p_n$ and conclusion q is valid when $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \to q$ is a tautology. Using a truth table to show the validity of an argument form can be very tedious (with n variables, the table has 2^n rows). This process can be simplified significantly by using rules of inference.

3.1.1 Table of Rules of Inference

Rules of Inference for Propositional Logic

Modus ponens:

$$\begin{array}{l} p \\ p \to q \\ \hline \therefore\ q \end{array} \qquad \text{Tautology: } [p \wedge (p \to q)] \to q$$

Modus tollens:

$$\begin{array}{l} \neg q \\ p \to q \\ \hline \therefore\ \neg p \end{array} \qquad \text{Tautology: } [\neg q \wedge (p \to q)] \to \neg p$$

Hypothetical syllogism:

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore\ p \to r \end{array} \qquad \text{Tautology: } [(p \to q) \wedge (q \to r)] \to (p \to r)$$

Disjunctive syllogism:

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore\ q \end{array} \qquad \text{Tautology: } [(p \vee q) \wedge \neg p] \to q$$

Addition:

$$\begin{array}{l} p \\ \hline \therefore\ p \vee q \end{array} \qquad \text{Tautology: } p \to (p \vee q)$$

Simplification:

$$\begin{array}{l} p \wedge q \\ \hline \therefore\ p \end{array} \qquad \text{Tautology: } (p \wedge q) \to p$$

Conjunction:

$$\begin{array}{l} p \\ q \\ \hline \therefore\ p \wedge q \end{array} \qquad \text{Tautology: } [(p) \wedge (q)] \to (p \wedge q)$$

Resolution:

$$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore\ q \vee r \end{array} \qquad \text{Tautology: } [(p \vee q) \wedge (\neg p \vee r)] \to (q \vee r)$$

Plus any logical equivalence.

3.1.2 Examples

Example. Show that the hypotheses "it is not snowing or Jasmine is skiing" and "it is snowing or Bart is playing hockey" imply that "Jasmine is skiing or Bart is playing hockey".

Solution. Let s: "it is snowing", k: "Jasmine is skiing", h: "Bart is playing hockey". The premises are $\neg s \vee k$ and $s \vee h$. By **resolution**:

$$\neg s \vee k, \quad s \vee h \ \vdash\ k \vee h.$$

Hence the conclusion "Jasmine is skiing or Bart is playing hockey" follows.

Example. Show that the hypotheses $(p \wedge q) \vee r$ and $r \to s$ imply the conclusion $p \vee s$.

Solution.

1. $(p \wedge q) \vee r$	Premise
2. $r \to s$	Premise
3. $\neg r \vee s$	From 2, equivalence (implication law)
4. $(p \wedge q) \vee s$	From 1, 3, resolution
5. $(p \vee s) \wedge (q \vee s)$	From 4, equivalence (distributive law)
6. $p \vee s$	From 5, simplification

Example. Show that the following premises lead to the conclusion.

Propositions:

- p: "it is sunny this afternoon."
- q: "it is colder than yesterday."
- r: "we will go swimming."
- s: "we will take a canoe trip."
- t: "we will be home by sunset."

Premises:

- "It is not sunny this afternoon and it is colder than yesterday": $\neg p \wedge q$
- "We will go swimming only if it is sunny": $r \to p$
- "If we do not go swimming, then we will take a canoe trip": $\neg r \to s$
- "If we take a canoe trip, then we will be home by sunset": $s \to t$

Conclusion: t: "we will be home by sunset".

Solution.

1. $\neg p \wedge q$	Premise
2. $\neg p$	From 1, simplification
3. $r \to p$	Premise
4. $\neg r$	From 2, 3, modus tollens
5. $\neg r \to s$	Premise
6. s	From 4, 5, modus ponens
7. $s \to t$	Premise
8. t	From 6, 7, modus ponens

Remark. We could have used a truth table to show that whenever each of the four premises is true, the conclusion is also true. However, because we are working with five propositional variables p, q, r, s, t, such a truth table would have $2^5 = 32$ rows.

3.1.3 Invalid Arguments

Example. Is the following argument valid?

Premises: $p \to r$, $q \to r$, $\neg(p \vee q)$. Conclusion: $\neg r$.

Solution. The argument is **invalid**. The truth assignment $p = F$, $q = F$, $r = T$ makes all the premises true:

- $p \to r$: $F \to T = T$
- $q \to r$: $F \to T = T$
- $\neg(p \vee q)$: $\neg(F \vee F) = \neg F = T$

but the conclusion $\neg r = \neg T = F$ is false. Hence it is possible for all premises to be true while the conclusion is false, so the argument is invalid.

3.2 Rules of Inference for Predicate Logic

Rules of Inference for Predicate Logic

Universal instantiation (UI):

$$\frac{\forall x\, P(x)}{\therefore\ P(c)}$$

Universal generalization (UG):

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore\ \forall x\, P(x)}$$

Existential instantiation (EI):

$$\frac{\exists x\, P(x)}{\therefore\ P(c) \text{ for some element } c}$$

Existential generalization (EG):

$$\frac{P(c) \text{ for some element } c}{\therefore\ \exists x\, P(x)}$$

Gödel's completeness theorem: These rules (together with the propositional rules) are enough to derive all valid conclusions.

3.2.1 Examples

Example. Show that the premises "everyone in this discrete mathematics class has taken a course in computer science" and "Joseph is a student in this class" imply the conclusion "Joseph has taken a course in computer science". Use the predicates $P(x)$: "x is a student in the discrete mathematics class" and $C(x)$: "x has taken a course in computer science".

Solution.

1. $\forall x\,(P(x) \to C(x))$	Premise
2. $P(\text{Joseph}) \to C(\text{Joseph})$	From 1, universal instantiation
3. $P(\text{Joseph})$	Premise
4. $C(\text{Joseph})$	From 2, 3, modus ponens

Example. Assume that "for all positive integers n, if n is greater than 4, then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2 < 2^{100}$.

Solution. Let $P(n)$: "$n > 4$" and $Q(n)$: "$n^2 < 2^n$". We are given $\forall n\,(P(n) \to Q(n))$. By universal instantiation, $P(100) \to Q(100)$. Since $100 > 4$, $P(100)$ is true. By modus ponens, $Q(100)$ is true, i.e., $100^2 < 2^{100}$.

Example. Show that the premises "a student in this class has not read the textbook" and "everyone in this class passed the course" imply the conclusion "someone who passed the course has not read the textbook". Use predicates $P(x)$: "x is a student in this class", $R(x)$: "x has read the textbook", $U(x)$: "x passed the course".

Solution.

1. $\exists x\,(P(x) \wedge \neg R(x))$	Premise
2. $P(a) \wedge \neg R(a)$ for some a	From 1, existential instantiation
3. $P(a)$	From 2, simplification
4. $\neg R(a)$	From 2, simplification
5. $\forall x\,(P(x) \to U(x))$	Premise
6. $P(a) \to U(a)$	From 5, universal instantiation
7. $U(a)$	From 3, 6, modus ponens
8. $U(a) \wedge \neg R(a)$	From 4, 7, conjunction
9. $\exists x\,(U(x) \wedge \neg R(x))$	From 8, existential generalization

3.3 Basic Proof Techniques

3.3.1 Some Terminology

Terminology

- A **theorem** is a statement that can be shown to be true.
- An **axiom** is a statement that is assumed to be true.
- A less important theorem that is helpful in the proof of other theorems is called a **lemma**.
- A **proof** is a valid argument that establishes the truth of a theorem. The statements used in a proof can include axioms, premises of the theorem, and previously proved theorems or lemmas. Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.
- A **corollary** is a theorem that can be established directly from a theorem that has been proved.
- A **conjecture** is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.

Remark. When a proof of a conjecture is found, the conjecture becomes a theorem. However, many conjectures are eventually found to be false.

3.3.2 The Axiomatic Method

Examples of axiomatic systems:

- **Euclidean geometry:** based on 5 axioms. Axiom 1: There is a straight line segment between every pair of points.
- **Peano axioms:** an axiomatic system for number theory, based on 5 axioms.
- **ZFC (Zermelo–Fraenkel with Choice):** 9 axioms, sufficient to derive essentially all of mathematics. But writing proofs using these axioms directly is not practical.

We will accept all familiar facts from high school math (unless specified otherwise).

Peano's Axioms.

1. $0 \in \mathbb{N}$
2. $\forall x\,(x \in \mathbb{N} \to S(x) \in \mathbb{N})$ $\quad (S = \text{successor})$
3. $\neg\exists x\,(S(x) = 0)$
4. $\forall x\,\forall y\,(S(x) = S(y) \to x = y)$
5. $\forall K \subseteq \mathbb{N}\,\big(0 \in K \wedge \forall x\,(x \in K \to S(x) \in K) \to K = \mathbb{N}\big)$ (a.k.a. induction; expressed in second-order logic)

Definition of Arabic numbers: $1 := S(0)$, $2 := S(S(0))$, ...

Definition of addition: $x + 0 := x$; $x + S(y) := S(x + y)$.

Example. A proof of $1 + 1 = 2$ from Peano's axioms.

Proof.

- By definition, $1 + 1 = 2$ is written as $S(0) + S(0) = S(S(0))$.
- By the recursive definition of addition: $S(0) + S(0) = S(S(0) + 0)$.
- By the base case of addition: $S(0) + 0 = S(0)$.
- Substituting back, we get $S(S(0) + 0) = S(S(0))$.

Hence $1 + 1 = 2$. ■

Example. Proving $\forall x \in \mathbb{N}\,(x + 0 = 0 + x)$.

Proof. Define $K := \{x \in \mathbb{N} \mid x + 0 = 0 + x\}$.

- $0 + 0 = 0 + 0$, so $0 \in K$.
- Prove $\forall x\,(x \in K \to S(x) \in K)$: Consider any $x \in K$, i.e., $x + 0 = 0 + x$. Then

$$\begin{aligned} S(x) + 0 &= S(x) && \text{(base definition of addition)} \\ &= S(x + 0) && \text{(base definition of addition)} \\ &= S(0 + x) && \text{(because } x \in K\text{)} \\ &= 0 + S(x) && \text{(recursive definition of addition)} \end{aligned}$$

  So $S(x) \in K$.
- By the induction axiom, $K = \mathbb{N}$, i.e., $\forall x \in \mathbb{N}\,(x + 0 = 0 + x)$. ■

3.3.3 Direct Proof

Direct Proof

A **direct proof** of a conditional statement $p \to q$: the first step is the assumption that p is true. Subsequent steps are constructed using axioms, definitions, previously proved theorems, and rules of inference, with the final step showing that q must also be true.

Example. Give a direct proof of the theorem "if n is an odd natural number, then n^2 is odd". (An integer n is odd if there exists a natural number k such that $n = 2k + 1$.)

Proof. Assume that the hypothesis of this implication is true; namely, suppose that n is odd. Then $n = 2k + 1$, where k is a natural number. It follows that

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is a natural number, n^2 is odd. ■

A formal version of the same proof. Theorem: $\forall n \left(\exists k\,(n = 2k + 1) \to \exists k'\,(n^2 = 2k' + 1)\right)$.

Proof.

(1) Let a be an arbitrary element in the domain ($\mathbb{N}$).
(2) Assume $\exists k\,(a = 2k + 1)$ is true.
(3) For some $b \in \mathbb{N}, a = 2b + 1$ (existential instantiation)
(4) $a^2 = (2b + 1)^2 = 2(2b^2 + 2b) + 1$ (high school math)
(5) $c := 2b^2 + 2b \in \mathbb{N}$ (high school math)
(6) $\exists k'\,(a^2 = 2k' + 1)$ (existential generalization)
(7) $\exists k\,(a = 2k + 1) \to \exists k'\,(a^2 = 2k' + 1)$
(8) $\forall n \left(\exists k\,(n = 2k + 1) \to \exists k'\,(n^2 = 2k' + 1)\right)$ (universal generalization)

Example. Give a direct proof that if m and n are both perfect squares, then mn is also a perfect square. (An integer a is a perfect square if there exists an integer b such that $a = b^2$.)

Proof. Assume m and n are both perfect squares. Then there exist integers s and t such that $m = s^2$ and $n = t^2$. It follows that

$$mn = s^2t^2 = (st)^2.$$

Since st is an integer, mn is a perfect square. ■

Example. Prove that the sum of two rational numbers is rational. (A real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called irrational.)

Proof. Let r and s be rational numbers. Then there exist integers p, q with $q \neq 0$ such that $r = p/q$, and integers t, u with $u \neq 0$ such that $s = t/u$. Then

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu}.$$

Since $pu + qt$ is an integer and $qu \neq 0$ is an integer, $r + s$ is rational. ■

Remark (Limitation of direct proofs). Direct proofs are useful but attempts at direct proofs sometimes lead to dead ends. There are other proof techniques. Proofs that are not direct proofs, i.e., that do not start with the hypothesis and end with the conclusion, are called **indirect proofs**. We will consider several types of indirect proofs.

3.3.4 Proof by Contraposition

Proof by Contraposition

A **proof by contraposition** makes use of the fact that the conditional statement $p \to q$ is equivalent to its contrapositive $\neg q \to \neg p$. This means that $p \to q$ can be proved by showing that $\neg q \to \neg p$ is true. To do so, we take $\neg q$ as a hypothesis, and using axioms, definitions, previously proved theorems, and rules of inference, we show that $\neg p$ must follow.

Example. Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Proof. Assume that the conclusion of this implication is false; namely, assume that n is even. Then $n = 2k$ for some integer k. It follows that

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

Since $3k + 1$ is an integer, $3n + 2$ is even. Thus the hypothesis "$3n + 2$ is odd" is false. By contraposition, the original implication is true. ■

Example. Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Proof. We prove the contrapositive: if $a > \sqrt{n}$ and $b > \sqrt{n}$, then $n \neq ab$. Suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Multiplying these inequalities gives $ab > \sqrt{n} \cdot \sqrt{n} = n$, so $ab \neq n$. By contraposition, if $n = ab$ then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. ■

3.3.5 Proof by Contradiction

Proof by Contradiction

Suppose we want to prove that a statement p is true. Instead, we assume p is false, i.e., $\neg p$ is true. Then, using axioms, definitions, previously proved theorems, and rules of inference, we derive a contradiction F. This means that our assumption that $\neg p$ is true is false. Consequently, p must be true.
An example of a contradiction is $r \wedge \neg r$, where r is any proposition.

Example. Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Proof. Suppose $\sqrt{2}$ is rational, i.e., there exist $a, b \in \mathbb{N}$ such that $\sqrt{2} = a/b$ and a and b have no common factors. We will show that this leads to a contradiction.

$$\sqrt{2} = \frac{a}{b} \implies 2 = \frac{a^2}{b^2} \implies 2b^2 = a^2 \implies a^2 \text{ is even} \implies a \text{ is even}.$$

Therefore there exists an integer c such that $a = 2c$. Substituting back:

$$2b^2 = a^2 \implies 2b^2 = (2c)^2 \implies 2b^2 = 4c^2 \implies b^2 = 2c^2.$$

Therefore b^2 is even, thus b is even. We have shown that 2 is a common factor of a and b, contradicting the assumption that a and b have no common factors. Hence $\sqrt{2}$ is irrational. ■

3.3.6 Proof by Contradiction for Conditional Statements

Proofs by contradiction can be used to prove conditional statements $p \to q$. We first assume that the negation of the conclusion is true. We then use the premises of the theorem and the negation of the conclusion to arrive at a contradiction. That is, we show $(p \wedge \neg q) \to F$. The validity of such proofs is based on the logical equivalence of $p \to q$ and $(p \wedge \neg q) \to F$.

We can rewrite a proof by contraposition of $p \to q$ as a proof by contradiction. In a proof by contraposition, we assume $\neg q$ is true and then show $\neg p$ must also be true. To rewrite as a proof by contradiction, we suppose that both p and $\neg q$ are true. Then, we use the steps from the proof of $\neg q \to \neg p$ to show that $\neg p$ is true. This leads to the contradiction $p \wedge \neg p$, completing the proof.

Example (Proof by contradiction vs. proof by contraposition). Prove by contradiction: "if $3n + 2$ is odd, then n is odd".

Proof. We assume $3n + 2$ is odd (p) and n is not odd, i.e., n is even $(\neg q)$. Following the same steps as in the contraposition proof: $n = 2k$ for some integer k, so $3n + 2 = 3(2k) + 2 = 2(3k + 1)$, which is even. Therefore $3n + 2$ is even $(\neg p)$, contradicting the assumption that $3n + 2$ is odd. The contradiction $p \wedge \neg p$ completes the proof. ■

Side-by-side comparison. Goal: Prove that if $3n + 2$ is odd (p), then n is odd (q).

By contraposition: Assume n is even $(\neg q)$. Then $n = 2k$ for some integer k. It follows that $3n + 2 = 3(2k) + 2 = 2(3k + 1)$. Therefore $3n + 2$ is even $(\neg p)$.

By contradiction: Assume $3n + 2$ is odd (p) and n is even $(\neg q)$. Then $n = 2k$ for some integer k. It follows that $3n + 2 = 2(3k + 1)$. Therefore $3n + 2$ is even $(\neg p)$, contradicting the assumption that $3n + 2$ is odd $(p \wedge \neg p)$.

3.3.7 Proving Biconditional Statements

To prove a theorem that is a biconditional statement of the form $p \leftrightarrow q$, we show that both $p \to q$ and $q \to p$ are true.

Example. Prove the theorem "if n is a positive integer, then n is odd if and only if n^2 is odd".

Proof.

- ($\Rightarrow$) If n is odd, then $n = 2k+1$ for some integer k, so $n^2 = (2k+1)^2 = 2(2k^2+2k)+1$ is odd. (Direct proof; done earlier.)
- ($\Leftarrow$) We prove the contrapositive of "n^2 is odd $\to$ n is odd", i.e., if n is even then n^2 is even. If n is even, then $n = 2k$, so $n^2 = 4k^2 = 2(2k^2)$ is even.

Since both directions hold, n is odd if and only if n^2 is odd. ■

Example. Show that these statements about the integer n are logically equivalent: p_1: "n is even"; p_2: "$n-1$ is odd"; p_3: "n^2 is even".

Solution. We prove the implications $p_1 \to p_2$, $p_2 \to p_3$, and $p_3 \to p_1$.

$p_1 \to p_2$**:** If n is even, then $n = 2k$ for some integer k. Then $n-1 = 2k-1 = 2(k-1)+1$ is odd.

$p_2 \to p_3$**:** If $n-1$ is odd, then $n-1 = 2k+1$ for some integer k, so $n = 2k+2 = 2(k+1)$ is even. Then $n^2 = 4(k+1)^2 = 2 \cdot 2(k+1)^2$ is even.

$p_3 \to p_1$**:** We prove the contrapositive: if n is odd, then n^2 is odd. If n is odd, $n = 2k+1$, so $n^2 = (2k+1)^2 = 2(2k^2+2k)+1$ is odd.

Since $p_1 \to p_2 \to p_3 \to p_1$, the three statements are logically equivalent. ■

3.3.8 Nonconstructive Existence Proof

Example. Show that there exist irrational numbers x and y such that x^y is rational.

Proof. We have shown that $\sqrt{2}$ is irrational. Consider $\sqrt{2}^{\sqrt{2}}$.

- If it is rational, we are done: take $x = y = \sqrt{2}$.
- If it is irrational, consider $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$. Then

$$x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2,$$

which is rational.

In either case, there exist irrational x, y with x^y rational. Note that the proof is **nonconstructive**: it does not tell us which specific pair works. ■

3.4 Some Comments on Proofs

- There are many other proof methods. We will cover some later; many will not be covered in this course.
- Constructing proofs is an art that can be learned only by trying various lines of attack. There are no fixed procedures for proving theorems.
- Many statements that appear to be theorems have resisted the persistent efforts of mathematicians for hundreds of years. For instance, **Goldbach's conjecture**: "every even positive integer greater than 2 is the sum of two primes" has not yet been proved, and no counterexample has been found.

3.4.1 Theorems and Proofs (Hilbert, Gödel)

- **Hilbert's program:** A computer program that, starting from a finite number of axioms and using the rules of inference, finds a proof of any true statement.
- **Gödel's incompleteness theorem:** In any consistent axiomatic system containing basic arithmetic, there are true statements that cannot be proved. In particular, its own consistency cannot be proved.
- Even for theorems with proofs, finding the proof in a huge search space is very computationally expensive.
- There are several automated theorem proving systems that can prove certain subsets of mathematics.
- Proofs require human insight!

4 Sets and Functions

Reading: Rosen 2.1, 2.2, 2.3, 2.5.

4.1 Sets

Definition. A *set* is an unordered collection of objects. The objects in a set are called the *elements* or *members* of the set, and the set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A, and $a \notin A$ to denote that a is not an element of A.

Roster method. List all elements of a set explicitly, using "..." when the pattern is obvious.

Example. The set of all odd positive integers less than 10 is $\{1, 3, 5, 7, 9\}$; the set of all odd positive integers less than 100 is $\{1, 3, 5, \ldots, 99\}$.

Set-builder notation. A set can be specified through a predicate $P(x)$:

$$\{x \in E \mid P(x)\} \quad \text{or} \quad \{x \in E : P(x)\},$$

which is the set of all $x \in E$ such that $P(x) = T$. For example, the set of odd positive integers less than 100 can be written as $\{x \mid x \text{ is an odd positive integer less than } 100\}$, or more precisely $\{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 100\}$.

Russell's paradox: always restrict the domain

Omitting the domain restriction $x \in E$ may lead to paradoxes. **Russell's paradox:** let $A = \{x \mid x \notin x\}$. Is $A \in A$? If $A \in A$, then by definition $A \notin A$; if $A \notin A$, then $A \in A$. Contradiction either way! This led to the development of axiomatic set theory.

Important sets

$\mathbb{N}$ = natural numbers = $\{0, 1, 2, 3, \ldots\}$,	$\mathbb{Z}$ = integers = $\{\ldots, -1, 0, 1, \ldots\}$,
$\mathbb{Z}^+$ = positive integers = $\{1, 2, 3, \ldots\}$,	$\mathbb{Z}^-$ = negative integers = $\{-1, -2, \ldots\}$,
$\mathbb{R}$ = real numbers,	$\mathbb{R}^+$ = positive real numbers,
$\mathbb{R}^-$ = negative real numbers,	$\mathbb{Q}$ = rational numbers.

Definition. The *empty set* (or *null set*), denoted by $\emptyset$ or $\{\}$, is the special set containing no elements.

Definition. A set with exactly one element is called a *singleton set*.

Example. Note that $\emptyset \neq \{\emptyset\}$: the latter is a singleton set whose unique element is the empty set. Yes, an element of a set can itself be a set — in pure set theory, *everything* is a set!

Definition. Two sets A and B are *equal*, denoted $A = B$, if and only if they have the same elements; that is, for every x, $x \in A$ if and only if $x \in B$. Formally: $A = B \iff \forall x\,(x \in A \leftrightarrow x \in B)$.

Example. $\{1, 2, 3\} = \{3, 1, 2\} = \{1, 2, 2, 2, 3, 3\}$. Order does not matter, and repetitions do not count.

Definition. A set A is a *subset* of a set B, denoted $A \subseteq B$, if and only if every element of A is also an element of B: $A \subseteq B \iff \forall x\,(x \in A \to x \in B)$.

Facts about subsets

For every set S: (a) $\emptyset \subseteq S$, and (b) $S \subseteq S$. Hence every nonempty set is guaranteed to have at least two subsets: the empty set and the set itself. Moreover,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

This is the standard strategy for proving two sets equal.

Definition. A set A is a *proper subset* of a set B, denoted $A \subset B$, if and only if $A \subseteq B$ but $A \neq B$. A Venn diagram can be used to illustrate the subset relationship.

Definition. Let S be a finite set. The *cardinality* of S, denoted $|S|$, is the number of (distinct) elements in S. A set is *infinite* if it is not finite.

4.2 Power Set

Definition. Given a set S, the *power set* of S, denoted $P(S)$, is the set of all subsets of S.

Size of the power set

If a set has n elements, where n is a nonnegative integer, then its power set has 2^n elements: $|S| = n \Rightarrow |P(S)| = 2^n$.

Example. The power set of $\{a, b, c\}$ is

$$P(\{a, b, c\}) = \big\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\big\}.$$

It has $2^3 = 8$ elements.

Example. What is the power set of the empty set? What is the power set of $\{\emptyset\}$?

Solution. The only subset of $\emptyset$ is $\emptyset$ itself, so $P(\emptyset) = \{\emptyset\}$ — a set with one element, consistent with $2^0 = 1$. The subsets of $\{\emptyset\}$ are $\emptyset$ and $\{\emptyset\}$, so $P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$, with $2^1 = 2$ elements.

4.3 Ordered Tuples and Cartesian Products

Definition. The *ordered n-tuple* $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, $\ldots$, and a_n as its n-th element. An ordered 2-tuple is more commonly called an *ordered pair*.

Definition. Two ordered n-tuples $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_m)$ are equal if and only if $n = m$ and $a_i = b_i$ for all $i = 1, 2, \ldots, n$.

Definition. Let A and B be two sets. The *Cartesian product* of A and B, denoted $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Example. Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Definition. Let A and B be two sets. A subset R of $A \times B$ is called a *relation* from the set A to the set B.

Example. With $A = \{1, 2\}$ and $B = \{a, b, c\}$ as above, $R = \{(1, a), (1, c), (2, a), (2, b)\} \subset A \times B$ is a relation from A to B.

Definition. The *Cartesian product* of the sets $A_1, A_2, \ldots, A_n$, denoted $A_1 \times A_2 \times \cdots \times A_n$, is the set of ordered n-tuples $(a_1, a_2, \ldots, a_n)$, where $a_i \in A_i$ for $i = 1, 2, \ldots, n$:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \ldots, n\}.$$

4.4 Set Operations

Definition. Let A and B be two sets. The *union* of A and B, denoted $A \cup B$, is the set containing those elements that are in A, in B, or in both: $A \cup B = \{x \mid x \in A \vee x \in B\}$.

Definition. Let $A_1, A_2, \ldots, A_n$ be n sets. The *union of the collection* is $\bigcup_{i=1}^{n} A_i$, the set of elements that are members of at least one set in the collection.

Definition. Let A and B be two sets. The *intersection* of A and B, denoted $A \cap B$, is the set containing those elements in both A and B: $A \cap B = \{x \mid x \in A \wedge x \in B\}$.

Definition. Let $A_1, A_2, \ldots, A_n$ be n sets. The *intersection of the collection* is $\bigcap_{i=1}^{n} A_i$, the set of elements that are members of all the sets in the collection.

Definition. Two sets A and B are *disjoint* if and only if $A \cap B = \emptyset$.

Inclusion–exclusion for two sets

Theorem. Let A and B be two finite sets. Then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Intuitively: adding $|A|$ and $|B|$ counts each element of $A \cap B$ twice, so we subtract $|A \cap B|$ once to correct the double counting.

Definition. Let A and B be two sets. The *difference* of A and B, denoted $A - B$ or $A \setminus B$, is the set containing those elements that are in A but not in B: $A - B = \{x \mid x \in A \wedge x \notin B\}$. It is also called the *complement of B with respect to A*.

Example. $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$.

Definition. Let U be the universal set. The *complement* of a set A, denoted $\overline{A}$, is the complement of A with respect to U; in other words, $\overline{A} = U - A$.

Set identities

$A \cap U = A, \quad A \cup \emptyset = A$	Identity laws
$A \cup U = U, \quad A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A, \quad A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A, \quad A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$	Associative laws
$A \cap (B \cap C) = (A \cap B) \cap C$	
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	
$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U, \quad A \cap \overline{A} = \emptyset$	Complement laws

Set identities follow from logic equivalences

All these set identities follow from corresponding logical equivalences: just replace $\cap$ with $\wedge$, $\cup$ with $\vee$, and $\overline{\ }$ with $\neg$. For example, De Morgan's law $\overline{A \cap B} = \overline{A} \cup \overline{B}$ corresponds to $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

4.5 Functions

Definition. Let A and B be nonempty sets. A *function* f from A to B is an assignment of *exactly one* element of B to each element of A. We write $f(a) = b$ if b is the unique element of B assigned by f to the element a of A. If f is a function from A to B, we write $f : A \to B$.

Remark. Functions are sometimes also called *mappings* or *transformations*. If $f : A \to B$, we say that f *maps* A to B.

Definition. If $f : A \to B$, then A is the *domain* of f and B is the *codomain* of f. If $f(a) = b$, then b is the *image* of a and a is a *preimage* of b. The *range* of f is the set of all images of elements of A.

Example. Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$. Then the domain of f is the set of all bit strings of length 2 or greater, and both the codomain and the range are the set $\{00, 01, 10, 11\}$.

Example. Let $f : \mathbb{Z} \to \mathbb{Z}$ assign the square of an integer to this integer. Then $f(x) = x^2$, where the domain of f is the set of all integers, we take the codomain of f to be the set of all integers, and the range of f is the set of all integers that are perfect squares, namely $\{0, 1, 4, 9, \ldots\}$.

4.6 Injective, Surjective, Bijective Functions

Definition. A function f is *injective* (or *one-to-one*) if and only if $f(a) = f(b)$ implies $a = b$ for all a and b in the domain of f. An injective function is also called an *injection*.

Example. Is the function $f(x) = x + 1$ from the set of real numbers to the set of real numbers injective?

Solution. Yes. Suppose $f(a) = f(b)$, i.e., $a + 1 = b + 1$. Subtracting 1 from both sides gives $a = b$. Hence $f(a) = f(b) \Rightarrow a = b$, so f is injective.

Example. Is the function $f(x) = x^2$ from the set of integers to the set of integers injective?

Solution. No. Take $a = 1$ and $b = -1$: then $f(1) = 1 = f(-1)$ but $1 \neq -1$. Since $f(a) = f(b)$ does not imply $a = b$, f is not injective.

Definition. A function $f : A \to B$ is *surjective* (or *onto*) if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A surjective function is also called a *surjection*.

Example. Is the function $f(x) = x + 1$ from the set of integers to the set of integers surjective?

Solution. Yes. Given any integer b, take $a = b - 1$, which is also an integer; then $f(a) = (b - 1) + 1 = b$. So every $b \in \mathbb{Z}$ has a preimage, and f is surjective.

Example. Is the function $f(x) = x^2$ from the set of integers to the set of integers surjective?

Solution. No. Consider $b = 2 \in \mathbb{Z}$: there is no integer a with $a^2 = 2$, so 2 has no preimage. Hence f is not surjective.

Definition. A function f is a *bijection* (or *one-to-one correspondence*) if it is both one-to-one and onto. Note: by convention, the "if" immediately following a definition means "iff". Beware the terminology: *one-to-one* means injection, while *one-to-one correspondence* means bijection.

4.7 Inverse Functions and Composition

Definition. Let f be a one-to-one correspondence from the set A to the set B. The *inverse function* of f is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$. The inverse function of f is denoted f^{-1}. Hence $f^{-1}(b) = a$ when $f(a) = b$. A one-to-one correspondence is called *invertible*.

Example. Let $f : \mathbb{Z} \to \mathbb{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Solution. As shown above, f is both injective and surjective on $\mathbb{Z}$, so it is a bijection and hence invertible. To find the inverse, solve $y = x + 1$ for x: $x = y - 1$. Therefore $f^{-1}(y) = y - 1$.

Example. Let $f : \mathbb{R} \to \mathbb{R}$ be such that $f(x) = x^2$. Is f invertible?

Solution. No. Since $f(-1) = f(1) = 1$, the function is not injective; and since no real number maps to -1, it is not surjective either. Not being a bijection, f has no inverse function $\mathbb{R} \to \mathbb{R}$.

Definition. Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The *composition* of the functions f and g, denoted $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$

Example. Let f and g be functions from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f?

Solution. Compute each composition step by step:

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 4 + 3 = 6x + 7.$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 9 + 2 = 6x + 11.$$

Note that $f \circ g \neq g \circ f$ in general — composition is *not commutative*.

4.8 Cardinality of Infinite Sets

Hilbert's Grand Hotel. Suppose a hotel has infinitely many rooms, numbered $1, 2, 3, \ldots$, and all rooms are occupied. A new guest arrives ... and can still be accommodated: move the guest in room n to room $n+1$ for every n, freeing room 1. This shows $|\{1, 2, 3, \ldots\}| = |\{2, 3, 4, \ldots\}|$ — an infinite set can have the same cardinality as a proper subset of itself!

Definition. Two sets A and B have the *same cardinality* if and only if there is a one-to-one correspondence (bijection) from A to B.

Countability

$$|\mathbb{N}| = \aleph_0 \qquad (\text{“aleph null”}),$$

where $\mathbb{N}$ is the set of natural numbers $\{0, 1, 2, \ldots\}$. A set S is

- *finite*, if $|S| = n$ for some $n \in \mathbb{N}$;
- *infinitely countable*, if $|S| = \aleph_0$;
- *uncountable*, otherwise.

A set is called *countable* if it is finite or infinitely countable.

Example. $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$ is countable.

Solution. Let $f : \mathbb{N} \to \mathbb{Z}^+$ be $f(x) = x + 1$. This is a bijection: it is injective since $x + 1 = y + 1 \Rightarrow x = y$, and surjective since every positive integer m is hit by $x = m - 1 \in \mathbb{N}$. Hence $|\mathbb{Z}^+| = |\mathbb{N}| = \aleph_0$.

Example. The set of all nonnegative even integers is countable.

Solution. Let $f : \mathbb{N} \to \{0, 2, 4, \ldots\}$ be $f(x) = 2x$. This is a bijection, so the set is countable.

Example. The set of all integers $\mathbb{Z}$ is countable.

Solution. List all integers in the order $0, 1, -1, 2, -2, 3, -3, \ldots$ and define $f : \mathbb{N} \to \mathbb{Z}$ by

$$f(x) = \begin{cases} -\dfrac{x}{2}, & \text{if } x \text{ is even,} \\ \dfrac{x+1}{2}, & \text{if } x \text{ is odd.} \end{cases}$$

Check: $f(0) = 0$, $f(1) = 1$, $f(2) = -1$, $f(3) = 2$, $f(4) = -2$, ... — every integer appears exactly once, so f is a bijection. Hence $\mathbb{Z}$ is countable.

Example. $\mathbb{Q}$ is countable.

Solution. We only show that $\mathbb{Q}^+$ is countable; the whole of $\mathbb{Q}$ then follows as for $\mathbb{Z}$. Recall: a rational number can be expressed as the ratio of two integers p and q with $q \neq 0$. Arrange all pairs (p, q) with $p, q \in \mathbb{Z}^+$ in an infinite grid and traverse it along diagonals (increasing $p + q$), listing p/q and skipping any value already listed (e.g., skip $2/2$ after $1/1$):

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \ldots$$

Every positive rational appears exactly once, so this defines a bijection $f : \mathbb{N} \to \mathbb{Q}^+$. Note: the function f has no closed form, but it is well defined.

Corollary. $\mathbb{N} \times \mathbb{N}$ is countable (the same diagonal traversal lists all pairs).

Corollary. $\mathbb{N}^k$ is countable for any $k \in \mathbb{Z}^+$.

The set of strings over any alphabet is countable

Theorem. Fix an alphabet Σ (e.g., $\Sigma = \{a, b, \ldots, z\}$ for English words, or $\Sigma = \{0, 1\}$ for machine code of computer programs). The set of all strings over Σ is countable.

Proof. (using $\Sigma = \{0, 1\}$ as example) List the strings in groups by length, and lexicographically within each group:
- Strings of length 0: the empty string λ;
- Strings of length 1: $0, 1$;
- Strings of length 2, in lexicographical order: $00, 01, 10, 11$;
- Strings of length 3, in lexicographical order: $000, 001, 010, 011, 100, 101, 110, 111$;
- $\ldots$

Each group is finite, and every string appears exactly once, so this enumeration is a bijection with $\mathbb{N}$. ■

Cantor's diagonalization: $(0, 1)$ is uncountable

Theorem. The interval $(0, 1)$ is uncountable; hence $\mathbb{R}$ is uncountable.

Proof. By contradiction. Suppose $(0, 1)$ is countable; then the real numbers in $(0, 1)$ can be listed as $r_1, r_2, r_3, \ldots$, with decimal expansions

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}\ldots \\ r_2 &= 0.d_{21}d_{22}d_{23}\ldots \\ r_3 &= 0.d_{31}d_{32}d_{33}\ldots \\ &\vdots \end{aligned}$$

Form a new real number $r = 0.d_1d_2d_3\ldots$, where

$$d_i = \begin{cases} 4, & \text{if } d_{ii} \neq 4, \\ 5, & \text{if } d_{ii} = 4. \end{cases}$$

Key observation: $r \neq r_i$ for any i, because r differs from r_i in the i-th decimal digit. So r is a real number in $(0, 1)$ not in the list — a contradiction. Hence $(0, 1)$ is uncountable. ■

Technical note: a real number may have two decimal representations ($1 = 0.\dot{9}$, $0.353 = 0.352\dot{9}$); we always choose the former (non-terminating-9s-free) one. Since r uses only digits 4 and 5, it has a unique representation, so the argument is watertight. This is a *diagonalization argument* (Georg Cantor, 1845–1918).

4.9 Comparing Infinite Sets

Definition. We write $|S| \leq |T|$ if there is an injection from S to T. We write $|S| < |T|$ if there exists an injection but no bijection from S to T.

Example. We have shown that there is no bijection from $\mathbb{N}$ to $(0, 1)$. An injection from $\mathbb{N}$ to $(0, 1)$ is, e.g., $f(x) = \dfrac{e^x}{e^x + 1}$ (its range is $(1/2, 1) \subset (0, 1)$). So

$$\text{any finite set} \;<\; |\mathbb{N}| \;<\; |(0, 1)| = |\mathbb{R}|.$$

(A bijection between $(0, 1)$ and $\mathbb{R}$: $f(x) = \tan(\pi x - \pi/2)$.)

Corollary. There is no computer number format that can represent all real numbers: the set of finite strings of bits is countable, while $\mathbb{R}$ is uncountable.

Theorem. For any infinite set S, $|\mathbb{N}| \leq |S|$. In plain language: $\mathbb{N}$ is the smallest infinite set.

Proof. We will show an injection from $\mathbb{N}$ to S. Since S is infinite, it is not empty, so we can find an element $a_1 \in S$. The set $S - \{a_1\}$ is still infinite, so we can find another element $a_2 \in S - \{a_1\}$. Continuing this process, we can find $a_i \in S$ for every $i \in \mathbb{N}$, all distinct. Then the function $f(i) = a_i$ is an injection from $\mathbb{N}$ to S. ■

Cantor's theorem: $|S| < |P(S)|$

Theorem. For any set S, $|S| < |P(S)|$.

Proof. An injection $S \to P(S)$ is easy: $f(a) = \{a\}$. The difficult part is to show that there is no bijection. Let g be any function from S to $P(S)$; we will show that g is not surjective, hence not bijective. Consider

$$A = \{x \in S \mid x \notin g(x)\}.$$

Note $A \subseteq S$, so $A \in P(S)$. We will show that for no $a \in S$ does $g(a) = A$. Suppose for some $a \in S$ that $g(a) = A$; then

$$a \in A \iff a \in \{x \in S \mid x \notin g(x)\} \iff a \notin g(a) \iff a \notin A,$$

a contradiction. So g is not surjective, and no bijection exists. ■

Corollary. $P(\mathbb{N})$ is uncountable.

The Continuum Hypothesis

We know: any finite set $< |\mathbb{N}| < |P(\mathbb{N})| = |\mathbb{R}|$ (the last equality is proved in tutorial). There is nothing between the first two ($\mathbb{N}$ is the smallest infinite set). The *Continuum Hypothesis* states that there is nothing between $|\mathbb{N}|$ and $|\mathbb{R}|$ — Hilbert's first problem, 1900. It *cannot be proved or disproved* in ZFC (an axiomatic set theory): Kurt Gödel, 1940; Paul Cohen, 1963.

Schröder–Bernstein theorem

Theorem. If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. (No easy proof exists for this seemingly obvious fact!)

Example. Show that $|(0,1)| = |(0,1]|$.

Proof. By Schröder–Bernstein, it suffices to give two injections:

- An injection $(0,1) \to (0,1]$: $f(x) = x$;
- An injection $(0,1] \to (0,1)$: $f(x) = \dfrac{x}{2}$ (its range is $(0, 1/2] \subset (0,1)$).

Since both injections exist, $|(0,1)| = |(0,1]|$. ■

4.10 Uncomputable Functions

Definition. A function is *computable* if there is a computer program that finds the value of the function on any input; otherwise it is *uncomputable*.

Theorem. There exist functions from $\mathbb{N}$ to $\{0,1\}$ that are uncomputable.

Proof. Compare cardinalities:

- |the set of computer programs| = |$\mathbb{N}$|, countable — every program is a finite string over a finite alphabet;
- |the set of functions $\mathbb{N} \to \{0,1\}$| = |$P(\mathbb{N})$|, uncountable — every function $g : \mathbb{N} \to \{0,1\}$ corresponds to a subset $S \subseteq \mathbb{N}$ via $S = \{x \in \mathbb{N} \mid g(x) = 1\}$ (e.g., $g = 0,1,0,1,1,\ldots$ corresponds to $S = \{0,3,4,\ldots\}$ — this is the characteristic-function correspondence).

Since there are strictly more functions $\mathbb{N} \to \{0,1\}$ than programs, some (in fact, almost all) such functions cannot be computed by any program. ■

A concrete uncomputable function: the *halting problem* (Alan Turing).

5 Mathematical Induction

Reading: Rosen 5.1, 5.2. Induction is one of the most powerful techniques in mathematics and computer science.

5.1 Mathematical Induction: An Axiom

The domain is $\mathbb{N}$, or any set with a bijection to $\mathbb{N}$. Mathematical induction has two equivalent forms (recall that $k+1 = S(k)$, the successor of k):

The principle of mathematical induction

Set form: for every $K \subseteq \mathbb{N}$,

$$\big[0 \in K \ \wedge\ \forall k\,(k \in K \to k+1 \in K)\big] \to K = \mathbb{N}.$$

Predicate form: for every predicate P,

$$\big[P(0) \ \wedge\ \forall k\,(P(k) \to P(k+1))\big] \to \forall n\, P(n).$$

Why is it true? Well, it is obvious — it is an *axiom* according to Peano. It is equivalent to the *well-ordering property*:

Definition (Well-ordering property). Every nonempty subset $S \subseteq \mathbb{N}$ has a least element:

$$\forall S \subseteq \mathbb{N},\ S \neq \emptyset \ \Rightarrow\ \exists x \in S\, \forall y \in S\,(x \leq y),$$

where $x \leq y$ is defined by $\exists z \in \mathbb{N}\,(x + z = y)$.

Theorem. The well-ordering property implies the principle of mathematical induction.

Proof. Consider any predicate P. We are given (i) $P(0)$, and (ii) $\forall k\,(P(k) \to P(k+1))$. We prove $\forall n\, P(n)$ by contradiction.

Assume $\forall n\, P(n)$ is false. Let $S = \{i \in \mathbb{N} \mid P(i) \text{ is false}\}$; then S is nonempty. By the well-ordering principle, S has a least element, say x.

- $x \neq 0$, since $P(0)$ is true.
- Then there is a $y \in \mathbb{N}$ such that $x = y + 1$. This means that $y < x$, so $y \notin S$, i.e., $P(y)$ is true.
- Now $P(y+1)$ is false (since $y + 1 = x \in S$) while $P(y)$ is true. This contradicts $\forall k\,(P(k) \to P(k+1))$.

Therefore the assumption is wrong, and $\forall n\, P(n)$ holds. ■

5.2 Variants of Induction

Variants of induction

Original version (domain $\mathbb{N}$):

$$\forall P\,\big[P(0) \wedge \forall k\,(P(k) \to P(k+1)) \to \forall n\,P(n)\big].$$

Variant (domain $\mathbb{Z}^+$):

$$\forall P\,\big[P(1) \wedge \forall k\,(P(k) \to P(k+1)) \to \forall n\,P(n)\big].$$

Variant (domain $\{a, a+1, \ldots\}$):

$$\forall P\,\big[P(a) \wedge \forall k\,(P(k) \to P(k+1)) \to \forall n\,P(n)\big].$$

The basis case is always the smallest element of the domain.

5.3 Proving Summations

Example. Show that $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Solution. Let $P(n)$ be the proposition that $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Basis: $P(1)$ is true, since $\dfrac{1(1+1)}{2} = 1 = \displaystyle\sum_{i=1}^{1} i$.

Inductive step: Assume $P(k)$ is true for an arbitrary positive integer k; i.e., the inductive hypothesis is

$$\sum_{i=1}^{k} i = \frac{k(k+1)}{2}.$$

Under this assumption,

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^{k} i + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2},$$

which is exactly $P(k+1)$. Hence $P(k) \to P(k+1)$, and by induction $P(n)$ holds for all positive integers n. ■

Example. Show that $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

Proof. Let $P(n)$ be the proposition that $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

Basis: $P(1)$ is true, since $1^2 = 1$.

Inductive step: We show $P(k) \to P(k+1)$ for every positive integer k. The induction hypothesis $P(k)$ states:

$$1 + 3 + 5 + \cdots + (2k-1) = k^2.$$

So, assuming $P(k)$, it follows that:

$$\begin{aligned} 1 + 3 + \cdots + (2k+1) &= \big[1 + 3 + \cdots + (2k-1)\big] + (2k+1) \\ &= k^2 + 2k + 1 \quad \text{(by the inductive hypothesis)} \end{aligned}$$

and $k^2 + 2k + 1 = (k+1)^2$, which is $P(k+1)$. ■

The good and bad of induction

The good: proofs using induction are often mechanical. **The bad:** a proof by induction needs a valid conjecture first, and offers less insight into the problem.

5.4 Proving Inequalities

Example. Use mathematical induction to prove that $n < 2^n$ for all positive integers n.

Solution. Let $P(n)$ be the proposition that $n < 2^n$.

Basis: $P(1)$ is true, since $1 < 2^1 = 2$.

Inductive step: Assume $P(k)$ holds, i.e., $k < 2^k$, for an arbitrary positive integer k. We must show that $P(k+1)$ holds. Since by the inductive hypothesis $k < 2^k$, it follows that:

$$k + 1 < 2^k + 1 \leq 2^k + 2^k = 2 \cdot 2^k = 2^{k+1},$$

where the second step uses $1 \leq 2^k$ (true for $k \geq 1$). Therefore $n < 2^n$ holds for all positive integers n. ■

Example. Use mathematical induction to prove that $2^n < n!$ for every integer $n \geq 4$.

Solution. Let $P(n)$ be the proposition that $2^n < n!$.

Basis: $P(4)$ is true, since $2^4 = 16 < 4! = 24$. (Note the domain is $\{4, 5, 6, \ldots\}$, so the basis starts at 4.)

Inductive step: Assume $P(k)$ holds, i.e., $2^k < k!$, for an arbitrary integer $k \geq 4$. To show that $P(k+1)$ holds:

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &< 2 \cdot k! && \text{(by the inductive hypothesis)} \\ &< (k+1) \cdot k! && \text{(since } k \geq 4, \text{ so } 2 < k+1) \\ &= (k+1)!. \end{aligned}$$

Hence $2^n < n!$ for every integer $n \geq 4$. ■

5.5 Proving Divisibility Results

Example. Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n.

Solution. Let $P(n)$ be the proposition that $3 \mid (n^3 - n)$.

Basis: $P(1)$ is true, since $1^3 - 1 = 0$ and $3 \mid 0$.

Inductive step: Assume $P(k)$ holds, i.e., $3 \mid (k^3 - k)$. To show that $P(k+1)$ is true:

$$\begin{aligned} (k+1)^3 - (k+1) &= (k^3 + 3k^2 + 3k + 1) - (k+1) \\ &= (k^3 - k) + 3(k^2 + k). \end{aligned}$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3. The second term $3(k^2 + k)$ is obviously divisible by 3. A sum of two multiples of 3 is a multiple of 3, so $3 \mid \big[(k+1)^3 - (k+1)\big]$, i.e., $P(k+1)$ is true. ■

5.6 Number of Subsets of a Finite Set

Example. Use mathematical induction to show that if S is a finite set with n elements, then S has 2^n subsets.

Solution. Let $P(n)$ be the proposition that a set with n elements has 2^n subsets.

Basis: $P(0)$ is true, because the empty set has only itself as a subset, and $2^0 = 1$.

Inductive step: Assume $P(k)$ is true for an arbitrary nonnegative integer k. Let T be a set with $k+1$ elements. Pick some arbitrary element $a \in T$, and let $S = T - \{a\}$. We have $|S| = k$.

For each subset X of S, there are exactly two subsets of T, namely X and $X \cup \{a\}$; moreover, every subset of T arises this way (it either contains a or not). By the inductive hypothesis, S has 2^k subsets. Since there are two subsets of T for each subset of S, the number of subsets of T is $2 \cdot 2^k = 2^{k+1}$. So $P(k+1)$ is true. ■

5.7 Odd Pie Fight Problem

Odd pie fight problem

An odd number of people stand in a yard. Each person throws a pie at their nearest neighbor (assuming all mutual distances are distinct). Show that at least one person survives.

Proof. Let $P(n)$ be the statement that there is a survivor when there are $2n+1$ people.

Basis: $P(1)$ is true. There are $2n+1 = 3$ people: A, B, C. Without loss of generality, suppose the distance between A and B is the smallest among the 3 pairs. Then A and B throw at each other (each is the other's nearest neighbor), and C survives (C throws at someone, but nobody throws at C).

Inductive step: Assume $P(k)$ is true, i.e., when there are $2k+1$ people there is a survivor, for an arbitrary positive integer k. We want to prove $P(k+1)$ is true, i.e., when there are $2k+3$ people there is a survivor.

Idea: we need to reduce the problem from $2k+3$ to $2k+1$. Can we remove two arbitrary people? Not quite — we remove a carefully chosen pair. Let A and B be the pair of people with the smallest mutual distance among all pairs. Then A and B must throw at each other.

Case 1: No one else throws at A or B. Remove A and B; the remaining $2k+1$ people all throw among themselves exactly as before (nobody's nearest neighbor was A or B). Apply the induction hypothesis on the remaining $2k+1$ people: someone survives. Done.

Case 2: Someone else throws at A or B. There are $2k+1$ people left (other than A and B). At most $2k$ pies are thrown at them (one pie from each of the $2k+1$ remaining people, minus at least one pie that went to A or B — precisely, of the $2k+1$ pies thrown by the remaining people, at least one was directed at A or B, so at most $2k$ land on the remaining people). Since $2k$ pies are distributed among $2k+1$ people, at least one person is hit by no pie and survives. ■

5.8 Tiling Checkerboards

Example. Show that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using L-shaped tiles (an L-tile covers 3 squares).

Solution. Let $P(k)$ be the proposition that every $2^k \times 2^k$ checkerboard with one square removed can be tiled by L-shaped tiles.

Basis: $P(1)$ is true trivially: a 2×2 board with one square removed is exactly one L-shaped tile.

Inductive step: Assume $P(k)$ is true. We want to show that $P(k+1)$ is true. Consider a $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed. Divide it into four quadrants, each a $2^k \times 2^k$ board. Look at which quadrant contains the removed square; without loss of generality, assume it is in the right-bottom quadrant. Place one L-shaped tile at the center so that it covers one corner square of each of the *other three* quadrants. Now each of the four quadrants is a $2^k \times 2^k$ checkerboard with exactly one square removed (either the original removed square, or the square covered by the central tile). Apply the induction hypothesis on each of the four quadrants to complete the tiling. ■

5.9 Strong Induction

Strong induction

Original version:

$$\forall P \left[P(0) \wedge \forall k \left(P(k) \to P(k+1) \right) \to \forall n \, P(n) \right].$$

Strong induction:

$$\forall P \left[P(0) \wedge \forall k \left((P(0) \wedge \cdots \wedge P(k)) \to P(k+1) \right) \to \forall n \, P(n) \right].$$

It is "strong" because the induction hypothesis is stronger — we may assume $P(0), \ldots, P(k)$ *all* hold, making it easier to prove the inductive step. In fact the two principles are equivalent (both are equivalent to the well-ordering property), but strong induction provides more convenience for proofs.

Example (Fundamental theorem of arithmetic, existence part). Theorem. Every positive integer greater than 1 can be written as the product of primes.

Proof. Let $P(n)$ be the proposition that n can be written as a product of primes. We use strong induction.

Basis: $P(2)$ is true, since 2 itself is prime (a product of one prime).

Inductive step: The inductive hypothesis is that $P(j)$ is true for all integers j with $2 \leq j \leq k$. We want to show that $P(k+1)$ is true.

- If $k+1$ is prime, then $P(k+1)$ is true immediately ($k+1$ is a product of one prime).
- Otherwise, $k+1$ is composite, i.e., $k+1 = ab$ with $2 \leq a \leq b \leq k$. By the inductive hypothesis, a and b can each be written as the product of primes. So $k+1 = ab$ can also be written as the product of those primes.

Hence $P(n)$ holds for all $n \geq 2$. Note that ordinary induction fails here: $k+1 = ab$ tells us nothing about $P(k)$, but strong induction gives us $P(a)$ and $P(b)$. ■

Two piles of matches problem

We have two piles of the same number of matches. Two players take turns to remove any positive number of matches from one of the two piles. The one who removes the last match from the two piles wins. Show that the second player can always win.

Proof. Let $P(k)$ be the statement that the second player can win when both piles have k matches. We use strong induction.

Basis: $P(1)$ is true trivially: the first player must remove one match from one pile (or both matches of one pile); either way the second player takes the remaining match(es)

and wins. Explicitly: if player 1 takes one match from a pile, one pile has 1 match and the other has 1 — wait, both piles start with 1 match. If the first player removes 1 match from one pile, that pile is empty; the second player removes the 1 match from the other pile and wins. If the first player could only remove from one pile, this is the only move, so $P(1)$ holds.

Inductive step: Assume $P(1), \ldots, P(k)$ are all true. We want to show that $P(k+1)$ is true. Suppose the first player first removes r matches from one pile, where r can be any integer with $1 \le r \le k+1$.

Case 1: If $r = k+1$, the first pile is emptied; the second player wins by taking all $k+1$ matches from the other pile.

Case 2: If $r \le k$, then the second player takes r matches from the other pile. Now there are $k+1-r$ matches left in both piles, with $1 \le k+1-r \le k$. By the induction hypothesis, $P(k+1-r)$ is true: in the resulting game the player who moves second wins — and from now on the second player plays exactly that winning strategy. So the second player can win. ■

5.10 Induction with Two Variables

Example. Prove $\forall x \, \forall y \, \big[S(x) + y = S(x+y)\big]$ from Peano's axioms (the domain is all natural numbers; S is the successor function, and addition is defined by $x + 0 = x$ and $x + S(y) = S(x+y)$).

Solution. We do induction on y. Let

$$P(y) := \forall x \, \big[S(x) + y = S(x+y)\big].$$

The goal is to prove $\forall y \, P(y)$.

Base case $P(0)$: $\forall x \, \big[S(x)+0 = S(x+0)\big]$. Both sides equal $S(x)$ by the base definition of addition ($a+0=a$): the left side $S(x)+0 = S(x)$, and the right side $S(x+0) = S(x)$. So $P(0)$ is true.

Inductive step: Assume $P(k)$: $\forall x \, \big[S(x) + k = S(x+k)\big]$. We need to prove $P(S(k))$: $\forall x \, \big[S(x) + S(k) = S(x+S(k))\big]$. For any x:

$$\begin{aligned} S(x) + S(k) &= S\big(S(x) + k\big) && \text{(definition of addition)} \\ &= S\big(S(x+k)\big) && \text{(induction hypothesis } P(k)\text{)} \\ &= S\big(x + S(k)\big) && \text{(definition of addition, applied backwards).} \end{aligned}$$

Hence $P(S(k))$ holds, and by induction $\forall y \, P(y)$. ■

6 Recursion

Reading: Rosen 5.3, 5.4, 8.1.

6.1 Recursively Defined Functions

A function is defined *recursively* by specifying (i) one or more *base cases* (its values at small arguments) and (ii) a *recursive step* that defines its value at a larger argument in terms of values at smaller arguments.

Example. Give a recursive definition of $n! = 1 \cdot 2 \cdot \cdots \cdot n$.

Solution.

- Base case: $1! = 1$ (equivalently, $0! = 1$).
- Recursive step: $(n+1)! = (n+1) \cdot n!$.

Example. What does this define: $f(x,0) = x$, and $f(x,y+1) = f(x,y)+1$?

Solution. Starting from $f(x,0) = x$, each increment of the second argument adds 1: $f(x,1) = f(x,0)+1 = x+1$, $f(x,2) = f(x,1)+1 = x+2$, ... So $f(x,y) = x+y$ — this is *addition*.

Example. What does this define: $g(x,0) = 0$, and $g(x,y+1) = f(g(x,y),x)$, where f is the addition function above?

Solution. $g(x,0) = 0$; $g(x,1) = f(g(x,0),x) = 0+x = x$; $g(x,2) = f(g(x,1),x) = x+x = 2x$; $g(x,3) = 3x$, ... So $g(x,y) = x \cdot y$ — this is *multiplication* (repeated addition).

Example. Suppose f is defined by $f(0) = 3$ and $f(n+1) = 2f(n)+3$. Find $f(1)$, $f(2)$, $f(3)$, $f(4)$.

Solution. Compute step by step:

- $f(1) = 2f(0)+3 = 2\cdot 3+3 = 9$;
- $f(2) = 2f(1)+3 = 2\cdot 9+3 = 21$;
- $f(3) = 2f(2)+3 = 2\cdot 21+3 = 45$;
- $f(4) = 2f(3)+3 = 2\cdot 45+3 = 93$.

Example (Recursion and induction). Suppose f is defined by $f(0) = 3$ and $f(n+1) = 2f(n)+3$. Show that $f(n) = 3(2^{n+1}-1)$.

Proof. By induction on n.

Basis: $f(0) = 3$ by definition, and the formula gives $3(2^{0+1}-1) = 3(2-1) = 3$. So the claim holds for $n = 0$.

Inductive step: Assume $f(k) = 3(2^{k+1}-1)$. We want to show $f(k+1) = 3(2^{k+2}-1)$:

$$\begin{aligned} f(k+1) &= 2f(k)+3 && \text{(recursive definition)} \\ &= 2\cdot 3\big(2^{k+1}-1\big)+3 && \text{(inductive hypothesis)} \\ &= 3\cdot 2^{k+2}-6+3 = 3\cdot 2^{k+2}-3 = 3\big(2^{k+2}-1\big). \end{aligned}$$

Hence the formula holds for all $n \in \mathbb{N}$. ■

Definition. A recursively defined function over $\mathbb{N}$ is called a *recurrence*. We can solve recurrences by (i) *guess and induction* (as above), or (ii) *direct methods* (as below).

6.2 Solving First-Order Linear Recurrences

First-order linear recurrence

Solve the recurrence $f(0) = a$, $f(n) = bf(n-1)+c$ for $n > 0$, where a, b, c are constants. Unfolding:

$$\begin{aligned} f(n) &= bf(n-1)+c \\ &= b\big(bf(n-2)+c\big)+c = b^2 f(n-2)+bc+c \\ &= b^2\big(bf(n-3)+c\big)+bc+c = b^3 f(n-3)+b^2 c+bc+c \\ &\ \ \vdots \\ &= b^n f(0)+b^{n-1}c+b^{n-2}c+\cdots+c \\ &= ab^n + c\left(b^{n-1}+b^{n-2}+\cdots+1\right) \end{aligned}$$

Summing the geometric series $1 + b + \cdots + b^{n-1} = \dfrac{b^n - 1}{b - 1}$ (for $b \neq 1$):

$$\boxed{f(n) = ab^n + c \cdot \frac{b^n - 1}{b - 1}}$$

(If $b = 1$, then $f(n) = a + nc$.)

Example (Mortgage calculation). Initial loan amount A; monthly interest rate p (banks usually list the p.a. rate — dividing that by 12 gives the monthly rate); monthly payment M; number of months to clear the loan n. The remaining debt satisfies the recurrence:

$$f(0) = A, \qquad f(n) = (1 + p)f(n - 1) - M.$$

This is a first-order linear recurrence with $b = 1 + p$ and $c = -M$. Solving:

$$f(n) = A(1 + p)^n - M \cdot \frac{(1 + p)^n - 1}{p}.$$

We want the loan cleared after n months, i.e., $f(n) = 0$, so

$$A(1 + p)^n = M \cdot \frac{(1 + p)^n - 1}{p} \quad \Longrightarrow \quad M = \frac{Ap(1 + p)^n}{(1 + p)^n - 1}.$$

6.3 Fibonacci Numbers

Definition. The *Fibonacci numbers* are defined as follows:

$$f_1 = 1, \qquad f_2 = 1, \qquad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 3.$$

This definition corresponds to *strong induction* (each value depends on the two previous ones, so proofs need two base cases).

Example. Find f_3, f_4, f_5, f_6.

Solution. $f_3 = f_2 + f_1 = 1 + 1 = 2$; $f_4 = f_3 + f_2 = 2 + 1 = 3$; $f_5 = f_4 + f_3 = 3 + 2 = 5$; $f_6 = f_5 + f_4 = 5 + 3 = 8$.

This is a *second-order linear recurrence*; see textbook Section 8.2 for methods to solve it exactly.

Example. Show that whenever $n \geq 3$, $f_n > \alpha^{n-2}$, where $\alpha = \dfrac{1 + \sqrt{5}}{2}$.

Solution. Let $P(n)$ be the statement $f_n > \alpha^{n-2}$. We use strong induction to show that $P(n)$ is true whenever $n \geq 3$.

Basis: $P(3)$ holds: $f_3 = 2 > \alpha \approx 1.618$; and $P(4)$ holds: $f_4 = 3 > \alpha^2 \approx 2.618$. (Two base cases are needed because the recurrence looks two steps back.)

Inductive step: Assume that $P(j)$ holds for all $3 \leq j \leq k$ (where $k \geq 4$). We want to show that $P(k + 1)$ is true, i.e., $f_{k+1} > \alpha^{k-1}$.

Note that α is a solution of $x^2 - x - 1 = 0$, so $\alpha^2 = \alpha + 1$. Hence

$$\alpha^{k-2} + \alpha^{k-3} = (\alpha + 1)\alpha^{k-3} = \alpha^2 \cdot \alpha^{k-3} = \alpha^{k-1}.$$

By the induction hypothesis (applied to $j = k$ and $j = k - 1$, both in range since $k \geq 4$): $f_k > \alpha^{k-2}$ and $f_{k-1} > \alpha^{k-3}$. So

$$f_{k+1} = f_k + f_{k-1} > \alpha^{k-2} + \alpha^{k-3} = \alpha^{k-1},$$

which is $P(k+1)$. ■

Example (Counting bit strings). How many bit strings of length n have no two consecutive 0's?

Solution. Let a_n be the number of such strings.

Base cases: $a_1 = 2$ (the strings 0 and 1 both qualify), and $a_2 = 3$ (the strings $01, 10, 11$ qualify; only 00 does not).

Recurrence: For $n \geq 3$, partition the valid strings of length n by their ending:

- Strings ending in 1: the first $n-1$ bits form any valid string of length $n-1$ — a_{n-1} choices;
- Strings ending in 0: the previous bit must be 1 (no two consecutive 0's), and the first $n-2$ bits form any valid string — a_{n-2} choices.

Therefore $a_n = a_{n-1} + a_{n-2}$ — the Fibonacci recurrence, shifted: $a_n = f_{n+2}$.

6.4 Recursively Defined Sets

Sets can also be defined recursively: give a *basis step* listing initial elements, and a *recursive step* giving rules to form new elements from old.

Example. Subset of integers S defined by:

Basis: $3 \in S$.

Recursive step: if $x \in S$ and $y \in S$, then $x + y \in S$.

This defines all positive multiples of 3: $3 \in S$; $3 + 3 = 6 \in S$; $3 + 6 = 9 \in S$; ...

Example. Fix a set A. Basis: for any $x \in A$, $\{x\} \in S$. Recursive step: if $x \in S$ and $y \in S$, then $x \cup y \in S$.

This defines $P(A) - \{\emptyset\}$, the set of all nonempty subsets of A (singletons are in; unions of smaller nonempty sets build up every nonempty finite subset).

Example (Full binary trees). The set of *full binary trees* is defined recursively:

- A single vertex r is a full binary tree with root r.
- Let T_1 and T_2 be two disjoint full binary trees with roots r_1 and r_2, respectively. Create a new vertex r, and make r_1 and r_2 the left and right child of r, respectively. The resulting structure is also a full binary tree T with root r. The trees T_1 and T_2 are called the *subtrees* of T.

6.5 Structural Induction

Example. Show that the set S defined below is the set of all positive integers that are multiples of 3:

Basis: $3 \in S$. Recursive step: if $x \in S$ and $y \in S$, then $x + y \in S$.

Proof. Let A be the set of all positive integers divisible by 3. We need to show $A \subseteq S$ and $S \subseteq A$.

(i) $A \subseteq S$**:** Let $P(n)$ be the statement that $3n \in S$; we use ordinary induction.

Basis: $P(1)$ is true since $3 \cdot 1 = 3 \in S$.

Inductive step: Assume $P(k)$ is true, i.e., $3k \in S$. Then $3(k+1) = 3k+3$. Since $3k \in S$ (induction hypothesis) and $3 \in S$ (basis of the definition), by the recursive definition with $x = 3k$ and $y = 3$ we have $3k + 3 \in S$, i.e., $P(k+1)$ is true. Hence every $3n \in S$.

(ii) $S \subseteq A$**:** by *structural induction.*

Basis: The element defined in the basis step of the definition, namely 3, belongs to A (since $3 \mid 3$).

Inductive step: Assume each of the elements used to define the new element in the recursive step of the definition belongs to A, i.e., $3 \mid x$ and $3 \mid y$. Then $3 \mid (x + y)$, so the new element also belongs to A.

Proof completed. The proof of $S \subseteq A$ is an example of structural induction. What is the $P(n)$ used implicitly? It is: "every element constructed after n steps of the recursive step belongs to A." ■

Structural induction framework

The validity of structural induction follows from the principle of mathematical induction: the $P(n)$ used implicitly is "the statement holds for any element constructed after n steps of the recursive step." It is used to show that a statement holds for all elements defined by a recursive definition.

- **Basis:** show that the statement holds for every element defined in the basis step of the recursive definition.
- **Recursive step:** show that if the statement holds for each of the elements used to construct new elements in the recursive step of the definition, then the statement holds for the new elements.

Example (Full binary trees). The *height* $h(T)$ and *size* $n(T)$ of a full binary tree T are defined recursively:

- If T consists of a single vertex, then $h(T) = 0$ and $n(T) = 1$.
- If r has two subtrees T_1, T_2, then

$$h(T) = \max\big(h(T_1), h(T_2)\big) + 1, \qquad n(T) = n(T_1) + n(T_2) + 1.$$

Theorem. If T is a full binary tree, then $n(T) \leq 2^{h(T)+1} - 1$.

Proof. By structural induction.

Basis: For a full binary tree consisting of a single vertex, the theorem holds: $n(T) = 1 = 2^{0+1} - 1 = 2^{h(T)+1} - 1$.

Inductive step: Consider the inductive step in the definition. Assume the theorem holds for T_1 and T_2, i.e.,

$$n(T_1) \leq 2^{h(T_1)+1} - 1, \qquad n(T_2) \leq 2^{h(T_2)+1} - 1.$$

We want to show that it holds for T:

$$\begin{aligned} n(T) &= n(T_1) + n(T_2) + 1 \\ &\leq \big(2^{h(T_1)+1} - 1\big) + \big(2^{h(T_2)+1} - 1\big) + 1 && \text{(inductive hypothesis)} \\ &\leq 2 \cdot \max\big(2^{h(T_1)+1}, 2^{h(T_2)+1}\big) - 1 \\ &= 2 \cdot 2^{\max(h(T_1),h(T_2))+1} - 1 \\ &= 2 \cdot 2^{h(T)} - 1 = 2^{h(T)+1} - 1, \end{aligned}$$

since $h(T) = \max(h(T_1), h(T_2)) + 1$. ■

6.6 Well-Formed Formulae

Definition. The set of *well-formed formulae* in propositional logic is defined as follows:

- T and F are well-formed formulae;
- For any propositional variable s, s is a well-formed formula;

- If E_1 and E_2 are well-formed formulae, then $(\neg E_1)$, $(E_1 \wedge E_2)$, $(E_1 \vee E_2)$, $(E_1 \to E_2)$, $(E_1 \leftrightarrow E_2)$ are well-formed formulae.

Examples. $((p \vee q) \to (q \wedge F))$ is a well-formed formula; $pq\wedge$ is not a well-formed formula. Is $\neg p \wedge q$ a well-formed formula? No — under this restricted definition, every compound formula must be fully parenthesized; $\neg p \wedge q$ lacks parentheses and is not generated by the rules.

Note. This is a restricted version of the definition; another version may add $\neg E_1$, $E_1 \wedge E_2$, $E_1 \vee E_2$, $E_1 \to E_2$, $E_1 \leftrightarrow E_2$ (without outer parentheses) to the recursive rule.

Example (Structural induction). Theorem. Any well-formed formula contains matching parentheses (equal numbers of left and right parentheses, properly nested).

Proof. By structural induction.

Basis: Each of T, F and any propositional variable s contains no parentheses, so trivially matching parentheses.

Inductive step: Assume E_1 and E_2 contain matching parentheses. Consider $(\neg E_1)$:

- The first '(' and the last ')' match each other;
- By the induction hypothesis, every parenthesis inside E_1 has a match within E_1;
- Hence all parentheses in $(\neg E_1)$ are matched.

The cases with $(E_1 \wedge E_2)$, $(E_1 \vee E_2)$, $(E_1 \to E_2)$, $(E_1 \leftrightarrow E_2)$ are similar: the outer pair matches, and the inner parentheses match by the induction hypothesis. ■

6.7 Strings

Definition. If $\Sigma = \{0, 1\}$, then Σ^* is the set of all bit strings defined over the alphabet Σ, i.e., $\lambda, 0, 1, 00, 01, 10, 11, \ldots$

Recursive definition of Σ^*

The set Σ^* of strings over the alphabet Σ:
- **Basis:** $\lambda \in \Sigma^*$ (where λ is the empty string);
- **Recursive step:** if $w \in \Sigma^*$ and $x \in \Sigma$, then $wx \in \Sigma^*$.

Example. If $\Sigma = \{a, b\}$, show that $aab \in \Sigma^*$.

Solution. Build up step by step using the recursive definition:

- Since $\lambda \in \Sigma^*$ and $a \in \Sigma$, $a \in \Sigma^*$;
- Since $a \in \Sigma^*$ and $a \in \Sigma$, $aa \in \Sigma^*$;
- Since $aa \in \Sigma^*$ and $b \in \Sigma$, $aab \in \Sigma^*$.

6.8 String Concatenation

Definition. Two strings can be combined via the operation of *concatenation*. Let Σ be the alphabet and Σ^* the set of strings over Σ. The concatenation of two strings, denoted by $\cdot$, is defined recursively as follows:

- **Basis:** if $w \in \Sigma^*$, then $w \cdot \lambda = w$;
- **Recursive step:** if $w_1 \in \Sigma^*$, $w_2 \in \Sigma^*$, $x \in \Sigma$, then $w_1 \cdot (w_2 x) = (w_1 \cdot w_2)x$.

Often $w_1 \cdot w_2$ is written simply as $w_1 w_2$. If $w_1 = abra$ and $w_2 = cadabra$, the concatenation is $w_1 w_2 = abracadabra$.

Example (Balanced parentheses). Give a recursive definition of the set of balanced parentheses P.

Solution.

- $\lambda \in P$;
- If $w \in P$, then $(w) \in P$;
- If $w_1 \in P$ and $w_2 \in P$, then $w_1 w_2 \in P$.

Example. Show that $(()())$ is in P.

Solution. $\lambda \in P$ (basis); hence $(\lambda) = () \in P$ (second rule); hence $()() \in P$ (third rule with $w_1 = ()$, $w_2 = ()$); hence $(()()) \in P$ (second rule with $w = ()()$).

Example. Why is $))((\,$ not in P?

Solution. By structural induction on the definition of P, every string $w \in P$ is balanced: every prefix of w contains at least as many left as right parentheses (and w has equal numbers of each). Basis: λ trivially satisfies this. Recursive step: if w satisfies it, then so does (w); if w_1, w_2 satisfy it, then so does $w_1 w_2$. But the string $))((\,$ starts with ')', so its first prefix already has more ')' than '(' — hence $))((\,$ cannot be generated, i.e., $))((\notin P$.

6.9 Length of a String

Example. Give a recursive definition of $l(w)$, the length of the string w.

Solution. The length of a string $w \in \Sigma^*$ can be recursively defined as:

$$l(\lambda) = 0; \qquad \text{for any } w \in \Sigma^*,\ x \in \Sigma:\ l(wx) = l(w) + 1.$$

Example (Structural induction on strings). Show that for any $u, v \in \Sigma^*$, $l(uv) = l(u) + l(v)$.

Proof. Let $P(v)$ be the statement $\forall u \left[l(uv) = l(u) + l(v) \right]$. We do structural induction on v (recall Σ^* is built up by appending one character at a time).

Basis: $P(\lambda)$ is true, since for any u,

$$l(u\lambda) = l(u) = l(u) + 0 = l(u) + l(\lambda),$$

using $u \cdot \lambda = u$ and $l(\lambda) = 0$.

Inductive step: Assume $P(w)$ is true, i.e., $\forall u \left[l(uw) = l(u) + l(w) \right]$. We need to show $P(wx)$ is true for any $x \in \Sigma$. For any u:

$$\begin{aligned} l\big(u(wx)\big) &= l\big((uw)x\big) && \text{(definition of concatenation)} \\ &= l(uw) + 1 && \text{(recursive definition of } l) \\ &= l(u) + l(w) + 1 && \text{(induction hypothesis } P(w)) \\ &= l(u) + l(wx) && \text{(recursive definition of } l). \end{aligned}$$

Hence $P(v)$ holds for all $v \in \Sigma^*$. ■

7 Modular Arithmetic

Number theory is the part of mathematics devoted to the study of the integers and their properties. It has a very long history — for example, the Chinese Remainder Theorem

is about 1700 years old. For a long time it was regarded as pure mathematics with no practical use; G. H. Hardy famously claimed that pure mathematics is "beautiful" and "useless". However, in recent decades number theory has found numerous applications in computer science, most notably in cryptography.

7.1 Divisibility

Definition. Let a and b be integers with $a \neq 0$. Then a divides b if there exists an integer c such that $b = ac$. The notation $a \mid b$ denotes that a divides b; if a does not divide b, we write $a \nmid b$. When $a \mid b$ we say that a is a factor or divisor of b and that b is a multiple of a. Note that if $a \mid b$, then b/a is an integer.

Example. Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Solution. $3 \nmid 7$ because $7/3$ is not an integer. $3 \mid 12$ because $12 = 3 \cdot 4$, i.e., $12/3 = 4$ is an integer.

Properties of Divisibility

Theorem. Let a, b, and c be integers, where $a \neq 0$.
(i) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;
(ii) If $a \mid b$, then $a \mid bc$ for all integers c;
(iii) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof of (i). Suppose $a \mid b$ and $a \mid c$. Then it follows from the definition of divisibility that there are integers s and t with $b = as$ and $c = at$. Hence,

$$b + c = as + at = a(s + t).$$

Therefore $a \mid (b + c)$. (The proofs of (ii) and (iii) follow similarly from the definition.) ■

Corollary. If a, b, and c are integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Proof. By part (ii) of the theorem, we have $a \mid mb$ and $a \mid nc$. By part (i) of the theorem, we have $a \mid mb + nc$. ■

7.2 Euclid's Division Theorem

Euclid's Division Theorem

Theorem. For any $a \in \mathbb{Z}$, $d \in \mathbb{Z}^+$, there exist unique integers q and r, with $0 \leq r < d$, such that

$$a = dq + r.$$

Here d is called the divisor, a is called the dividend, q is called the quotient, and r is called the remainder. We write

$$q = a \,\mathbf{div}\, d, \qquad r = a \bmod d.$$

Example. $11 \,\mathbf{div}\, 3 = 3$, $11 \bmod 3 = 2$; $-11 \,\mathbf{div}\, 3 = -4$, $-11 \bmod 3 = 1$. Note that the remainder is always nonnegative, even for negative dividends: $-11 = 3 \cdot (-4) + 1$.

The theorem has two parts: existence of q and r, and their uniqueness. We give two proofs of existence — one by the well-ordering property, one by mathematical induction — and then prove uniqueness.

Proof of Existence (well-ordering). Let

$$S = \{\, x \mid x = a - dq,\ x \geq 0,\ q \in \mathbb{Z} \,\}.$$

The set S is nonempty: since $d \geq 1$, the quantity $-dq$ can be made as large as needed (choose q to be a very negative number), so some $a - dq \geq 0$ exists. By the well-ordering property, let r be the smallest integer in S. By the definition of S, there is a $q \in \mathbb{Z}$ such that

$$r = a - dq, \qquad \text{i.e.,} \qquad a = dq + r.$$

By the definition of S, we have $r \geq 0$. We must also have $r < d$. If not, i.e., if $r \geq d$, then there would be a smaller nonnegative integer in S, namely

$$a - d(q+1) = a - dq - d = r - d \geq 0,$$

contradicting the minimality of r. Hence $0 \leq r < d$. ■

Proof of Existence (by induction, for $a \geq 0$). Let d be any positive integer. Let $P(k)$ be the proposition

$$\exists q\, \exists r\, (0 \leq r < d \ \wedge\ k = dq + r).$$

Basis: $P(0)$ is true by taking $q = 0$, $r = 0$.

Inductive step: Assume $P(k)$ is true. We want to show $P(k+1)$. By the induction hypothesis, there exist q, r such that $0 \leq r < d$ and $k = dq + r$. So

$$k + 1 = dq + r + 1.$$

We distinguish two cases.

- Case 1: $r < d - 1$. Then we are done, since we have found $q' = q$, $r' = r + 1$ such that $0 \leq r' < d$ and $k + 1 = dq' + r'$.
- Case 2: $r = d - 1$. Let $q' = q+1$, $r' = 0$. Then $k+1 = dq+(d-1)+1 = d(q+1) = dq'+r'$, and $0 \leq r' < d$.

This proves the theorem for all $a \geq 0$. ■

How about $a < 0$? We cannot use induction directly on $\mathbb{Z}$ because $\mathbb{Z}$ does not satisfy the well-ordering property. The idea is to prove the theorem separately on $\{0\} \cup \mathbb{Z}^+$ (done above) and on $\{0\} \cup \mathbb{Z}^-$ (done below).

Proof of Existence (by induction, for $a \leq 0$). Let $P(k)$ be the proposition

$$\exists q\, \exists r\, (0 \leq r < d \ \wedge\ -k = dq + r).$$

Basis: $P(0)$ is true by taking $q = 0$, $r = 0$.

Inductive step: Assume $P(k)$ is true; we show $P(k+1)$. By the induction hypothesis, there exist q, r such that $0 \leq r < d$ and $-k = dq + r$. So

$$-(k+1) = dq + r - 1.$$

- Case 1: $r > 0$. Then we are done, since we have found $q' = q$, $r' = r - 1$ such that $0 \leq r' < d$ and $-(k+1) = dq' + r'$.
- Case 2: $r = 0$. Let $q' = q-1$, $r' = d-1$. Then $-(k+1) = dq-1 = d(q-1)+(d-1) = dq' + r'$, and $0 \leq r' < d$.

This proves the theorem for all $a \leq 0$, and hence for all $a \in \mathbb{Z}$. ■

Proof of Uniqueness. Suppose there are q_1, r_1, q_2, r_2 such that

$$a = dq_1 + r_1, \qquad 0 \leq r_1 < d, \tag{1}$$
$$a = dq_2 + r_2, \qquad 0 \leq r_2 < d. \tag{2}$$

Subtracting (2) from (1):

$$0 = d(q_1 - q_2) + r_1 - r_2 \quad \Longrightarrow \quad d(q_1 - q_2) = r_2 - r_1.$$

So $d \mid (r_2 - r_1)$. But $0 \le r_1 < d$ and $0 \le r_2 < d$ imply $-d < r_2 - r_1 < d$. The only multiple of d strictly between $-d$ and d is 0, so $r_2 - r_1 = 0$, i.e., $r_1 = r_2$. Then $d(q_1 - q_2) = 0$ and $d \neq 0$ give $q_1 = q_2$. ■

7.3 Modular Arithmetic

Lemma. For any $a, k \in \mathbb{Z}$, $m \in \mathbb{Z}^+$,

$$a \bmod m = (a + km) \bmod m.$$

Proof. By Euclid's Division Theorem, there exist unique q, r with $0 \le r < m$ such that

$$a = mq + r. \tag{1}$$

Similarly, there exist unique q', r' with $0 \le r' < m$ such that

$$a + km = mq' + r'. \tag{2}$$

Adding km to both sides of (1):

$$a + km = m(q + k) + r.$$

Since $0 \le r < m$, this is a valid quotient–remainder representation of $a + km$. By the uniqueness in the Division Theorem applied to (2), we have $r = r'$. By the definition of mod, $a \bmod m = r = r' = (a + km) \bmod m$. ■

Example. Prove the property $(a \bmod mn) \bmod n = a \bmod n$ (for $m, n \in \mathbb{Z}^+$, $a \in \mathbb{Z}$).

Solution. By Euclid's Division Theorem, write

$$a = qmn + s, \qquad 0 \le s < mn,$$

so $a \bmod mn = s$. Then write

$$s = pn + r, \qquad 0 \le r < n,$$

so $(a \bmod mn) \bmod n = s \bmod n = r$. On the other hand, substituting back,

$$a = qmn + pn + r = (qm + p)\, n + r, \qquad 0 \le r < n,$$

so $a \bmod n = r$ as well. The equation is proved. ■

Addition and Multiplication Modulo m

Theorem. For any $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$,

$$(a + b) \bmod m = \big((a \bmod m) + (b \bmod m)\big) \bmod m,$$

$$(a \cdot b) \bmod m = \big((a \bmod m) \cdot (b \bmod m)\big) \bmod m.$$

Proof (addition). By Euclid's Division Theorem, there exist unique q_1, q_2 such that

$$a = q_1 m + (a \bmod m), \qquad b = q_2 m + (b \bmod m).$$

Adding these two equations and taking modulo m:

$$(a + b) \bmod m = \big((q_1 + q_2)m + (a \bmod m) + (b \bmod m)\big) \bmod m.$$

The theorem then follows from the previous Lemma (adding a multiple of m does not change the remainder). ■

The proof for multiplication is similar. These theorems can be used in two directions:

$$(a+b) \bmod m = \big((a \bmod m) + b\big) \bmod m = \big(a + (b \bmod m)\big) \bmod m,$$

$$(a \cdot b) \bmod m = \big((a \bmod m) \cdot b\big) \bmod m = \big(a \cdot (b \bmod m)\big) \bmod m,$$

i.e., we may reduce either operand (or both) modulo m before adding or multiplying.

7.4 Modular Arithmetic on $\mathbb{Z}_m$

Definition. $\mathbb{Z}_m = \{0, 1, \ldots, m-1\}$, the set of nonnegative integers less than m.

Definition. For $a, b \in \mathbb{Z}_m$, define the operations $+_m$ and $\cdot_m$ on $\mathbb{Z}_m$ by

$$a +_m b = (a+b) \bmod m, \qquad a \cdot_m b = (a \cdot b) \bmod m.$$

Example. Working in $\mathbb{Z}_{11}$:

$$7 +_{11} 9 = (7+9) \bmod 11 = 16 \bmod 11 = 5,$$

$$7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8.$$

Properties of Arithmetic Modulo m

For $a, b, c \in \mathbb{Z}_m$:

- **Closure:** $a +_m b$ and $a \cdot_m b$ belong to $\mathbb{Z}_m$.
- **Associativity:** $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
- **Commutativity:** $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$.
- **Distributivity:** $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$.

Proof of Associativity of $+_m$.

$$\begin{aligned}(a +_m b) +_m c &= \big((a+b) \bmod m +_m c\big) = \big((a+b) \bmod m + c\big) \bmod m \\ &= (a+b+c) \bmod m \qquad \text{(by the addition theorem)} \\ &= \big(a + (b+c) \bmod m\big) \bmod m \qquad \text{(same theorem, other side)} \\ &= a +_m (b +_m c).\end{aligned}$$

The proofs of the other properties are similar. ■

7.4.1 Additive and Multiplicative Inverses

Identity elements. The elements 0 and 1 are identity elements for addition and multiplication modulo m, respectively: for $a \in \mathbb{Z}_m$,

$$a +_m 0 = a \qquad \text{and} \qquad a \cdot_m 1 = a.$$

Additive inverses. For $a \in \mathbb{Z}_m$, $(-a \bmod m)$ is the additive inverse of a:

$$a +_m (-a \bmod m) = \big(a + (-a \bmod m)\big) \bmod m = (a + (-a)) \bmod m = 0,$$

where the second equality uses the addition theorem.

Example. What is the additive inverse of 27 in $\mathbb{Z}_{58}$?

Solution. $-27 \bmod 58 = 58 - 27 = 31$. Check: $27 +_{58} 31 = 58 \bmod 58 = 0$. So the answer is 31.

Multiplicative inverses. For $a \in \mathbb{Z}_m$, b is its multiplicative inverse if $a \cdot_m b = 1$.

Example. Let $m = 4$. What is the multiplicative inverse of 3? What is the multiplicative inverse of 2?

Solution. For 3: $3 \cdot_4 3 = 9 \bmod 4 = 1$, so the multiplicative inverse of 3 in $\mathbb{Z}_4$ is 3 itself. For 2: $2 \cdot_4 0 = 0$, $2 \cdot_4 1 = 2$, $2 \cdot_4 2 = 0$, $2 \cdot_4 3 = 2$ — none equals 1, so 2 has no multiplicative inverse in $\mathbb{Z}_4$. (We will see in the next chapter that a has an inverse in $\mathbb{Z}_m$ iff $\gcd(a, m) = 1$.)

7.5 Congruences

Congruence Modulo m

Definition. Let $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$. Then a is congruent to b modulo m if $a \bmod m = b \bmod m$. We write

$$a \equiv b \pmod{m}.$$

Such a statement $a \equiv b \pmod{m}$ is called a congruence, and m is its modulus. Equivalent definitions:

(i) $m \mid (a - b)$;

(ii) $\exists t \in \mathbb{Z}\ (a = b + mt)$.

Example. $17 \equiv 5 \pmod{6}$ because $17 \bmod 6 = 5 \bmod 6 = 5$ (equivalently $6 \mid (17-5) = 12$). On the other hand $24 \not\equiv 14 \pmod{6}$ because $24 \bmod 6 = 0 \neq 2 = 14 \bmod 6$.

Note on notation. In the expression $a \bmod m$, mod is a binary operator producing an integer in $\{0, \ldots, m-1\}$. In $a \equiv b \pmod{m}$, the notation denotes an equivalence relationship between a and b.

7.5.1 Modular Arithmetic and Congruences

Congruences provide another way to express modular arithmetic: replace $+_m$, $\cdot_m$, $=$ by $+$, $\cdot$, $\equiv$, and add $(\bmod m)$ at the end. For any integers a, b, c and positive integer m:

Associativity:

$$(a + b) + c \equiv a + (b + c) \pmod{m}, \qquad (a \cdot b) \cdot c \equiv a \cdot (b \cdot c) \pmod{m}.$$

Commutativity:

$$a + b \equiv b + a \pmod{m}, \qquad a \cdot b \equiv b \cdot a \pmod{m}.$$

Distributivity:

$$a(b + c) \equiv ab + ac \pmod{m}.$$

Difference between the two views. The operations $+_m$ and $\cdot_m$ are defined only on elements of $\mathbb{Z}_m$, and produce results in $\mathbb{Z}_m$. Congruences are defined over all of $\mathbb{Z}$, and numbers on the two sides of $\equiv$ need not lie in $\mathbb{Z}_m$.

Example. $6 +_8 7 = 5$. In congruence notation we may write $6 + 15 \equiv 5 \pmod{8}$ and also $6 + 15 \equiv 21 \pmod{8}$ (both 5 and 21 are congruent to $6 + 15 = 21$ modulo 8). But we cannot write $6 +_8 7 = 13$ or $6 +_8 15 = 5$, because $+_8$ requires both operands and the result to be in $\mathbb{Z}_8$.

Adding and Multiplying Congruences

Theorem. Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m} \qquad \text{and} \qquad ac \equiv bd \pmod{m}.$$

Example. Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5},$$
$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}.$$

Corollary. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a - c \equiv b - d \pmod{m}.$$

Warning. Division does not work: $a/c \not\equiv b/d \pmod{m}$ in general.

Corollary. If $a \equiv b \pmod{m}$, then for any $c \in \mathbb{Z}$,

$$a + c \equiv b + c \pmod{m}, \qquad a - c \equiv b - c \pmod{m}, \qquad ac \equiv bc \pmod{m}.$$

(This follows from the theorem because $c \equiv c \pmod{m}$.)

7.6 Applications of Modular Arithmetic

7.6.1 Parity Bits

Digital information is transmitted as a bit stream, e.g.,

$$00010010101111010101010101010101.$$

Denote the bits as $x_1, x_2, \ldots, x_n$. At the end of the stream, we often add a parity bit

$$x_{n+1} = (x_1 + x_2 + \cdots + x_n) \bmod 2.$$

This can detect one wrong bit — or, more generally, an odd number of wrong bits, since any odd number of flips changes the parity. It cannot detect an even number of wrong bits, since they leave the parity unchanged.

7.6.2 Hash Functions

Example. During exam checking, how should we organize the exam papers so that the TA can quickly find the paper for any given student?

Solution. Use a hash function such as

$$h(id) = id \bmod 10,$$

or hash by the initials of the student names. In general, a hash function maps a universe of keys to a small set of locations (buckets).

7.6.3 Hash Table

Example. Build a student database where each student has an id, name, address, phone, GPA, etc. Given a student id, we want to retrieve the record quickly.

Solution. Suppose there are n students. Create an array H of size m, and put the student with id x at location $H[h(x)]$. Resolving collisions: $H[i]$ stores a linked list of all elements x with $h(x) = i$.

7.6.4 Hashing Strings

Each character is an integer between 0 and 255, and we need to take all characters into account. A commonly used hash function for a string s whose characters are accessed as $s[0], s[1], \ldots$ is

$$h(s) = \big(s[0] \cdot 31^{n-1} + s[1] \cdot 31^{n-2} + s[2] \cdot 31^{n-3} + \cdots\big) \bmod n,$$

i.e., the string is treated as a number in base 31. This is the hash function used in the Java string library (note that integer overflows there are equivalent to modular arithmetic):

```
int h(String s, int n) {
    int hash = 0;
    for (int i = 0; i < s.length(); i++)
        hash = ((31 * hash) + s[i]);
    return hash % n;
}
```

The loop computes the polynomial hash by Horner's rule, one character at a time. (Why the constant 31? This is analyzed in the next chapter.)

7.6.5 Pseudorandom Numbers

Random numbers are needed for many purposes: computer games, computer simulation, gambling, and so on. But how does a computer "generate" random numbers (e.g., `rand()`)?

Pseudorandom numbers are generated by deterministic methods, so they are not truly random. A standard approach is the linear congruential method: given a modulus m, a multiplier a, an increment c, and a seed x_0 (the seed is usually given by the user, often the system time; the other parameters are hard-coded), generate a sequence of pseudorandom numbers by

$$x_{n+1} = (ax_n + c) \bmod m.$$

Example. Let $m = 9$, $a = 7$, $c = 4$, $x_0 = 3$. Compute the sequence.

Solution.

$x_1 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7,$ $x_2 = (7 \cdot 7 + 4) \bmod 9 = 53 \bmod 9 = 8,$
$x_3 = (7 \cdot 8 + 4) \bmod 9 = 60 \bmod 9 = 6,$ $x_4 = (7 \cdot 6 + 4) \bmod 9 = 46 \bmod 9 = 1,$
$x_5 = (7 \cdot 1 + 4) \bmod 9 = 11 \bmod 9 = 2,$ $x_6 = (7 \cdot 2 + 4) \bmod 9 = 18 \bmod 9 = 0,$
$x_7 = (7 \cdot 0 + 4) \bmod 9 = 4,$ $x_8 = (7 \cdot 4 + 4) \bmod 9 = 32 \bmod 9 = 5,$
$x_9 = (7 \cdot 5 + 4) \bmod 9 = 39 \bmod 9 = 3 = x_0.$

So the sequence is

$$3, 7, 8, 6, 1, 2, 0, 4, 5, 3, 7, 8, 6, 1, 2, 0, 4, 5, 3, \ldots$$

which repeats with period 9.

Remarks. The linear congruential method always generates repeating patterns. It has been found that with $m = 2^{31} - 1$, $a = 7^5 = 16807$, $c = 0$, it generates $2^{31} - 2$ different numbers before repeating. If random numbers in a certain range are needed, take another mod at the end. The method generates uniformly distributed numbers, but they are not truly random — do not use them for lotteries! There is a whole theory about pseudorandom number generators; `http://www.random.org` provides truly random numbers generated from atmospheric noise.

8 Greatest Common Divisor

8.1 Primes and the Fundamental Theorem of Arithmetic

Definition. A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p. A positive integer that is greater than 1 and is not prime is called composite.

The Fundamental Theorem of Arithmetic

Theorem. Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, up to the order of the prime factors.

8.2 Greatest Common Divisor

Definition. Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of a and b, denoted by $\gcd(a, b)$.

One can find the gcd by prime factorizations: write

$$a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad b = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$$

(with zero exponents allowed), then

$$\gcd(a, b) = p_1^{\min(a_1,b_1)} p_2^{\min(a_2,b_2)} \cdots p_k^{\min(a_k,b_k)}.$$

Example. Find $\gcd(120, 500)$.

Solution. Prime factorize:

$$120 = 2^3 \cdot 3 \cdot 5, \qquad 500 = 2^2 \cdot 5^3.$$

Taking the minimum exponent of each prime:

$$\gcd(120, 500) = 2^{\min(3,2)} \cdot 3^{\min(1,0)} \cdot 5^{\min(1,3)} = 2^2 \cdot 3^0 \cdot 5^1 = 20.$$

Definition. The integers a and b are relatively prime if $\gcd(a, b) = 1$.

Example. 17 and 22 are relatively prime: 17 is prime and $17 \nmid 22$, so $\gcd(17, 22) = 1$.

8.3 The Euclidean Algorithm

Factoring large numbers is hard — no efficient algorithms exist. Hence the prime-factorization method for gcd is impractical for large integers. The Euclidean algorithm computes gcds efficiently, based on the following lemma.

GCD Lemma

Lemma. Let $a = bq + r$, where a, b, q, r are integers. Then

$$\gcd(a, b) = \gcd(b, r).$$

Proof. Suppose that d divides both a and b. Then d also divides $a - bq = r$. Hence, any common divisor of a and b must also be a common divisor of b and r.

Conversely, suppose that d divides both b and r. Then d also divides $bq + r = a$. Hence, any common divisor of b and r must also be a common divisor of a and b.

The two pairs (a, b) and (b, r) therefore have exactly the same set of common divisors, so their greatest common divisors are equal: $\gcd(a, b) = \gcd(b, r)$. ■

The algorithm. Idea: to obtain maximum efficiency, choose the smallest r, i.e., $r = a \bmod b$ (suppose $a > b$), and iterate. The correctness of the algorithm follows from the lemma.

```
gcd(a, b):
    x := a
    y := b
    while y != 0
        r := x mod y
        x := y
        y := r
    return x
```

Example. Compute $\gcd(287, 91)$.

Solution. Applying the lemma repeatedly:

$$\begin{aligned}\gcd(287, 91) &= \gcd(91, 287 \bmod 91) = \gcd(91, 14) = \gcd(14, 91 \bmod 14) \\ &= \gcd(14, 7) = \gcd(7, 0) = 7.\end{aligned}$$

(Indeed $287 = 3 \cdot 91 + 14$, $91 = 6 \cdot 14 + 7$, $14 = 2 \cdot 7$.)

8.3.1 Analysis: Lamé's Theorem

Given a, b with $a > b$, how many steps (mod's) does Euclid's algorithm take to compute $\gcd(a, b)$?

Lemma. If the algorithm takes n steps to compute $\gcd(a, b)$, then $a \geq f_{n+2}$ and $b \geq f_{n+1}$, where f_n is the n-th Fibonacci number, i.e., $f_1 = 1$, $f_2 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 3$.

Proof (by induction). Let $P(n)$ be the statement of the lemma.

Basis: $P(1)$ is true. If the algorithm takes 1 step, then $b \neq 0$ and $b \mid a$. The smallest such a, b with $a > b$ are $a = 2$, $b = 1$. And indeed $f_2 = 1 \leq b$ and $f_3 = 2 \leq a$.

Inductive step: Assume $P(k)$ is true. We will show $P(k + 1)$, i.e., if $\gcd(a, b)$ takes $k + 1$ steps, then $a \geq f_{k+3}$ and $b \geq f_{k+2}$.

After the first step, we get $\gcd(b, r)$ where $r = a \bmod b$. Then it takes k steps to compute $\gcd(b, r)$. By the induction hypothesis,

$$b \geq f_{k+2}, \qquad r \geq f_{k+1}.$$

Since $a = bq + r$ with $q \geq 1$ (because $a > b$), we have

$$a \geq b + r \geq f_{k+2} + f_{k+1} = f_{k+3}.$$

Together with $b \geq f_{k+2}$, this establishes $P(k + 1)$. ■

Consequence. Since the Fibonacci numbers grow exponentially, $b \geq f_{n+1} > \alpha^{n-1}$ where $\alpha = \dfrac{1 + \sqrt{5}}{2}$ is the golden ratio. Taking logarithms,

$$n < 1 + \log_\alpha b.$$

So the number of steps of the Euclidean algorithm is logarithmic in b — the algorithm is very efficient even for huge inputs.

8.4 Bézout's Theorem and the Extended Euclidean Algorithm

Bézout's Theorem

Theorem. If a and b are positive integers, then there exist integers s and t such that

$$\gcd(a, b) = sa + tb.$$

The integers s and t are called Bézout coefficients of a and b.

Example. $\gcd(6, 14) = 2 = (-2) \cdot 6 + 1 \cdot 14$.

Instead of proving the theorem directly, we give an algorithm — the extended Euclidean algorithm — to find such s and t: run the Euclidean algorithm, then work backwards through the divisions.

Example. Express $\gcd(252, 198)$ as a linear combination of 252 and 198.

Solution. First find $\gcd(252, 198)$ by the Euclidean algorithm:

$$252 = 1 \cdot 198 + 54, \tag{1}$$
$$198 = 3 \cdot 54 + 36, \tag{2}$$
$$54 = 1 \cdot 36 + 18, \tag{3}$$
$$36 = 2 \cdot 18. \tag{4}$$

The last nonzero remainder is 18, so $\gcd(252, 198) = 18$.

Now rewrite each division to isolate the remainder:

$$54 = 252 - 1 \cdot 198, \qquad 36 = 198 - 3 \cdot 54, \qquad 18 = 54 - 1 \cdot 36.$$

Substituting backwards:

$$\begin{aligned} 18 &= 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198 \\ &= 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 \\ &= 4 \cdot 252 - 5 \cdot 198. \end{aligned}$$

So $\gcd(252, 198) = 18 = 4 \cdot 252 + (-5) \cdot 198$.

Puzzle: Water Measuring. Given two bottles, one of volume 500 ml and the other 700 ml, and an infinite water supply, get exactly 100 ml of water.

Solution. This follows exactly from

$$100 = \gcd(500, 700) = 3 \times 500 - 2 \times 700 :$$

fill the 500 ml bottle and pour into the 700 ml bottle three times, emptying the 700 ml bottle whenever it is full (twice in total); what remains at the end is exactly 100 ml. Corollary: any multiple of the gcd can be obtained this way.

8.5 Multiplicative Inverses

Definition. The (multiplicative) inverse of a modulo m is some b such that $ab \equiv 1 \pmod{m}$. By default, "inverse" means "multiplicative inverse".

Example: inverse tables. Below are the inverses a^{-1} for each a in $\mathbb{Z}_5, \mathbb{Z}_6, \mathbb{Z}_7, \mathbb{Z}_8$ (X means "does not exist"):

$\mathbb{Z}_5$

a	1	2	3	4
a^{-1}	1	3	2	4

$\mathbb{Z}_7$

a	1	2	3	4	5	6
a^{-1}	1	4	5	2	3	6

$\mathbb{Z}_6$

a	1	2	3	4	5
a^{-1}	1	X	X	X	5

$\mathbb{Z}_8$

a	1	2	3	4	5	6	7
a^{-1}	1	X	3	X	5	X	7

For example, in $\mathbb{Z}_7$: $2 \cdot 4 = 8 \equiv 1$, $3 \cdot 5 = 15 \equiv 1$, $6 \cdot 6 = 36 \equiv 1 \pmod 7$. Notice the pattern: inverses exist exactly for the elements relatively prime to the modulus.

Existence and Uniqueness of Inverses

Theorem. For any $a \in \mathbb{Z}_m$, $m > 1$, if $\gcd(a, m) = 1$ then a has a unique inverse in $\mathbb{Z}_m$.
Corollary. For any prime p, every nonzero $a \in \mathbb{Z}_p$ has a multiplicative inverse.

Proof of existence. Since $\gcd(a, m) = 1$, by Bézout's Theorem there are integers s and t such that $sa + tm = 1$. Hence

$$sa = -tm + 1.$$

Since $tm \equiv 0 \pmod m$, it follows that

$$sa \equiv 1 \pmod m.$$

Since s may not be in $\mathbb{Z}_m$, we write $(s \bmod m) \cdot_m a = 1$. Consequently, $s \bmod m$ is the inverse of a in $\mathbb{Z}_m$. ■

Proof of uniqueness. Suppose b, c are both inverses of a, i.e.,

$$ab \equiv 1 \pmod m, \tag{1}$$

$$ac \equiv 1 \pmod m. \tag{2}$$

Multiply both sides of (1) by c: $abc \equiv c \pmod m$. Multiply both sides of (2) by b: $abc \equiv b \pmod m$. So $b \equiv c \pmod m$, i.e., a has a unique inverse in $\mathbb{Z}_m$, written a^{-1}. ■

Note. It is also true that if $\gcd(a, m) \neq 1$, then a^{-1} does not exist. (Left as an exercise.)

8.5.1 Finding Inverses

Given a, m such that $\gcd(a, m) = 1$, how do we find the inverse of a in $\mathbb{Z}_m$? Look at the proof of the previous theorem: use the extended Euclidean algorithm to find s and t such that $sa + tm = 1$; then $s \bmod m$ is the multiplicative inverse of a in $\mathbb{Z}_m$.

Example. Find an inverse of 3 modulo 7.

Solution. Using the extended Euclidean algorithm: $7 = 2 \cdot 3 + 1$. Working backwards, we get

$$(-2) \cdot 3 + 1 \cdot 7 = 1,$$

so $s = -2$. Then $-2 \bmod 7 = 5$ is the inverse of 3 in $\mathbb{Z}_7$. Check: $3 \cdot 5 = 15 \equiv 1 \pmod 7$.

Example. Find the inverse of 101 modulo 4620.

Solution. First run the Euclidean algorithm:

$$\begin{aligned} 4620 &= 45 \cdot 101 + 75, \\ 101 &= 1 \cdot 75 + 26, \\ 75 &= 2 \cdot 26 + 23, \\ 26 &= 1 \cdot 23 + 3, \\ 23 &= 7 \cdot 3 + 2, \\ 3 &= 1 \cdot 2 + 1, \\ 2 &= 2 \cdot 1. \end{aligned}$$

Since the last nonzero remainder is 1, $\gcd(101, 4620) = 1$, and the inverse exists. Now work backwards:

$$\begin{aligned} 1 &= 3 - 1 \cdot 2 \\ &= 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3 \\ &= -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23 \\ &= 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75 \\ &= 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75 = 26 \cdot 101 - 35 \cdot 75 \\ &= 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101) = -35 \cdot 4620 + 1601 \cdot 101. \end{aligned}$$

So $1601 \cdot 101 \equiv 1 \pmod{4620}$, and 1601 is an inverse of 101 modulo 4620.

8.6 Solving Linear Congruences

A linear congruence is an equation

$$ax \equiv b \pmod{m}.$$

Given a, b, m such that $\gcd(a, m) = 1$, how do we find x? Solution: find a^{-1} and multiply it on both sides.

Example. Solve $3x \equiv 4 \pmod{7}$.

Solution. From the earlier example, $3^{-1} = 5$ in $\mathbb{Z}_7$. Multiply both sides by 5:

$$3^{-1} \cdot 3x \equiv 3^{-1} \cdot 4 \pmod{7} \quad \Longrightarrow \quad x \equiv 5 \cdot 4 \equiv 6 \pmod{7}.$$

Check: $3 \cdot 6 = 18 \equiv 4 \pmod{7}$. So $x \equiv 6 \pmod{7}$.

Unique Solution when $\gcd(a, m) = 1$

Corollary. If $\gcd(a, m) = 1$, the linear congruence

$$ax \equiv b \pmod{m}$$

has a unique solution in $\mathbb{Z}_m$.

Proof. Existence has already been proved by construction: $x \equiv a^{-1}b \pmod{m}$ is a solution.

Uniqueness: Suppose the congruence has two solutions x_1, x_2:

$$ax_1 \equiv b \pmod{m}, \qquad ax_2 \equiv b \pmod{m}.$$

Multiply both by a^{-1}:

$$x_1 \equiv ba^{-1} \pmod{m}, \qquad x_2 \equiv ba^{-1} \pmod{m}.$$

So $x_1 \equiv x_2 \pmod{m}$. ■

Note. If $\gcd(a, m) \neq 1$, the linear congruence $ax \equiv b \pmod{m}$ may have no solution or multiple solutions in $\mathbb{Z}_m$.

Example. (a) $2x \equiv 1 \pmod{6}$ has no solution in $\mathbb{Z}_6$: the left side $2x$ is always even, while $1+6k$ is always odd. (b) $2x \equiv 4 \pmod{6}$ has two solutions in $\mathbb{Z}_6$: $x = 2$ and $x = 5$, since $2 \cdot 2 = 4 \equiv 4$ and $2 \cdot 5 = 10 \equiv 4 \pmod{6}$.

8.7 Solving Systems of Congruences

Systems of linear congruences are solved the same way as in the real domain. Just remember: there is no division — multiply by the inverse instead.

Example. Solve the system

$$\begin{aligned} 2x + 3y &\equiv 5 \pmod{7}, \\ 3x - 4y &\equiv 1 \pmod{7}. \end{aligned}$$

Solution. Eliminate one variable using inverses in $\mathbb{Z}_7$ (recall $2^{-1} = 4$, $3^{-1} = 5$, $4^{-1} = 2 \bmod 7$). Multiply the first equation by 3 and the second by 2:

$$\begin{aligned} 6x + 9y \equiv 6x + 2y \equiv 15 \equiv 1 \pmod{7}, \\ 6x - 8y \equiv 6x - y \equiv 2 \pmod{7}. \end{aligned}$$

Subtracting the second from the first: $3y \equiv -1 \equiv 6 \pmod{7}$. Multiply by $3^{-1} = 5$: $y \equiv 5 \cdot 6 = 30 \equiv 2 \pmod{7}$. Substituting back into the first equation: $2x + 6 \equiv 5 \pmod{7}$, so $2x \equiv -1 \equiv 6 \pmod{7}$, and multiplying by $2^{-1} = 4$: $x \equiv 24 \equiv 3 \pmod{7}$. Check: $2 \cdot 3 + 3 \cdot 2 = 12 \equiv 5$, $3 \cdot 3 - 4 \cdot 2 = 1 \pmod{7}$. So $(x, y) \equiv (3, 2) \pmod{7}$.

8.8 Revisiting the String Hash Function

Recall the string hash from the previous chapter. Consider the case with only 3 characters:

$$h(s) = \big(s[0] \cdot 31^2 + s[1] \cdot 31 + s[2]\big) \bmod 2^{32} \bmod n.$$

Note that $\gcd(31, 2^{32}) = 1$ since 31 is odd and 2^{32} is a power of two. Given any $s[2]$ and b, the congruence

$$31x + s[2] \equiv b \pmod{2^{32}}$$

has a unique solution x (because 31 has an inverse modulo 2^{32}). This means, given $s[2]$, $h(s)$ depends on $x = s[0] \cdot 31 + s[1]$ and every value b is possible. Similarly, given any $s[1]$, x can possibly take any value depending on $s[0]$. So changing any single character changes the hash — a desirable property.

Other reasons for choosing 31: performance — $x \cdot 31 = (x \ll 5) - x$, which is a cheap shift-and-subtract; and empirically, using 31 produces more balanced hashes over English text.

8.9 Checksums: the HKID Check Digit

Example. HKID numbers are of the format X123456(Y), where X is one or two letters and Y is the check digit, 0 to 9 or A. How is it computed?

Solution. Replace the first two characters (letters, or a letter and a space) as follows:

A=10	B=11	C=12	D=13	E=14	F=15	G=16	H=17	I=18	J=19
K=20	L=21	M=22	N=23	O=24	P=25	Q=26	R=27	S=28	T=29
U=30	V=31	W=32	X=33	Y=34	Z=35	empty=36			

Denote the resulting two numbers and the 6 digits as $x_1, \ldots, x_8$. Compute

$$c = \big(9x_1 + 8x_2 + 7x_3 + 6x_4 + \cdots + 2x_8\big) \bmod 11.$$

The check digit is $x_9 = 11 - c$; if $x_9 = 11$, the check digit is 0; if $x_9 = 10$, the check digit is A.

8.9.1 Single Error Detection

Note that for a valid HKID, we have

$$9x_1 + 8x_2 + 7x_3 + \cdots + 2x_8 + x_9 \equiv 0 \pmod{11}.$$

Suppose x_3 is mistyped as $x_3' \neq x_3$. Suppose the checksum is still correct, i.e.,

$$9x_1 + 8x_2 + 7x_3' + \cdots + 2x_8 + x_9 \equiv 0 \pmod{11}.$$

Subtracting one congruence from the other:

$$7(x_3 - x_3') \equiv 0 \pmod{11}.$$

Since $\gcd(7, 11) = 1$, 7 has an inverse modulo 11. Multiply both sides by 7^{-1}:

$$x_3 - x_3' \equiv 0 \pmod{11}.$$

This contradicts the assumption that $x_3' \neq x_3$ and they are both in $\{0, \ldots, 9\}$ (two distinct digits in $\{0, \ldots, 9\}$ differ by at most $10 < 11$, so they cannot be congruent mod 11). Hence every single-digit error is detected.

Note. If the first or second letter is wrong, the error may not be detected! (The letters take values in $\{10, \ldots, 36\}$, and two different letter codes can be congruent mod 11, e.g., $\mathrm{A} = 10 \equiv 21 = \mathrm{L} \pmod{11}$.)

8.9.2 Transposition Error Detection

Suppose x_3 and x_5 are swapped, and $x_3 \neq x_5$. Suppose the checksum is still correct, i.e.,

$$9x_1 + 8x_2 + 7x_5 + 6x_4 + 5x_3 + \cdots + 2x_8 + x_9 \equiv 0 \pmod{11}.$$

Subtracting the valid congruence from this one:

$$7(x_5 - x_3) + 5(x_3 - x_5) \equiv 0 \pmod{11} \quad \Longrightarrow \quad 2(x_5 - x_3) \equiv 0 \pmod{11}.$$

Since $\gcd(2, 11) = 1$, 2 has an inverse modulo 11. Multiply both sides by 2^{-1}:

$$x_5 - x_3 \equiv 0 \pmod{11}.$$

This contradicts the assumption $x_3 \neq x_5$ with both in $\{0, \ldots, 9\}$. The same argument works for swapping any two positions i, j: the difference of weights $(10-i)-(10-j) = j-i$ is a nonzero element of $\mathbb{Z}_{11}$ (since $1 \leq |i - j| \leq 8 < 11$), hence invertible, forcing $x_i \equiv x_j \pmod{11}$, a contradiction. So every transposition of two digits is detected.

8.10 The Chinese Remainder Theorem

Sun-Tsu's Problem. There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?

This is the system of linear congruences

$$x \equiv 2 \pmod{3}, \qquad x \equiv 3 \pmod{5}, \qquad x \equiv 2 \pmod{7}.$$

The Chinese Remainder Theorem

Theorem. Let $m_1, m_2, \ldots, m_n$ be pairwise relatively prime positive integers greater than one, and $a_1, a_2, \ldots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \quad x \equiv a_2 \pmod{m_2}, \quad \ldots, \quad x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$.

Proof (existence, by construction). Let

$$M_k = \frac{m}{m_k}, \qquad k = 1, 2, \ldots, n,$$

i.e., M_k is the product of all the moduli except m_k. Since the moduli are pairwise relatively prime, $\gcd(m_k, M_k) = 1$, so M_k has an inverse y_k modulo m_k:

$$M_k y_k \equiv 1 \pmod{m_k}.$$

We claim that this is a solution:

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_n M_n y_n.$$

Check that $x \equiv a_k \pmod{m_k}$ for each k:

- If $j \neq k$, then $m_k \mid M_j$ (since M_j contains m_k as a factor), so $M_j \equiv 0 \pmod{m_k}$, hence $a_j M_j y_j \equiv 0 \pmod{m_k}$.
- For $j = k$, $M_k y_k \equiv 1 \pmod{m_k}$ implies $a_k M_k y_k \equiv a_k \pmod{m_k}$.

Summing up, $x \equiv a_k \pmod{m_k}$ for every k. (The uniqueness proof is left as an exercise.) ■

Example (Sun-Tsu's problem solved). Consider the 3 congruences from Sun-Tsu's problem:

$$x \equiv 2 \pmod{3}, \qquad x \equiv 3 \pmod{5}, \qquad x \equiv 2 \pmod{7}.$$

Let

$$m = 3 \cdot 5 \cdot 7 = 105, \qquad M_1 = m/3 = 35, \qquad M_2 = m/5 = 21, \qquad M_3 = m/7 = 15.$$

We see that

- 2 is an inverse of $M_1 = 35$ modulo 3: $35 \cdot 2 = 70 \equiv 1 \cdot 2 = 2 \cdot 35 \equiv 1 \pmod{3}$ since $35 \equiv 2 \pmod{3}$ and $2 \cdot 2 = 4 \equiv 1 \pmod{3}$;
- 1 is an inverse of $M_2 = 21$ modulo 5: $21 \equiv 1 \pmod{5}$, so $21 \cdot 1 \equiv 1 \pmod{5}$;
- 1 is an inverse of $M_3 = 15$ modulo 7: $15 \equiv 1 \pmod{7}$, so $15 \cdot 1 \equiv 1 \pmod{7}$.

Hence

$$\begin{aligned} x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\ &= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \\ &= 140 + 63 + 30 = 233 \equiv 23 \pmod{105}. \end{aligned}$$

So the smallest positive answer is $x = 23$. Check: $23 = 3 \cdot 7 + 2$, $23 = 5 \cdot 4 + 3$, $23 = 7 \cdot 3 + 2$. Indeed $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, $23 \equiv 2 \pmod{7}$.

9 Cryptography

Cryptography is the study of methods for sending and receiving secret messages through insecure channels. We cover: secret key cryptography, key exchange, and public key cryptography with RSA.

9.1 Secret Key Cryptography

In secret key cryptography, the sender (Bob) and the receiver (Alice) first agree on a common secret key in advance. The plaintext is encrypted with the secret key into ciphertext, which travels over the insecure channel; the receiver decrypts it with the same key.

9.1.1 Caesar Cipher (Shift Cipher)

Encryption. The secret key k is a number from $\mathbb{Z}_{26}$.

1. Replace each letter by an integer from $\mathbb{Z}_{26}$ (A= 0, B= 1, ..., Z= 25).
2. The encryption function is $f(p) = (p + k) \bmod 26$. Replace each integer p by $f(p)$.
3. Replace each integer by the corresponding letter.

Decryption. Just replace $f(p)$ by $f^{-1}(p) = (p - k) \bmod 26$ in the procedure above.

Example. Encrypt the message "MEET YOU IN THE PARK" using $k = 3$.

Solution. Replace letters by numbers:

12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10.

Replace each of these numbers p by $f(p) = (p + 3) \bmod 26$:

15 7 7 22 1 17 23 11 16 22 10 7 18 3 20 13.

Translating the numbers back to letters gives the ciphertext:

"PHHW BRX LQ WKH SDUN."

9.1.2 Affine Ciphers

The shift cipher is easy to break: just try all 26 possible keys! Affine ciphers make it (a little bit) safer by using both multiplications and additions:

$$f(p) = (ap + b) \bmod 26,$$

where the pair (a, b) is the secret key. Now there are $26^2 = 676$ possible secret keys.

However, there is a problem: suppose $a = 13$, $b = 1$. Then

$$f(1) = f(3) = f(5) = f(7) = \cdots = 14 \pmod{26},$$

since $13 \cdot 1 + 1 = 14$ and $13 \cdot 3 + 1 = 40 \equiv 14 \pmod{26}$ (in general $f(p+2) - f(p) = 26 \equiv 0 \pmod{26}$). If we receive a 14, which number does it decrypt to? The function is not injective, so decryption is ambiguous.

How to fix it. Choose a such that $\gcd(a, 26) = 1$, e.g., $a = 7$. Then a has an inverse modulo 26, so the congruence

$$ax + b \equiv y \pmod{26}$$

has a unique solution $x \equiv a^{-1}(y - b) \pmod{26}$ for every y, and decryption is well-defined.

9.1.3 Block Ciphers

Each character is a number between 0 and 255 (a byte = 8 bits). Partition the message into blocks of k characters and treat each block as a big number of $8k$ bits. Use arithmetic modulo 2^{8k}.

Example. Choose $k = 10$. Then each block is 80 bits, and:

Encryption: $f(x) = (ax + b) \bmod 2^{80}$, Decryption: $f^{-1}(y) = a^{-1}(y - b) \bmod 2^{80}$.

Now there are $\dfrac{2^{80}}{2} \times 2^{80} = 2^{159}$ different keys (the factor $1/2$ comes from requiring a odd so that $\gcd(a, 2^{80}) = 1$). There are libraries for arbitrary-precision arithmetic to carry out these computations. But affine ciphers are subject to known-plaintext attacks! (Given a few known plaintext–ciphertext block pairs, the key (a, b) can be solved from the resulting linear congruences.)

9.1.4 Advanced Encryption Standard (AES)

AES is the modern secret-key standard, used in Transport Layer Security (TLS, previously known as Secure Sockets Layer, SSL), providing security for https, email, etc. It is a block cipher with block size 128 bits and key lengths 128, 192, or 256 bits. It applies complicated operations (substitutions, permutations, mixing over finite fields, over multiple rounds) that make it very difficult to break.

9.2 Key Exchange

Problems with secret-key cryptography. How to send the secret key in the first place? In World War II keys had to be transmitted in physical form; even in modern days, electronic keys still have to be delivered physically for maximum security. This is the key exchange problem.

The Key Exchange Puzzle (lockbox riddle). Alice wants to send a valuable item to Bob, but the postman cannot be trusted. Alice can put an (unbreakable) lock on the box, but Bob cannot open it without the key. Solution:

1. Alice puts her lock on the box, and sends it to Bob.
2. Bob, after receiving the box, puts his lock on the box as well, and returns it to Alice.
3. Alice, after receiving the box, takes off her lock, and sends it back to Bob.
4. Bob takes off his lock and opens the box.

"Locks" in cryptography are one-way functions: easy to compute $f(x)$ from x, but given y, it is hard to find an x such that $f(x) = y$.

9.2.1 Modular Exponentiation

Lemma. For any $a \in \mathbb{Z}_n$ and any nonnegative integers i, j:

- $a^i \bmod n = \big((a \bmod n)(a \bmod n) \cdots (a \bmod n)\big) \bmod n$ (i factors);
- $a^{i+j} \bmod n = \big((a^i \bmod n)(a^j \bmod n)\big) \bmod n$;
- $a^{ij} \bmod n = (a^i \bmod n)^j \bmod n$.

Proof. Directly from the multiplication theorem of L07: $(xy) \bmod n = \big((x \bmod n)(y \bmod n)\big) \bmod n$, applied repeatedly. ■

Examples. Working modulo 7:

$$3^4 \equiv 81 \equiv 4 \pmod 7 \qquad (\text{since } 81 = 11 \cdot 7 + 4),$$

$$3^{2+4} \equiv 3^2 \cdot 3^4 \equiv 2 \cdot 4 \equiv 1 \pmod 7,$$

$$3^{4\cdot 2} \equiv (3^4)^2 \equiv 4^2 \equiv 2 \pmod 7.$$

9.2.2 Fast Modular Exponentiation: Repeated Squaring

How to compute $a^n \bmod m$ efficiently for large n? Use the repeated squaring method:

- Compute $a^2 \bmod m$, then $a^{2^2} \bmod m = a^4 \bmod m = (a^2 \bmod m)^2 \bmod m$, then $a^{2^3} \bmod m = a^8 \bmod m = (a^4 \bmod m)^2 \bmod m$, and so on. Each new power of two is obtained by squaring the previous one.
- Write n in binary: $n = (b_k \dots b_1 b_0)_2$. Then

  $$a^n \equiv a^{b_0 \cdot 1} \cdot a^{b_1 \cdot 2} \cdot a^{b_2 \cdot 2^2} \cdots \pmod m,$$

  i.e., multiply together exactly those powers a^{2^i} for which the bit $b_i = 1$.

Example. Let $n = 50 = (110010)_2$, i.e., $50 = 2 + 16 + 32 = 2^1 + 2^4 + 2^5$. Then

$$a^{50} \equiv a^{2^1} \cdot a^{2^4} \cdot a^{2^5} \pmod m,$$

so only 5 squarings plus 2 multiplications are needed, instead of 49 multiplications.

9.2.3 A Hard Problem: Discrete Logarithm

The discrete logarithm problem is the inverse of modular exponentiation: given a prime p (potentially very large) and $r, a \in \mathbb{Z}_p$, find x such that

$$a^x \equiv r \pmod p.$$

Modular exponentiation and discrete logarithm are believed to form a one-way function pair: exponentiation is easy (repeated squaring), while discrete logarithm is hard. Indeed, if you could solve discrete logarithm efficiently, you could break current crypto systems. In 2015 it was reported that for 512-bit primes, the problem can be solved with a few thousand CPUs in a week; the estimated cost to break 1024-bit primes is about US$100 million.

9.2.4 Diffie–Hellman Key Exchange

Fix a prime p and $a \in \mathbb{Z}_p$ (e.g., hardcoded in the TLS library). The protocol:

1. Alice chooses a secret integer k_1 and Bob chooses a secret integer k_2.
2. Alice sends $a^{k_1} \bmod p$ to Bob. This is secure against eavesdropping: even if this value is known to attackers, they cannot compute k_1 (discrete logarithm!).
3. Bob computes $(a^{k_1})^{k_2} \bmod p$.
4. Bob sends $a^{k_2} \bmod p$ to Alice.
5. Alice computes $(a^{k_2})^{k_1} \bmod p$.

The shared key is

$$(a^{k_1})^{k_2} \bmod p = (a^{k_2})^{k_1} \bmod p = a^{k_1 k_2} \bmod p.$$

Both parties now share the same secret key, which was never transmitted.

Man-in-the-middle attack. If an attacker intercepts all traffic between the two parties, the Diffie–Hellman protocol can be compromised: the attacker communicates with Alice pretending to be Bob, and communicates with Bob pretending to be Alice, establishing a separate shared key with each and relaying (and reading) all messages.

9.3 Public Key Cryptography and RSA

All previous ciphers need a common secret key: encryption and decryption use the same key, which has to be communicated physically in secret, or using the DH protocol (secure against eavesdropping but not against man-in-the-middle). In public key cryptography, encryption and decryption use different keys. Everyone has a public key, shared with everyone else, and a private key, kept secret.

9.3.1 The RSA Cryptosystem

RSA was introduced in 1976 by Rivest, Shamir and Adleman. In fact, Clifford Cocks, working secretly for the UK government, discovered it 3 years earlier; this was made known to the public only in 1997.

Another one-way function: multiplication and factoring. Let $n = pq$, where p and q are large primes (e.g., 1024 bits or longer). Multiplying $p \cdot q$ is easy; the factoring problem — given n, find p and q — is believed to be hard. On the other hand, it is known how to find random large primes efficiently.

RSA keys.

- **Public key:** n and e, such that e is relatively prime to $(p-1)(q-1)$. In practice: pick e randomly and check.
- **Private key:** p and q (actually, only d, defined below, is needed).

Everyone uses a different set of keys.

RSA Encryption. Let $x < n$ be the message to be encrypted. Alice encrypts it as

$$C = x^e \bmod n,$$

where (n, e) is Bob's public key, and sends C to Bob. C may be eavesdropped. Security: exponentiation can be computed efficiently (time proportional to the length of the key, by repeated squaring), but computing x from C, n, e is believed to be difficult — this is known as the RSA problem.

RSA Decryption. Bob receives C and decrypts using his private key (p, q): find d, the inverse of e modulo $(p-1)(q-1)$, i.e.,

$$de \equiv 1 \pmod{(p-1)(q-1)},$$

and compute

$$C^d \bmod n.$$

We will show later that $C^d \equiv (x^e)^d \equiv x^{de} \equiv x \pmod{n}$. Security: it is hard to find d without knowing (p, q), since computing $(p-1)(q-1)$ requires factoring n.

Example (full RSA computation). Let

$$p = 43, \qquad q = 59, \qquad n = 43 \cdot 59 = 2537, \qquad e = 13, \qquad d = 937.$$

(Check: $(p-1)(q-1) = 42 \cdot 58 = 2436$, and $13 \cdot 937 = 12181 = 5 \cdot 2436 + 1 \equiv 1 \pmod{2436}$.) Translate the message STOP into numerical equivalents (S= 18, T= 19, O= 14, P= 15, two digits per letter) and group into blocks of 4 digits so that $x < n$: 1819 and 1415.

Encrypt: compute $1819^{13} \bmod 2537$ by repeated squaring. Since $13 = 2^0 + 2^2 + 2^3$:

$$\begin{aligned} 1819^{2^0} &\equiv 1819 \pmod{2537}, \\ 1819^{2^1} &\equiv 1819^2 \equiv 513 \pmod{2537}, \\ 1819^{2^2} &\equiv 513^2 \equiv 1858 \pmod{2537}, \\ 1819^{2^3} &\equiv 1858^2 \equiv 1844 \pmod{2537}. \end{aligned}$$

Therefore

$$1819^{13} \equiv 1819^{2^0+2^2+2^3} \equiv 1819 \cdot 1858 \cdot 1844 \equiv 2081 \pmod{2537}.$$

Similarly, $1415^{13} \equiv 2182 \pmod{2537}$. The ciphertext is 2081 2182.

Decrypt: $2081^{937} \equiv 1819 \pmod{2537}$ and $2182^{937} \equiv 1415 \pmod{2537}$, recovering the message blocks 1819 1415 = STOP.

9.4 Fermat's Little Theorem

Lemma. Let p be a prime number. For any nonzero $a \in \mathbb{Z}_p$, the sequence

$$1 \cdot_p a, \quad 2 \cdot_p a, \quad \ldots, \quad (p-1) \cdot_p a$$

is a permutation of the set $\{1, 2, \ldots, p-1\}$.

Proof sketch. (i) None of them is 0: if $x \cdot_p a = 0$ then $p \mid xa$, but p is prime and both $x, a \in \{1, \ldots, p-1\}$, impossible. (ii) They are all distinct: if $x \cdot a \equiv y \cdot a \pmod{p}$, then multiplying both sides by a^{-1} (which exists since p is prime and $a \neq 0$), $x \cdot a \cdot a^{-1} \equiv y \cdot a \cdot a^{-1}$, so $x \equiv y \pmod{p}$. A list of $p-1$ distinct nonzero elements of $\mathbb{Z}_p$ must be exactly $\{1, 2, \ldots, p-1\}$. ■

Fermat's Little Theorem

Theorem. Let p be a prime number. Then for any nonzero $a \in \mathbb{Z}_p$,

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. From the lemma, the two lists $\{1, 2, \ldots, p-1\}$ and $\{1 \cdot_p a, 2 \cdot_p a, \ldots, (p-1) \cdot_p a\}$ are the same set, so their products are equal modulo p. Let

$$x \equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \equiv (1 \cdot_p a)(2 \cdot_p a) \cdots ((p-1) \cdot_p a) \pmod{p}.$$

Collecting the $p-1$ factors of a:

$$x \equiv \big(1 \cdot 2 \cdot 3 \cdots (p-1)\big) \cdot a^{p-1} \pmod{p}, \qquad \text{i.e.,} \qquad x \equiv x \cdot a^{p-1} \pmod{p}.$$

Since $x = (p-1)! \bmod p \neq 0$ and p is prime, x^{-1} exists in $\mathbb{Z}_p$; multiplying both sides by x^{-1} gives

$$1 \equiv a^{p-1} \pmod{p}.$$ ■

Example: the power table of $\mathbb{Z}_7$. Each row cycles with period dividing $6 = 7 - 1$, and column a^6 is all 1's, confirming Fermat's Little Theorem:

a	a^0	a^1	a^2	a^3	a^4	a^5	a^6
1	1	1	1	1	1	1	1
2	1	2	4	1	2	4	1
3	1	3	2	6	4	5	1
4	1	4	2	1	4	2	1
5	1	5	4	6	2	3	1
6	1	6	1	6	1	6	1

Corollary. If p is prime and a is an integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$.

Examples. $9^6 \equiv 2^6 \equiv 1 \pmod{7}$ (since $9 \equiv 2 \pmod{7}$). Note $14^6 \equiv 0 \pmod{7}$ — the theorem does not apply when $p \mid a$.

Fermat's Little Theorem is useful in computing the remainders of large powers.

Example. Find 7^{222} mod 11.

Solution. By the theorem, since 11 is prime and $11 \nmid 7$, we know that $7^{10} \equiv 1 \pmod{11}$, and so $(7^{10})^k \equiv 1 \pmod{11}$ for any positive integer k. Write $222 = 22 \cdot 10 + 2$. Therefore

$$7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} \cdot 7^2 \equiv 1^{22} \cdot 49 \equiv 5 \pmod{11}.$$

9.5 RSA Correctness

Proof plan. We want to show $C^d = x^{de} \equiv x \pmod{n}$, where $n = pq$.
Step 1: Show that $x^{de} \equiv x \pmod{p}$ and $x^{de} \equiv x \pmod{q}$.
Step 2: Combine to get $x^{de} \equiv x \pmod{pq}$.

Proof of Step 1. We know d is the inverse of e modulo $(p-1)(q-1)$, so for some integer k,

$$de = 1 + k(p-1)(q-1).$$

It follows that

$$C^d \equiv (x^e)^d \equiv x^{de} \equiv x^{1+k(p-1)(q-1)} \pmod{p}.$$

Case 1: x is not a multiple of p. Applying Fermat's Little Theorem, $x^{p-1} \equiv 1 \pmod{p}$, so

$$x^{k(p-1)(q-1)} = \left(x^{p-1}\right)^{k(q-1)} \equiv 1^{k(q-1)} \equiv 1 \pmod{p},$$

and therefore

$$x^{1+k(p-1)(q-1)} \equiv x \cdot 1 \equiv x \pmod{p}.$$

Case 2: x is a multiple of p. Then $x \equiv 0 \pmod{p}$, and any positive power of x is also $\equiv 0 \pmod{p}$. Thus in this case

$$x^{1+k(p-1)(q-1)} \equiv x \equiv 0 \pmod{p}.$$

The proof for $x^{de} \equiv x \pmod{q}$ is symmetric. ■

Proof of Step 2. We already have

$$x^{de} \equiv x \pmod{p}, \qquad x^{de} \equiv x \pmod{q}.$$

Because $\gcd(p, q) = 1$, by the Chinese Remainder Theorem (treating x^{de} as the unknown and x as given, with moduli $m_1 = p$, $m_2 = q$), the system has a unique solution in $\mathbb{Z}_{pq}$. Since both x^{de} and x satisfy the system,

$$x^{de} \equiv x \pmod{pq}.$$ ■

9.6 RSA in Use

Sending secret keys (e.g., for use in AES). Alice encodes the secret key using Bob's public key; Bob decodes the secret key using his private key. (Public key crypto is slow, so it is used to exchange a key; bulk data is then encrypted with fast secret key crypto like AES.)

Digital signatures (authentication). Alice encodes her message using her private key d and her modulus n: she computes

$$C = x^d \bmod n,$$

and sends (C, x) to Bob. Bob decodes using Alice's public key (n, e):

$$C^e \bmod n = x^{de} \bmod n.$$

He will know the message indeed came from Alice if $C^e \bmod n = x$, since only Alice knows d. (How to ensure both secrecy and authenticity? Encrypt with the receiver's public key and sign with the sender's private key.)

Certificate authorities. How to prevent man-in-the-middle attacks? How to make sure that Alice's public key indeed belongs to Alice? Use a certificate authority (CA): a small number of trusted third parties — Comodo, Symantec, GoDaddy, GlobalSign, ... — that sign certificates binding public keys to identities. How to make sure that a CA's public key indeed belongs to that CA? Those keys are built into Internet browsers. How can you trust your browser and the CAs? Well, you have to .... Note also that RSA is gradually being replaced by new public-key encryption methods, e.g., Elliptic Curve Cryptography.

Bitcoin transactions. When you register a bitcoin wallet, you generate a (public key, private key) pair. The public key is the id of your wallet; you keep your private key in a safe place. All blockchain nodes (miners) maintain the account balance under each public key. To send BTC to Bob, send a transaction message to all nodes, containing:

- your public key, Bob's public key, and the amount;
- the signature of the message, encoded with your private key.

The nodes decode your signature and check that it matches the message, check that the balance under your public key is at least the amount, and then add the transaction to the blockchain.

10 Analysis of Algorithms

10.1 Measuring Running Time

An algorithm is a finite set of precise instructions for performing a computation or for solving a problem.

To measure the running time of an algorithm without executing it on a physical machine, we count the number of elementary operations as a function of the input size n, ignoring lower-order terms and constant coefficients. A constant is any quantity that does not depend on n.

Example (Selection Sort). Consider the selection sort algorithm:

```
(1) for i = 1 to n - 1
(2)   for j = i + 1 to n
(3)     if (A[i] > A[j])
(4)       swap A[i] and A[j]
(5)     endif
(6)   endfor
(7) endfor
```

We analyze how many times line (3) is executed.

- When $i = 1$, j iterates from 2 to n, making $n - 1$ comparisons. After this pass, $A[1]$ contains the smallest element of $A[1..n]$.

- When $i = 2$, j iterates from 3 to n, making $n - 2$ comparisons. After this pass, $A[2]$ contains the second smallest element.
- ...
- When $i = n-1$, j iterates from n to n, making 1 comparison. After this pass, $A[n-1]$ contains the $(n-1)$-th smallest element.

Therefore the total number of comparisons is

$$(n-1) + (n-2) + \cdots + 1 = \frac{n(n-1)}{2}.$$

Hence the running time of selection sort is $\Theta(n^2)$.

10.2 Growth of Functions and Big-Theta

For large input sizes, we compare algorithms by the growth rate of their running-time functions.

Example. Suppose Algorithm 1 has running time $T_1(n) = 3n^3 + 6n^2 - 4n + 17 = \Theta(n^3)$ and Algorithm 2 has $T_2(n) = 7n^2 - 8n + 20 = \Theta(n^2)$. For large n, Algorithm 2 is clearly better.

Big-Theta Notation

Let f and g be functions from the set of positive real numbers to the set of positive real numbers. We say that $f(x)$ is $\Theta(g)$ if there are positive constants C_1, C_2, and k such that

$$C_1 g(x) \leq f(x) \leq C_2 g(x)$$

whenever $x > k$. This is read as "$f(x)$ is big-Theta of $g(x)$" or "$f(x)$ is asymptotically the same as $g(x)$." The constants C_1, C_2, k are called witnesses.

Usually we write $f(n) = \Theta(g(n))$, although the more mathematically correct notation is $f(n) \in \Theta(g(n))$.

Example (Using the Definition). Show that $T_1(n) = 3n^3 + 6n^2 - 4n + 17 = \Theta(n^3)$.

Solution. Choose $C_1 = 2$ and $C_2 = 4$. We want a k such that for all $n > k$,

$$2n^3 \leq 3n^3 + 6n^2 - 4n + 17 \leq 4n^3.$$

Subtracting $3n^3$ from all parts gives

$$-n^3 \leq 6n^2 - 4n + 17 \leq n^3.$$

Since $6n^2 - 4n + 17$ is dominated by n^3 for sufficiently large n, such a k must exist; there is no need to find it explicitly.

10.3 Comparing Algorithms via Limits

When n is large, we are interested in

$$\lim_{n \to \infty} \frac{T_1(n)}{T_2(n)}.$$

Three cases arise:

- If the limit is 0, Algorithm 1 is better.
- If the limit is ∞, Algorithm 2 is better.
- If the limit is a constant C with $0 < C < \infty$, or if the ratio oscillates, then Θ-notation cannot distinguish the algorithms and a more careful analysis is needed.

If $T_1(n) = \Theta(g_1(n))$ and $T_2(n) = \Theta(g_2(n))$, it is sufficient to consider $\lim_{n\to\infty} g_1(n)/g_2(n)$.

10.4 Important Examples of Growth Rates

Example (Logarithm bases). $\log_{10} n = \dfrac{\log_2 n}{\log_2 10} = \Theta(\log_2 n) = \Theta(\log n)$. The base of the logarithm does not matter inside Θ.

Example (Constants). $999999999999 = \Theta(1)$.

Example (Exponentials). 2^{10n} is not $\Theta(2^n)$, and 3^n is not $\Theta(2^n)$.

Example (Sum of squares). Show that $\sum_{i=1}^{n} i^2 = \Theta(n^3)$.

Solution. For the upper bound,

$$\sum_{i=1}^{n} i^2 \le n^2 \cdot n = n^3.$$

For the lower bound, consider the terms from $i = \lceil n/2 \rceil$ to n:

$$\sum_{i=1}^{n} i^2 \ge \left(\frac{n}{2}\right)^2 \cdot \frac{n}{2} = \frac{1}{8}n^3.$$

Thus $\dfrac{1}{8}n^3 \le \sum_{i=1}^{n} i^2 \le n^3$, so $\sum_{i=1}^{n} i^2 = \Theta(n^3)$.

Example (Geometric series). Let c be a constant. Then

$$\sum_{i=0}^{n} c^i = \frac{c^{n+1}-1}{c-1} = \begin{cases} \Theta(c^n), & c > 1, \\ \Theta(n), & c = 1, \\ \Theta(1), & c < 1. \end{cases}$$

Derivation of the Geometric Series Formula. Let $S(n) = 1 + c + c^2 + \cdots + c^n$. Then

$$S(n+1) = S(n) \cdot c + 1$$

and also

$$S(n+1) = S(n) + c^{n+1}.$$

Equating the two expressions,

$$S(n) \cdot c + 1 = S(n) + c^{n+1} \implies (c-1)S(n) = c^{n+1} - 1.$$

If $c \neq 1$, then

$$S(n) = \frac{c^{n+1}-1}{c-1}.$$

Example (Factorial logarithm). Show that $\log(n!) = \Theta(n \log n)$.

Solution. For the upper bound,

$$\log(n!) = \log n + \log(n-1) + \cdots + \log 1 \le n \log n.$$

For the lower bound, keep only the first $n/2$ terms:

$$\log(n!) \geq \log n + \log(n-1) + \cdots + \log\left(\frac{n}{2}\right) \geq \frac{n}{2}\log\frac{n}{2}.$$

Since $\frac{n}{2}\log\frac{n}{2} = \frac{n}{2}(\log n - 1) = \Theta(n\log n)$, we conclude $\log(n!) = \Theta(n\log n)$.

Example (Harmonic series). $\displaystyle\sum_{i=1}^{n}\frac{1}{i} = \Theta(\log n)$.

Example (Dominance). For constants $c_1 > 1$, $c_2 > 0$, $c_3 > 0$, c_1^n dominates n^{c_2}, which dominates $\log^{c_3} n$.

- $n^{0.1} + \log^{10} n = \Theta(n^{0.1})$.
- $1.1^n + n^{100} = \Theta(1.1^n)$.

10.5 Limitations of Big-Theta

Some functions cannot be described by Θ.

Example. The number of 1's in the binary representation of n oscillates between 1 and $\log n$, so it is not Θ of any simple function.

Some functions are hard to describe by Θ. For example, Stirling's formula states

$$n! = \Theta\left(\left(\frac{n}{e}\right)^n\sqrt{n}\right),$$

but this is difficult to derive.

10.6 Big-Oh and Big-Omega

Big-Oh Notation

Let f and g be functions from the set of positive real numbers to the set of positive real numbers. We say that $f(x)$ is $O(g)$ if there are positive constants C_2 and k such that

$$f(x) \leq C_2 g(x)$$

whenever $x > k$. This is read as "$f(x)$ is big-Oh of $g(x)$" or "$f(x)$ is asymptotically dominated by $g(x)$."

Big-Omega Notation

Let f and g be functions from the set of positive real numbers to the set of positive real numbers. We say that $f(x)$ is $\Omega(g)$ if there are positive constants C_1 and k such that

$$C_1 g(x) \leq f(x)$$

whenever $x > k$. This is read as "$f(x)$ is big-Omega of $g(x)$" or "$f(x)$ asymptotically dominates $g(x)$."

Relationship Between Θ, O, and Ω

$$f(x) = \Theta(g(x)) \iff f(x) = O(g(x)) \text{ and } f(x) = \Omega(g(x)).$$

The following table summarizes the meaning of each notation:

Notation	Meaning for large n	Phrase
$f(n) = O(g(n))$	Upper bounded by $c \cdot g(n)$	"$f(n)$ is dominated by $g(n)$"
$f(n) = \Omega(g(n))$	Lower bounded by $c \cdot g(n)$	"$f(n)$ dominates $g(n)$"
$f(n) = \Theta(g(n))$	Lower bounded by $c_1 g(n)$ and upper bounded by $c_2 g(n)$	"$f(n)$ grows asymptotically with $g(n)$"

Example. Let $f(n) = 32n^2 + 17n - 32$.

- $f(n)$ is $O(n^2)$, $O(n^3)$, $\Omega(n^2)$, $\Omega(n)$, and $\Theta(n^2)$.
- $f(n)$ is not $O(n)$, $\Omega(n^3)$, $\Theta(n)$, or $\Theta(n^3)$.

Example (Number of 1's in binary representation). Let $f(n)$ be the number of 1's in the binary representation of n.

- $f(n) = O(\log n)$, since there are at most $\log n + 1$ bits.
- $f(n) = \Omega(1)$, since there is at least one 1 for $n \geq 1$.
- $f(n)$ is not Θ of any simple function because it oscillates.

Example (Factorial bounds).

$$n! \leq n^n = O(n^n), \qquad n! \geq \left(\frac{n}{2}\right)^{n/2} = \Omega\left(\left(\frac{n}{2}\right)^{n/2}\right).$$

10.7 A Warning: Big-O Inside Induction

Do not use asymptotic notation inside induction proofs. Here is a false "proof" that $\sum_{i=1}^{n} i = O(n)$:

Incorrect Proof. Let $P(n)$ be the statement $\sum_{i=1}^{n} i = O(n)$.

Basis: $P(1)$ is $1 = O(1)$, trivially true.

Inductive step: Assume $P(k)$ is true, i.e., $\sum_{i=1}^{k} i = O(k)$. We show $P(k+1)$:

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^{k} i + (k+1) = O(k) + (k+1) = O(k+1).$$

This is invalid because $O(k)$ and $O(k+1)$ hide different constants; the induction does not establish a single uniform constant for all n.

Lesson: Never use asymptotic notation inside induction proofs.

10.8 Insertion Sort

The insertion sort algorithm is:

```
Insertion-Sort(A):
  for j = 2 to n do
    key = A[j]
    i = j - 1
```

```
    while i >= 1 and A[i] > key do
      A[i+1] = A[i]
      i = i - 1
    endwhile
    A[i+1] = key
  endfor
```

Example (Trace). Sort the array $(4, 1, 8, 2, 5)$.

- 1st iteration ($key = 1$): $(4, 1, 8, 2, 5) \to (4, 4, 8, 2, 5) \to (1, 4, 8, 2, 5)$.
- 2nd iteration ($key = 8$): $(1, 4, 8, 2, 5)$ unchanged.
- 3rd iteration ($key = 2$): $(1, 4, 8, 2, 5) \to (1, 4, 8, 8, 5) \to (1, 4, 4, 8, 5) \to (1, 2, 4, 8, 5)$.
- 4th iteration ($key = 5$): $(1, 2, 4, 8, 5) \to (1, 2, 4, 8, 8) \to (1, 2, 4, 5, 8)$.

10.9 Best-Case, Worst-Case, and Average-Case Analysis

The running time of an algorithm depends not only on n but also on the actual input.

Example (Insertion Sort).

- If the input array is already sorted, the running time is $\Theta(n)$.
- If the input array is inversely sorted, the running time is $\Theta(n^2)$.

For algorithm analysis we usually consider:

- **Worst-case analysis** (the default): guarantees an upper bound for all inputs.
- Best-case analysis: rarely useful.
- Average-case analysis: requires a probability distribution over inputs.

To show that an algorithm has worst-case running time $\Theta(n^2)$, we must:

1. Show it is $O(n^2)$ (upper bound for all inputs).
2. Show it is $\Omega(n^2)$ (find at least one input that forces $\Omega(n^2)$ time).

For insertion sort, the worst case is $O(n^2)$ because the inner while loop executes at most $j - 1 \leq n - 1$ times for each j, giving $\sum_{j=2}^{n}(j-1) = O(n^2)$. It is $\Omega(n^2)$ because the inversely sorted array forces exactly $j - 1$ shifts at iteration j, totaling $\sum_{j=2}^{n}(j-1) = \frac{n(n-1)}{2} = \Omega(n^2)$. Hence the worst-case running time is $\Theta(n^2)$.

10.10 Searching Algorithms

10.10.1 Linear Search

Problem. Given an array A of unordered elements and a target x, find x or report that x does not exist in A.

```
procedure linear search(x: integer, a1, a2, ..., an: distinct integers)
  i := 1
  while (i <= n and x != a_i)
    i := i + 1
  if i <= n then location := i
  else location := 0
  return location
```

Analysis of Linear Search.

- Running time is $O(n)$: the loop has at most n iterations.
- Worst-case running time is $\Omega(n)$: when x does not exist in A, the loop executes exactly n iterations.
- Therefore the worst-case running time is $\Theta(n)$.

10.10.2 Binary Search

Problem. Given an array A of ordered elements and a target x, find x or report that x does not exist in A.

```
procedure binary search(x: integer, a1, a2, ..., an: increasing integers)
  i := 1  {left endpoint}
  j := n  {right endpoint}
  while i <= j
    m := floor((i + j) / 2)
    if x = a_m then return m
    if x > a_m then i := m + 1
    else j := m - 1
  return -1  {x does not exist}
```

Analysis of Binary Search. Assume first that $n = 2^k$.

- The length of the search range $j - i + 1$ is halved in each iteration.
- The algorithm terminates when $i > j$, i.e., when the range length becomes ≤ 1.
- Starting from length $n = 2^k$, after k halvings the length becomes 1. Hence the loop executes at most $k = \log_2 n$ times.
- Running time is $O(\log n)$.
- When x does not exist in A, the loop executes exactly $\log n$ iterations, giving $\Omega(\log n)$.
- Therefore the worst-case running time is $\Theta(\log n)$.

If n is not a power of 2, find k such that $2^k < n < 2^{k+1}$. The running time is between $\Theta(k)$ and $\Theta(k+1)$, which is still $\Theta(\log n)$.

10.11 Recursive Algorithms

Recursively defined functions naturally give recursive algorithms.

Example (Factorial).

$$0! = 1, \qquad n! = n \cdot (n-1)! \text{ for } n \geq 1.$$

Recursive version:

```
Factorial(n):
  if n = 0 then return 1
  else return Factorial(n - 1) * n
```

Iterative version (tail recursion eliminated):

```
Factorial(n):
  s = 1
  for i = 1 to n
    s = s * i
  return s
```

10.11.1 Euclid's GCD Algorithm

Recursive Definition. $\gcd(a, b) = \gcd(b, a \bmod b)$.

Recursive version:

```
gcd(a, b):
  if b = 0 then return a
  else return gcd(b, a mod b)
```

Iterative version:

```
gcd(a, b):
  x = a
  y = b
  while y != 0
    r = x mod y
    x = y
    y = r
  return x
```

10.11.2 Fibonacci Numbers

The Fibonacci numbers are defined by

$$f_1 = 1, \quad f_2 = 1, \quad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 3.$$

Naive recursive algorithm:

```
Fib(n):
  if n = 1 or n = 2 then return 1
  else return Fib(n - 1) + Fib(n - 2)
```

The recursion tree has size at least $f_n \geq \alpha^{n-2}$ for some constant $\alpha > 1$, so the naive algorithm is exponential.

Efficient iterative algorithm (dynamic programming):

```
Fib(n):
  allocate an array A of size n
  A[1] = 1
  A[2] = 1
  for i = 3 to n
    A[i] = A[i-1] + A[i-2]
  return A[n]
```

This runs in $\Theta(n)$ time.

10.11.3 The Tower of Hanoi

Problem. There are three pegs. The first peg has n disks of different sizes, stacked with larger disks below smaller ones. Move all disks to the second peg, moving one disk at a time and never placing a larger disk on top of a smaller one.

Recursive Solution.

1. Move the top $n - 1$ disks from peg 1 to peg 3 (using peg 2 as auxiliary).
2. Move the largest disk from peg 1 to peg 2.
3. Move the $n - 1$ disks from peg 3 to peg 2 (using peg 1 as auxiliary).

Analysis. Let $f(n)$ be the number of moves needed for n disks. The recursive algorithm gives

$$f(1) = 1, \qquad f(n) = 2f(n-1) + 1.$$

Solving this recurrence:

$$f(n) = 2^n - 1.$$

For $n = 64$, $f(64) > 10^{19}$. At one move per second, this would take more than 500 billion years.

Optimality. Let $g(n)$ be the minimum number of moves of any algorithm. Any algorithm must move the largest disk at least once; before that move, the top $n-1$ disks must be moved to another peg, requiring at least $g(n-1)$ moves. After moving the largest disk, the $n-1$ disks must be moved onto it, requiring at least $g(n-1)$ more moves. Hence

$$g(1) \geq 1, \qquad g(n) \geq 2g(n-1) + 1.$$

Therefore $g(n) \geq f(n)$, proving the recursive algorithm is optimal.

10.12 Summary of Recursive Algorithms

- If a recursive algorithm makes one recursive call at the end (tail recursion), it can be rewritten as an iterative algorithm. Good compilers can do this automatically.
- If the algorithm makes more than one recursive call but each call just returns a number, it can often be implemented using dynamic programming.
- The running time of a recursive algorithm is analyzed by solving recurrences.

11 Basics of Counting

Counting arises in computer science when determining whether there are enough IP addresses, analyzing the time or space complexity of algorithms, and computing probabilities of events.

The two basic counting principles are the product rule (sequential rule) and the sum rule (disjunctive rule).

11.1 The Product Rule

Product Rule

Suppose a procedure can be broken down into $m > 1$ tasks $T_1, T_2, \ldots, T_m$, carried out in sequence. If each task T_i can be done in n_i ways regardless of how the previous tasks were done, then there are

$$n_1 n_2 \cdots n_m$$

ways to carry out the procedure.

Example (Chairs). Ten chairs are numbered from 1 to 10. How many ways are there to assign different chairs to Joseph and Mary?

Solution. There are 10 ways to assign a chair to Joseph and 9 remaining ways to assign a chair to Mary. By the product rule, there are $10 \times 9 = 90$ ways.

Example (Functions between small sets). Let $U = \{1, 2, 3\}$ and $V = \{a, b\}$. How many different functions $f : U \to V$ can be defined?

Solution. Each of the three elements of U can be mapped to either a or b independently. By the product rule, there are $2 \times 2 \times 2 = 2^3 = 8$ possible functions. They are:

	$f(1)$	$f(2)$	$f(3)$
f_1	a	a	a
f_2	a	a	b
f_3	a	b	a
f_4	a	b	b
f_5	b	a	a
f_6	b	a	b
f_7	b	b	a
f_8	b	b	b

Example (Number of functions). Let U and V be finite sets with $|U| = m > 0$ and $|V| = n > 0$. How many different functions $f : U \to V$ can be defined?

Solution. Each of the m elements of the domain U can be mapped to any of the n elements of the codomain V. By the product rule, there are n^m different functions.

Example (Number of injective functions). Let U and V be finite sets with $|U| = m > 0$ and $|V| = n > 0$. How many different one-to-one (injective) functions $f : U \to V$ can be defined?

Solution.

- If $m > n$, no injective function exists (by the pigeonhole principle).
- If $m \leq n$: the first element of U can be mapped to any of the n elements of V; the second element to any of the remaining $n - 1$ elements; and so on. The k-th element of U can be mapped to $n - k + 1$ possible elements. By the product rule, the number of injective functions is

$$n(n-1)(n-2)\cdots(n-m+1) = \frac{n!}{(n-m)!}.$$

Example (Number of subsets). Let S be a finite set with $|S| = n > 0$. How many different subsets of S are there?

Solution. There is a one-to-one correspondence between subsets of S and bit strings of length n. Index the elements of S from 1 to n. A subset corresponds to the bit string with a 1 in the i-th position if the i-th element is in the subset, and 0 otherwise. For example, if $|S| = 6$, the subset $\{2, 3, 6\}$ corresponds to the bit string 011001. By the product rule, there are 2^n bit strings of length n, hence $2^{|S|}$ subsets.

Example (Matrix multiplication). Consider the following algorithm:

```
for i = 1 to m
  for j = 1 to n
    s = 0
    for k = 1 to p
      s = s + A[i,k] * B[k,j]
    C[i,j] = s
  end for
end for
```

How many multiplications are executed?

Solution. The multiplication is inside the k-loop. For fixed i and j, the k-loop executes p multiplications. There are n values of j, so for fixed i the inner two loops execute $n \cdot p$ multiplications. There are m values of i, so the total number of multiplications is $m \cdot n \cdot p$.

11.2 The Sum Rule

Sum Rule

Suppose a procedure can be carried out in one of $m > 1$ methods $T_1, T_2, \ldots, T_m$. If each method T_i can be done in n_i ways and no two methods share any way (the methods are disjoint), then there are

$$\sum_{i=1}^{m} n_i$$

different ways to carry out the procedure.

The sum rule cannot be applied if the tasks (or sets) are not disjoint; in that case we need the inclusion–exclusion principle.

Example (Class representative). There are 15 boys and 18 girls in a class. How many ways are there to choose a representative for the class?

Solution. There are 15 ways to choose a boy representative and 18 ways to choose a girl representative. Since no one is both a boy and a girl, the two cases are disjoint. By the sum rule, there are $15 + 18 = 33$ possible ways.

11.3 Using Both Product and Sum Rules

Example (Variable names). In a programming language, a variable name is a string of one or two alphanumeric characters which are not case-sensitive. An alphanumeric character is one of the 26 English letters or one of the 10 digits. A variable name must begin with a letter and must be different from five reserved two-character strings. How many different variable names are there?

Solution. Split into two disjoint cases by the sum rule.

Case 1: Variable names of length 1. The single character must be a letter, so there are 26 possibilities.

Case 2: Variable names of length 2. The first character must be a letter (26 choices) and the second character can be any alphanumeric character ($26 + 10 = 36$ choices). By the product rule, there are $26 \times 36 = 936$ two-character strings beginning with a letter. Subtracting the 5 reserved strings gives $936 - 5 = 931$ valid names.

By the sum rule, the total number of variable names is

$$26 + 931 = 957.$$

Example (Passwords). In a computer system, each user has a password of six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution. Let P be the total number of possible passwords, and let P_6, P_7, P_8 denote the numbers of valid passwords of length 6, 7, 8 respectively. By the sum rule, $P = P_6 + P_7 + P_8$.

Counting directly the passwords with at least one digit is complicated, because a valid password of length 6 could contain 1, 2, ..., or 6 digits in various positions. Instead, use the complement trick: count all passwords formed from the 36 alphanumeric characters, then subtract those with no digit (formed only from the 26 letters).

$$P_6 = 36^6 - 26^6 = 1{,}867{,}866{,}560,$$
$$P_7 = 36^7 - 26^7 = 70{,}332{,}353{,}920,$$
$$P_8 = 36^8 - 26^8 = 2{,}612{,}282{,}842{,}880.$$

Therefore
$P = P_6 + P_7 + P_8 = 1{,}867{,}866{,}560 + 70{,}332{,}353{,}920 + 2{,}612{,}282{,}842{,}880 = 2{,}684{,}483{,}063{,}360.$

(If this were a number of seconds, it would correspond to roughly 85,000 years.)

11.4 Tree Diagrams

Some counting problems can be solved more easily using tree diagrams.

Example (Bit strings without consecutive 1's). How many bit strings of length four do not have two consecutive 1's?

Solution. Build a tree: each node branches to 0 or 1, but a branch to 1 is allowed only if the parent bit is not 1. Enumerating the leaves of the tree gives the valid strings:

$$0000,\ 0001,\ 0010,\ 0100,\ 0101,\ 1000,\ 1001,\ 1010.$$

There are 8 such bit strings.

Example (Playoff). A playoff between two teams consists of at most five games. The first team that wins three games wins the playoff. In how many different ways can the playoff occur?

Solution. Draw a tree where each node represents the current score. The playoff ends as soon as one team reaches 3 wins. Counting the leaves of the tree:

- 3 games: 3-0 (either team sweeps): 2 ways.
- 4 games: 3-1: the loser wins exactly one of the first 3 games: $\binom{3}{1} = 3$ ways per winning team, giving $2 \times 3 = 6$ ways.
- 5 games: 3-2: the loser wins exactly two of the first 4 games: $\binom{4}{2} = 6$ ways per winning team, giving $2 \times 6 = 12$ ways.

Total: $2 + 6 + 12 = 20$ ways.

11.5 The Pigeonhole Principle

Pigeonhole Principle

If $k + 1$ or more objects (pigeons) are placed into k boxes (pigeonholes), where k is a positive integer, then there is at least one box containing two or more of the objects.

Proof. By contradiction. Suppose each of the k boxes contains at most one object. Then the total number of objects is at most k, contradicting the assumption that there are at least $k + 1$ objects. ■

Corollary

A function f from a set with $k + 1$ or more elements to a set with k elements is not one-to-one.

The challenge in applying the pigeonhole principle is to correctly identify what the objects (pigeons) and the boxes (pigeonholes) are.

Example (Birthdays). Among any group of 367 people, there must be at least two with the same birthday.

Solution. There are at most 366 possible birthdays (including February 29). Treat people as pigeons and birthdays as pigeonholes. Since $367 > 366$, at least two people share a birthday.

Example (English words). In any group of 27 English words, there must be at least two that begin with the same letter.

Solution. There are 26 letters in the English alphabet (pigeonholes) and 27 words (pigeons). By the pigeonhole principle, at least two words begin with the same letter.

Example (Exam scores). How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Solution. There are 101 possible scores $(0, 1, \ldots, 100)$. To guarantee a repetition, we need $101 + 1 = 102$ students.

Example (Points in a rectangle). Twenty-five points are marked inside the area bounded by a rectangle that is 6 cm long and 4 cm wide. Show that there are at least two points that are at most $\sqrt{2}$ cm apart.

Solution. Divide the 6×4 rectangle into 24 squares of side 1 cm (6 columns and 4 rows). Treat the 24 squares as pigeonholes and the 25 points as pigeons. By the pigeonhole principle, at least two points lie in the same 1×1 square. The maximum distance between two points inside a 1×1 square is the diagonal length $\sqrt{1^2 + 1^2} = \sqrt{2}$ cm. Hence two points are at most $\sqrt{2}$ cm apart.

Example (Odd or even product). Let $a_1, a_2, \ldots, a_{1997}$ be an arbitrary arrangement of the numbers $1, 2, \ldots, 1997$. Is the product $(a_1 - 1)(a_2 - 2) \cdots (a_{1997} - 1997)$ odd or even?

Solution. The product is odd only if every factor $(a_i - i)$ is odd, i.e., only if a_i and i have opposite parity for every i. Among $1, 2, \ldots, 1997$ there are 999 odd numbers and 998 even numbers. For all factors to be odd, each of the 999 odd values of i would need a_i to be even; but there are only 998 even numbers available to assign. Treat the 998 even numbers as pigeonholes and the 999 odd indices as pigeons. By the pigeonhole principle, at least one odd index i is assigned an odd a_i, making $(a_i - i)$ even. Therefore the product must be even.

11.6 The Generalized Pigeonhole Principle

Generalized Pigeonhole Principle

If n objects are placed into k boxes, then there is at least one box containing at least $\lceil n/k \rceil$ objects.

Proof. By contradiction. Suppose every box contains at most $\lceil n/k \rceil - 1$ objects. Then the total number of objects is at most

$$k\left(\left\lceil \frac{n}{k} \right\rceil - 1\right).$$

Since $\lceil n/k \rceil < n/k + 1$, we have

$$k\left(\left\lceil \frac{n}{k} \right\rceil - 1\right) < k\left(\frac{n}{k} + 1 - 1\right) = n,$$

which contradicts the fact that there are exactly n objects. ■

Example (Birth months). Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

Example (Grades). What is the minimum number of students required in a class to be sure that at least six will receive the same grade, if there are five possible grades A, B, C, D, and F?

Solution. We need the smallest integer n such that $\lceil n/5 \rceil = 6$. This is $n = 5 \times 5 + 1 = 26$: with 25 students it is possible that each grade is received by exactly 5 students; with 26 students, some grade must be received by at least $\lceil 26/5 \rceil = 6$ students.

Example (Cards). How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen? How many cards must be selected to guarantee that at least three hearts are selected?

Solution.

- Treat the selected cards as pigeons and the four suits as pigeonholes. To guarantee at least three cards of the same suit, we need $\lceil n/4 \rceil \geq 3$, i.e., $n = 2 \times 4 + 1 = 9$ cards.
- To guarantee at least three hearts, note that in the worst case we might draw all 13 clubs, all 13 diamonds, and all 13 spades first. After these $13 \times 3 = 39$ cards, the next 3 cards must be hearts. Hence we need $39 + 3 = 42$ cards.

Example (Phone numbers). Assume that phone numbers are of the form NXX-NXX-XXXX, where the first three digits form the area code, N represents a digit from 2 to 9 inclusive, and X represents any digit. What is the least number of area codes needed to guarantee that the 25 million phones in a state can be assigned distinct 10-digit phone numbers?

Solution. Discarding the area code, by the product rule there are

$$8 \times 10^6$$

different phone numbers of the form NXX-XXXX (8 choices for N, then six digits with 10 choices each: $8 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 8 \times 10^6$).

Treating the 25 million phones as pigeons and the 8×10^6 local numbers as pigeonholes, by the generalized pigeonhole principle,

$$\left\lceil \frac{25 \times 10^6}{8 \times 10^6} \right\rceil = \lceil 3.125 \rceil = 4,$$

so at least one local number would be shared by 4 phones. To give all 25×10^6 phones distinct numbers, we therefore need 4 area codes, so that phones sharing the same local NXX-XXXX number can be distinguished by different area codes.

12 Permutations and Combinations

12.1 Permutations

Example 1 (Selecting students for a photo). In how many ways can we select three students from a group of five students to stand in line for a photo?

Solution. Here the order in which the students are selected matters. There are 5 ways to select the first student to stand at the start of the line. Once this student has been selected, there are 4 ways to select the second student from the remaining 4 students. After the first and second students have been selected, there are 3 ways to select the third student. By the product rule, there are

$$5 \times 4 \times 3 = 60$$

ways.

Example 2 (Arranging all students). In how many ways can we arrange all five of the students above in a line for a photo?

Solution. This is similar to Example 1, except that we now select all 5 students. By the same reasoning, there are

$$5 \times 4 \times 3 \times 2 \times 1 = 120$$

ways.

Definition. A permutation of a set of distinct elements is an ordered arrangement of the elements of the set.

Example. Let $S = \{1, 2, 3\}$. The ordered arrangement $(3, 1, 2)$ is a permutation of S.

Definition. A k-permutation (or k-element permutation) of a set of n distinct elements is an ordered arrangement of $k \leq n$ elements of the set. An n-permutation of a set of n distinct elements is simply called a permutation of the set.

Example. Let $S = \{1, 2, 3\}$. The ordered arrangement $(3, 2)$ is a 2-permutation of S.

Number of k-Permutations

Let n and k be integers with $0 \leq k \leq n$. The number of k-permutations of a set with n distinct elements, denoted by $P(n, k)$, is

$$P(n, k) = n(n-1)\cdots(n-k+1) = \prod_{i=0}^{k-1}(n-i) = \frac{n!}{(n-k)!}.$$

Example 6 (Prize winners). How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution. The order matters because it matters which person wins which prize. The number of ways is the number of 3-permutations of a set of 100 elements:

$$P(100, 3) = 100 \times 99 \times 98 = 970{,}200$$

ways.

Example 7 (Saleswoman's route). Suppose a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wants. How many possible orders can she use?

Solution. Each travel order gives a different trip schedule, so order matters. Since the first city is fixed, she only needs to order the remaining 7 cities:

$$P(7, 7) = 7! = 5040$$

possible orders.

Example 8 (Permutations containing a substring). How many permutations of the letters ABCDEFGH contain the string ABC?

Solution. Since the string ABC must occur together as a block, treat it as a single super-letter S. We then need the number of permutations of the six objects S, D, E, F, G, H:

$$P(6,6) = 6! = 720.$$

Example (Round table). Twelve people sit down at a round table. We consider two seating arrangements equivalent if each person has the same person to the right in both seating charts. How many different seating charts are there?

Solution. In a circular arrangement, rotations of the same configuration are equivalent. Fix the position of the first person (this can be done in only one way, since all positions are equivalent when no one is seated). Then arrange the remaining 11 people in the remaining seats in

$$(12-1)! = 11! = 39{,}916{,}800$$

ways.

12.2 Combinations

Example 9 (Committee). How many different committees of three students can be formed from a group of four students?

Solution. We need the number of subsets containing 3 elements from a set of 4 students. There are 4 such subsets (each subset leaves out exactly one student), so there are 4 ways. Here the order in which the students are chosen does not matter.

Definition. A k-combination of a set of n distinct elements is an unordered selection of $k \leq n$ elements from the set.

Example 10. Let $S = \{1, 2, 3, 4\}$. Then $\{1, 3, 4\}$ is a 3-combination of S.

Number of k-Combinations

Let n and k be integers with $0 \leq k \leq n$. The number of k-combinations of a set with n distinct elements, denoted by $C(n,k)$ or $\binom{n}{k}$, is

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}.$$

Proof. The k-permutations of the set can be obtained by first forming the $C(n,k)$ different k-combinations of the set, and then ordering the elements in each k-combination, which can be done in $P(k,k) = k!$ ways. By the product rule,

$$P(n,k) = C(n,k) \cdot P(k,k).$$

Therefore

$$C(n,k) = \frac{P(n,k)}{P(k,k)} = \frac{n!/(n-k)!}{k!} = \frac{n!}{k!(n-k)!}.$$

■

Example 11 (Poker hands). How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Solution. The order does not matter, since we are only interested in the set of cards selected. Hence the number of 5-card hands is

$$C(52,5) = \frac{52!}{5!\,47!} = 2{,}598{,}960.$$

When 5 cards are selected, 47 cards remain. There is a one-to-one correspondence between sets of 5 cards and sets of 47 cards, so

$$C(52,47) = C(52,5) = \frac{52!}{5!\,47!} = 2{,}598{,}960.$$

Corollary: Symmetry of Binomial Coefficients

Let n and k be integers with $0 \le k \le n$. Then

$$C(n,k) = C(n,n-k).$$

Proof (algebraic). From the formula,

$$C(n,k) = \frac{n!}{k!(n-k)!}, \qquad C(n,n-k) = \frac{n!}{(n-k)!\,(n-(n-k))!} = \frac{n!}{(n-k)!\,k!}.$$

Hence $C(n,k) = C(n,n-k)$. ■

12.3 Combinatorial Proofs and the Bijection Principle

Definition. A combinatorial proof of an identity is a proof that uses counting arguments to show that both sides of the identity count the same objects, but in different ways.

Bijection Principle

Two sets U and V have the same size, i.e., $|U| = |V|$, if and only if there is a bijection f from U to V (one-to-one and onto).

Alternative (combinatorial) proof that $C(n,k) = C(n,n-k)$. Suppose S is a set with n distinct elements. Every subset A of S with k elements corresponds to its complement $S - A$, which has $n-k$ elements. This correspondence is a bijection between the k-element subsets and the $(n-k)$-element subsets. Consequently, $C(n,k) = C(n,n-k)$. ■

Example 12 (Bit strings with exactly k ones). How many bit strings of length n contain exactly k 1's?

Solution. A bit string of length n with exactly k ones is determined by choosing which k of the n positions contain 1's. For instance, with $n = 5$ and $k = 3$, the string 01011 corresponds to the position set $\{2,4,5\}$. Hence the number of such strings equals the number of k-combinations of the n positions:

$$C(n,k).$$

Example (Playoff revisited). In the playoff example (first to 3 wins in at most 5 games), a 5-game playoff corresponds to a string of length 5 with exactly 3 wins by one team: exactly three 0's if team 1 wins the playoff, or exactly three 1's if team 2 wins. The number of such strings is

$$2 \cdot C(5,3) = 2 \times 10 = 20.$$

Example 13 (Mixed committee). Suppose there are 40 faculty members in the computer science department and 30 in the mathematics department. How many ways are there to select a committee consisting of four faculty members from computer science and three from mathematics?

Solution. There are $C(40, 4)$ ways to select the 4 computer science members. For each such selection, there are $C(30, 3)$ ways to select the 3 mathematics members. By the product rule, the total number of ways is

$$C(40, 4) \times C(30, 3) = \frac{40!}{4!\,36!} \cdot \frac{30!}{3!\,27!} = 91{,}390 \times 4{,}060 = 371{,}043{,}400.$$

12.4 Permutations with Indistinguishable Objects

Example 15 (Reordering SUCCESS). How many different strings can be made by reordering the letters of the word SUCCESS?

Solution. Since some letters repeat, the answer is not 7!. The word SUCCESS contains 3 S's, 2 C's, 1 E, and 1 U (7 letters in total). Place the letters step by step:

- The three indistinguishable S's can be placed among the 7 positions in $C(7, 3)$ ways, leaving 4 positions free.
- The two C's can be placed in the remaining positions in $C(4, 2)$ ways, leaving 2 positions free.
- The E can be placed in $C(2, 1)$ ways, leaving 1 position.
- The U goes in the last position: $C(1, 1)$ way.

By the product rule, the number of different strings is

$$C(7, 3)\, C(4, 2)\, C(2, 1)\, C(1, 1) = \frac{7!}{3!\,4!} \cdot \frac{4!}{2!\,2!} \cdot \frac{2!}{1!\,1!} \cdot 1 = \frac{7!}{3!\,2!\,1!\,1!} = 420.$$

Permutations with Indistinguishable Objects

The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, ..., and n_k indistinguishable objects of type k, is

$$\frac{n!}{n_1!\, n_2! \cdots n_k!}.$$

12.5 Throwing Balls into Bins

Many counting problems can be modeled as placing balls into bins, where the order in which the balls are placed does not matter. The balls can be distinguishable (labeled) or indistinguishable (unlabeled), and the bins can also be distinguishable or indistinguishable. We first consider the case where both balls and bins are distinguishable.

Example 17 (Dealing cards). How many ways are there to distribute hands of five cards to each of four players from a standard deck of 52 cards?

Solution. The cards are distinguishable balls and the players are distinguishable bins.

- The first player can be dealt 5 cards in $C(52, 5)$ ways.
- The second player can be dealt 5 cards from the remaining 47 in $C(47, 5)$ ways.
- The third player: $C(42, 5)$ ways.
- The fourth player: $C(37, 5)$ ways.

By the product rule, the total number of ways is

$$C(52, 5)\, C(47, 5)\, C(42, 5)\, C(37, 5) = \frac{52!}{47!\,5!} \cdot \frac{47!}{42!\,5!} \cdot \frac{42!}{37!\,5!} \cdot \frac{37!}{32!\,5!} = \frac{52!}{5!\,5!\,5!\,5!\,32!}.$$

Distinguishable Balls into Distinguishable Bins (Fixed Counts)

The number of ways to throw r distinguishable balls into n distinguishable bins so that b_i balls are placed into bin i, with $b_1 + b_2 + \cdots + b_n = r$, is

$$\frac{r!}{b_1!\, b_2! \cdots b_n!}.$$

Remark. The solution of Example 17 equals the number of permutations of 52 positions where there are five indistinguishable objects of each of four types (the players' hands) and 32 of a fifth type (the undealt cards). Positions assigned to the first type correspond to cards dealt to the first player, and so on. This defines a bijection between such permutations and distributions of cards to players.

12.6 Indistinguishable Balls into Distinguishable Bins

Example (Selecting bills). How many ways are there to select five bills from the denominations \$1, \$2, \$5, \$10, \$20, \$50, and \$100? Assume there are enough bills of each denomination.

Solution. This is the same as distributing 5 indistinguishable balls into 7 distinguishable bins: each bin corresponds to a denomination, and the number of balls in a bin is the number of bills of that denomination selected. Imagine a cash box with 7 compartments separated by 6 bars; the selected bills are represented as stars placed in the compartments. Each selection corresponds to an arrangement of 6 bars and 5 stars in a row. The number of such arrangements is

$$C(11, 5) = \frac{11!}{5!\, 6!} = 462.$$

Stars and Bars

The number of ways to distribute r indistinguishable balls into n distinguishable bins is

$$C(n + r - 1,\ r).$$

Proof. Each distribution can be represented by a list of $n - 1$ bars and r balls (stars). The bars mark the boundaries between the n bins; the balls between two consecutive bars correspond to the balls in one bin. The list has $n - 1 + r$ positions in total, and we choose r of them to place the balls (equivalently, $n - 1$ positions to place the bars). Hence the number of distributions is $C(n + r - 1, r)$. ■

12.7 Combinations with Repetition

Example (Integer solutions). How many solutions does the equation

$$x_1 + x_2 + x_3 = 11$$

have, where x_1, x_2, x_3 are nonnegative integers?

Solution. Each solution corresponds to throwing 11 indistinguishable balls into 3 distinguishable bins: x_1 balls into bin 1, x_2 into bin 2, and x_3 into bin 3. By the stars-and-bars theorem, the number of solutions is

$$C(3 + 11 - 1,\ 11) = C(13, 11) = C(13, 2) = \frac{13 \cdot 12}{2} = 78.$$

Example (Cookies). Suppose that a cookie shop has 4 different kinds of cookies. How many different ways can 6 cookies be chosen?

Solution. The number of ways to choose 6 cookies is the number of ways to throw 6 indistinguishable balls into 4 distinguishable bins:

$$C(4+6-1,\ 6) = C(9,6) = C(9,3) = \frac{9\cdot 8\cdot 7}{3\cdot 2\cdot 1} = 84.$$

12.8 Summary of Balls and Bins

Summary: r Balls into n Bins

Balls	Bins	Formula
Distinguishable	Distinguishable	n^r
Distinguishable	Distinguishable, bin i must have b_i balls	$\dfrac{r!}{b_1!\,b_2!\cdots b_n!}$
Indistinguishable	Distinguishable	$C(r+n-1,r)$ (stars and bars)
Distinguishable	Indistinguishable	Stirling numbers of the second kind
Indistinguishable	Indistinguishable	Integer partition numbers

13 Binomial Coefficients

13.1 The Binomial Theorem

Definition. The number of k-combinations of a set with n elements, denoted $C(n,k)$ or $\binom{n}{k}$, is also called a binomial coefficient because it occurs as a coefficient in the expansion of a power of a binomial expression such as $(x+y)^n$.

Binomial Theorem

Let x and y be variables and n a nonnegative integer. Then

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k}y^k.$$

For example, in the expansion of $(x+y)^3$, the coefficient of xy^2 equals the number of length-3 lists having y in exactly 2 places, which is $\binom{3}{2} = 3$: indeed, $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$.

Proof (combinatorial). When we expand the product

$$(x+y)^n = (x+y)(x+y)\cdots(x+y) \quad (n \text{ factors}),$$

each term of the expansion is obtained by choosing either x or y from each factor. The terms are therefore of the form $x^{n-k}y^k$ for $k = 0, 1, \ldots, n$. To obtain a term $x^{n-k}y^k$, we must choose y from exactly k of the n factors (and x from the remaining $n-k$ factors). The number of ways to do this is $\binom{n}{k}$. Therefore the coefficient of $x^{n-k}y^k$ is $\binom{n}{k}$. ■

Example 1 (Plain binomial coefficient). What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

Solution. By the binomial theorem, the coefficient is

$$\binom{25}{13} = \frac{25!}{13!\,12!} = 5{,}200{,}300.$$

Example 2 (Coefficient with constants). What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?

Solution. By the binomial theorem, the term involving $x^{12}y^{13}$ is

$$\binom{25}{12}(2x)^{12}(-3y)^{13}.$$

Therefore the coefficient of $x^{12}y^{13}$ is

$$\binom{25}{12} \cdot 2^{12} \cdot (-3)^{13} = -\binom{25}{12} 2^{12}\, 3^{13}.$$

Example 3 (Constant term). What is the constant term in the expansion of $\left(x + \dfrac{1}{x^3}\right)^{12}$?

Solution. By the binomial theorem, the general term is

$$\binom{12}{a} x^a \left(\frac{1}{x^3}\right)^{12-a}.$$

The power of x in this term is $a - 3(12 - a)$. For the term to be constant, we need

$$a - 3(12 - a) = 0 \implies a - 36 + 3a = 0 \implies 4a = 36 \implies a = 9.$$

Therefore the constant term is

$$\binom{12}{9} = \frac{12!}{9!\,3!} = \frac{12 \cdot 11 \cdot 10}{3 \cdot 2 \cdot 1} = 220.$$

13.2 Corollaries of the Binomial Theorem

Corollary: Sum of Binomial Coefficients

Let n be a nonnegative integer. Then

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Proof. Using the binomial theorem with $x = y = 1$,

$$2^n = (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} 1^k = \sum_{k=0}^{n} \binom{n}{k}.$$

■

Alternative combinatorial proof. A set with n elements has 2^n subsets in total. Each subset has zero, one, two, ..., or n elements. There are $\binom{n}{0}$ subsets with zero elements,

$\binom{n}{1}$ subsets with one element, ..., and $\binom{n}{n}$ subsets with n elements. Therefore $\sum_{k=0}^{n}\binom{n}{k}$ counts the total number of subsets, which is 2^n. ■

Corollary: Alternating Sum

Let n be a positive integer. Then

$$\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0.$$

Proof. Using the binomial theorem with $x = 1$ and $y = -1$,

$$0 = (1+(-1))^n = \sum_{k=0}^{n}\binom{n}{k}1^{n-k}(-1)^k = \sum_{k=0}^{n}\binom{n}{k}(-1)^k.$$

■

Remark. This corollary implies

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \cdots,$$

i.e., the sum of the binomial coefficients with even lower index equals the sum with odd lower index.

Corollary: Weighted Sum

Let n be a positive integer. Then

$$\sum_{k=0}^{n}2^k\binom{n}{k} = 3^n.$$

Proof. Using the binomial theorem with $x = 1$ and $y = 2$,

$$3^n = (1+2)^n = \sum_{k=0}^{n}\binom{n}{k}1^{n-k}2^k = \sum_{k=0}^{n}\binom{n}{k}2^k.$$

■

13.3 Pascal's Identity and Pascal's Triangle

Pascal's Identity

Let n and k be integers with $0 < k < n$. Then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Example. Let $X = \{A, B, C, D, E\}$ and let S be the set of 2-combinations of X. Split S according to whether the subset contains E:

$$S = S_1 \cup S_2,$$

where

$$S_1 = \{\{A, E\}, \{B, E\}, \{C, E\}, \{D, E\}\}, \qquad |S_1| = \binom{4}{1} = 4,$$

(these are the 1-combinations of $\{A, B, C, D\}$ together with E) and

$$S_2 = \{\{A, B\}, \{A, C\}, \{A, D\}, \{B, C\}, \{B, D\}, \{C, D\}\}, \qquad |S_2| = \binom{4}{2} = 6,$$

(the 2-combinations of $\{A, B, C, D\}$). Hence

$$\binom{5}{2} = |S| = |S_1| + |S_2| = \binom{4}{1} + \binom{4}{2} = 4 + 6 = 10.$$

Proof (combinatorial). Suppose S is a set containing n elements. Let a be an element of S and $T = S - \{a\}$. There are $\binom{n}{k}$ subsets of S containing k elements. Each such subset either

- contains a together with $k-1$ elements of T — there are $\binom{n-1}{k-1}$ of these; or
- contains k elements of T and does not contain a — there are $\binom{n-1}{k}$ of these.

By the sum rule,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

■

The proof counts the k-combinations of $\{1, 2, \ldots, n\}$ by splitting them into the k-combinations not containing n (i.e., k-combinations of $\{1, \ldots, n-1\}$) and those containing n (i.e., $(k-1)$-combinations of $\{1, \ldots, n-1\}$ together with n).

Pascal's Triangle

Pascal's identity, together with the initial conditions $\binom{n}{0} = \binom{n}{n} = 1$ for all integers n, gives a geometric arrangement of the binomial coefficients in a triangle, called Pascal's triangle:

$$\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
\end{array}$$

Each interior entry is the sum of the two entries above it.

Remark (Yang Hui triangle). The same triangle was known in China as the Yang Hui triangle. Its entries also count the number of distinct paths to each square by a rook when only rightward and downward movements are allowed.

13.4 Vandermonde's Identity

Vandermonde's Identity (1772)

Let m, n, r be integers with $0 \le r \le m$ and $0 \le r \le n$. Then

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

(Discovered by Zhu Shijie in 1303.)

Proof (combinatorial). Suppose there are m elements in one set and n elements in a second set. The total number of ways to choose r elements from the union of the two sets is $\binom{m+n}{r}$. Another way to choose r elements from the union is to choose $r-k$ elements from the first set and then k elements from the second set, where k ranges over $0 \le k \le r$. By the product rule, for each k this can be done in $\binom{m}{r-k}\binom{n}{k}$ ways. Summing over all k by the sum rule,

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

■

Corollary

Let n be a nonnegative integer. Then

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

Proof. Using Vandermonde's identity with $m = n = r$,

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{n-k}\binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k}^2,$$

where the last step uses the symmetry $\binom{n}{n-k} = \binom{n}{k}$. ■

13.5 Counting Bit Strings

We can prove combinatorial identities by counting bit strings with different properties. Recall: the number of bit strings of length n containing exactly r ones equals the number of ways to choose r positions out of n positions, i.e., the number of r-combinations of n objects, $\binom{n}{r}$.

Theorem (Hockey-Stick / Upper Summation Identity)

Let n and r be integers with $0 \le r \le n$. Then

$$\binom{n+1}{r+1} = \sum_{k=r+1}^{n+1} \binom{k-1}{r} = \binom{r}{r} + \binom{r+1}{r} + \cdots + \binom{n}{r}.$$

Proof, Solution 1 (counting bit strings by the last 1). The left-hand side counts the number of bit strings of length $n+1$ containing exactly $(r+1)$ 1's.

The right-hand side counts the same objects by considering the possible locations of the last 1 in such a string: the last 1 must occur at location $r+1, r+2, \ldots,$ or $n+1$ (it cannot occur earlier, since we need room for r other 1's before it).

If the last 1 is at the k-th bit (with $r+1 \le k \le n+1$), then among the first $k-1$ positions there must be exactly r ones, which can be placed in $\binom{k-1}{r}$ ways. Summing over all possible k,

$$\sum_{k=r+1}^{n+1} \binom{k-1}{r}.$$

Since both sides count the same objects, they are equal. ■

Proof, Solution 2 (counting subsets by the largest element). The left-hand side counts the number of ways of selecting $r+1$ objects from $n+1$ objects labeled 1 to $n+1$.

The right-hand side considers the possible highest label among the selected $r+1$ objects: the highest label must be $r+1, r+2, \ldots,$ or $n+1$.

- If the highest label is $r+1$, the other r objects must be selected from the objects labeled 1 to r: $\binom{r}{r}$ ways.
- If the highest label is $r+2$, the other r objects must be selected from the objects labeled 1 to $r+1$: $\binom{r+1}{r}$ ways.
- ...
- If the highest label is $n+1$, the other r objects must be selected from the objects labeled 1 to n: $\binom{n}{r}$ ways.

Therefore the total number of ways is

$$\binom{r}{r} + \binom{r+1}{r} + \cdots + \binom{n}{r}.$$

Since both sides count the same objects, the identity holds. ■

14 Inclusion-Exclusion Principle

14.1 Two Finite Sets

Inclusion-exclusion principle for two sets

If A and B are finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

When we compute $|A| + |B|$, every element of $A \cap B$ is counted twice — once as an element of A and once as an element of B. Subtracting $|A \cap B|$ corrects this double counting, so each element of $A \cup B$ is counted exactly once.

Example. In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

Solution. Let A = the set of computer science majors and B = the set of mathematics majors. Then $|A| = 25$, $|B| = 13$, and $|A \cap B| = 8$. Since every student in the class is in $A \cup B$,

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30.$$

Hence there are 30 students in the class.

Example. How many positive integers not exceeding 1000 are divisible by 7 or 11?

Solution. Let A be the set of positive integers not exceeding 1000 that are divisible by 7, and B those divisible by 11.

- $|A| = \left\lfloor \dfrac{1000}{7} \right\rfloor = 142$ (the multiples of 7 are $7 \cdot 1, 7 \cdot 2, \ldots, 7 \cdot 142 = 994$).
- $|B| = \left\lfloor \dfrac{1000}{11} \right\rfloor = 90$ (the multiples of 11 are $11 \cdot 1, \ldots, 11 \cdot 90 = 990$).
- $A \cap B$ = the set of integers divisible by both 7 and 11. Since $\gcd(7, 11) = 1$, these are exactly the multiples of $7 \cdot 11 = 77$:

$$|A \cap B| = \left\lfloor \frac{1000}{77} \right\rfloor = 12.$$

By the inclusion-exclusion principle,

$$|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = 220.$$

So 220 positive integers not exceeding 1000 are divisible by 7 or 11.

14.2 Three Finite Sets

Inclusion-exclusion principle for three sets

If A, B, and C are finite sets, then

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Add the sizes of the individual sets, subtract the sizes of all pairwise intersections (elements in two sets were counted twice), and add back the size of the triple intersection (elements in all three sets were added 3 times and subtracted 3 times, so they must be added back once).

Example. A total of 1232 students have taken a course in Spanish; 879 have taken a course in French; 114 have taken a course in Russian; 103 have taken courses in both

Spanish and French; 23 have taken courses in both Spanish and Russian; 14 have taken courses in both French and Russian; and 2092 students have taken a course in at least one of Spanish, French, and Russian. How many students have taken courses in all three languages?

Solution. Let S, F, R be the sets of students who have taken Spanish, French, Russian, respectively. We are given

$$|S| = 1232, \quad |F| = 879, \quad |R| = 114,$$
$$|S \cap F| = 103, \quad |S \cap R| = 23, \quad |F \cap R| = 14, \quad |S \cup F \cup R| = 2092.$$

By the inclusion-exclusion principle,

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|,$$

so substituting the known values,

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|.$$

The right-hand side without the unknown term is $2225 - 140 = 2085$. Solving,

$$|S \cap F \cap R| = 2092 - 2085 = 7.$$

Therefore 7 students have taken courses in all three languages.

14.3 The General Inclusion-Exclusion Principle

Theorem. (Inclusion-Exclusion Principle) Let $A_1, A_2, \ldots, A_n$ be finite sets. Then

Inclusion-exclusion principle (n sets)

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|.$$

Note the alternating signs: the sum over single sets is added, the sum over pairs is subtracted, the sum over triples is added, and so on; the last term has sign $(-1)^{n+1}$.

Proof. We prove the formula by showing that every element of $A_1 \cup A_2 \cup \cdots \cup A_n$ is counted exactly once by the right-hand side of the equation.

Suppose a is a member of exactly r of the sets $A_1, \ldots, A_n$, where $1 \le r \le n$.

- In the first sum $\sum |A_i|$, the element a is counted once for each set containing it: $\binom{r}{1}$ times.
- In the second sum $\sum |A_i \cap A_j|$, a is counted once for each pair of sets both containing it: $\binom{r}{2}$ times.
- In general, in the sum involving intersections of m sets, a is counted $\binom{r}{m}$ times (for $m \le r$; for $m > r$ it is not counted at all).

Thus the total number of times a is counted by the right-hand side is

$$N = \binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r}.$$

We claim that $N = 1$. Recall the corollary of the binomial theorem obtained by setting

$x = 1, y = -1$ in $(x+y)^r$:

$$0 = (1-1)^r = \sum_{k=0}^{r} (-1)^k \binom{r}{k} = \binom{r}{0} - \binom{r}{1} + \binom{r}{2} - \binom{r}{3} + \cdots + (-1)^r \binom{r}{r}.$$

Moving all terms except $\binom{r}{0}$ to the other side,

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r} = \binom{r}{0} = 1.$$

Hence each element of the union is counted exactly once, and the formula holds. ■

14.4 Counting Onto Functions

Example. How many surjective (onto) functions are there from a set with six elements to a set with three elements?

Solution. Suppose the function is from A to B, where $|A| = 6$ and $B = \{b_1, b_2, b_3\}$. Altogether there are 3^6 functions from A to B; we subtract those that miss at least one element of B.

Let P_i be the set of functions that do not map any element of A to b_i $(i = 1, 2, 3)$. A function is not onto iff it belongs to $P_1 \cup P_2 \cup P_3$, so

$$\#\text{onto} = 3^6 - |P_1 \cup P_2 \cup P_3|.$$

Compute each intersection:

- $|P_i|$: functions using only the other 2 values: $|P_1| = |P_2| = |P_3| = 2^6 = 64$.
- $|P_i \cap P_j|$: functions using only the remaining 1 value: $|P_1 \cap P_2| = |P_1 \cap P_3| = |P_2 \cap P_3| = 1^6 = 1$.
- $|P_1 \cap P_2 \cap P_3|$: functions using no value at all: 0.

By inclusion-exclusion,

$$|P_1 \cup P_2 \cup P_3| = \binom{3}{1} 2^6 - \binom{3}{2} 1^6 + \binom{3}{3} \cdot 0 = 3 \cdot 64 - 3 \cdot 1 + 0 = 189.$$

Therefore the number of onto functions is

$$3^6 - 189 = 729 - 189 = 540.$$

Example. How many ways are there to assign five different jobs to four different employees if every employee is assigned at least one job?

Solution. Each assignment in which every employee gets at least one job is exactly an onto function from the set of 5 jobs to the set of 4 employees. Let P_i = the set of assignments in which employee i gets no job. Using inclusion-exclusion as above, the number of onto assignments is

$$4^5 - \binom{4}{1} 3^5 + \binom{4}{2} 2^5 - \binom{4}{3} 1^5 + \binom{4}{4} 0^5$$

$$= 1024 - 4 \cdot 243 + 6 \cdot 32 - 4 \cdot 1 + 0 = 1024 - 972 + 192 - 4 = 240.$$

So there are 240 such assignments.

Number of onto functions

The number of onto functions from a set with n elements to a set with k elements

$(n \geq k)$ is

$$\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n = k^n - \binom{k}{1}(k-1)^n + \binom{k}{2}(k-2)^n - \cdots + (-1)^{k-1}\binom{k}{k-1}1^n.$$

14.5 Distributing Objects into Boxes

Balls into boxes, no box empty

Distinguishable boxes: The number of ways to distribute n distinguishable balls into k distinguishable boxes so that no box is empty equals the number of onto functions:

$$\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n.$$

Indistinguishable boxes: The number of ways to distribute n distinguishable balls into k indistinguishable boxes so that no box is empty is

$$\frac{1}{k!}\sum_{i=0}^{k}(-1)^i\binom{k}{i}(k-i)^n.$$

(This is the Stirling number of the second kind $S(n,k)$.)

For indistinguishable boxes we divide by $k!$: when the boxes are distinguishable, each distribution into unlabeled boxes is counted once for each of the $k!$ ways to assign labels to the boxes.

14.6 Derangements

Example. (The Hatcheck Problem) A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats randomly. How many ways are there to return the hats so that no one receives the correct hat?

Remark. The answer is the number of permutations of the n hats in which no hat is in its original position.

Definition. A *derangement* is a permutation of objects that leaves no object in its original position.

Example. The permutation 21453 is a derangement of 12345, because no number is left in its original position. The permutation 21543 is not a derangement of 12345, because it leaves 4 unchanged (position 4 still contains 4).

Theorem. (Number of Derangements)

Number of derangements

The number of derangements of a set with n elements is

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n\frac{1}{n!}\right).$$

Proof. A permutation is a derangement iff it has no fixed point. Let P_i be the set of permutations in which element i remains unchanged $(i = 1, \ldots, n)$. The permutations

that are not derangements form $P_1 \cup P_2 \cup \cdots \cup P_n$, so

$$D_n = n! - |P_1 \cup P_2 \cup \cdots \cup P_n|.$$

We apply the inclusion-exclusion principle. If m specified elements must stay fixed, the remaining $n - m$ elements can be permuted freely, so

$$|P_i| = (n-1)!, \qquad |P_i \cap P_j| = (n-2)!, \qquad \ldots, \qquad |P_{i_1} \cap P_{i_2} \cap \cdots \cap P_{i_m}| = (n-m)!.$$

For each m there are $\binom{n}{m}$ ways to choose the m fixed elements. Hence

$$\begin{aligned} |P_1 \cup P_2 \cup \cdots \cup P_n| &= \binom{n}{1}(n-1)! - \binom{n}{2}(n-2)! + \cdots + (-1)^{n+1}\binom{n}{n}(n-n)! \\ &= \frac{n!}{1!(n-1)!}(n-1)! - \frac{n!}{2!(n-2)!}(n-2)! + \cdots + (-1)^{n+1}\frac{n!}{n!\,0!}0! \\ &= n!\left(\frac{1}{1!} - \frac{1}{2!} + \cdots + (-1)^{n+1}\frac{1}{n!}\right). \end{aligned}$$

Subtracting from the total number $n!$ of permutations,

$$\begin{aligned} D_n &= n! - n!\left(\frac{1}{1!} - \frac{1}{2!} + \cdots + (-1)^{n+1}\frac{1}{n!}\right) \\ &= n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^{n}\frac{1}{n!}\right). \end{aligned}$$

■

Example. A machine that inserts letters into envelopes goes haywire and inserts letters randomly into envelopes. In a group of 100 letters, in how many ways can this happen if

(a) No letter is put into the correct envelope?
(b) Exactly 1 letter is put into the correct envelope?
(c) Exactly 98 letters are put into the correct envelopes?

Solution.

(a) This is precisely the number of derangements of 100 letters: $\boldsymbol{D_{100}}$.

(b) First choose which one letter is correctly stuffed: $\binom{100}{1} = 100$ ways. The remaining 99 letters must all be inserted incorrectly, which is a derangement of 99 letters: D_{99} ways. Total: $\boldsymbol{100\,D_{99}}$.

(c) Exactly 98 correct means exactly 2 letters are misplaced. Choose the two letters to be misplaced: $\binom{100}{2}$ ways. Two letters can be deranged in only one way (swap them), since $D_2 = 1$. Total: $\boldsymbol{\binom{100}{2}}$.

Probability of a derangement. Since all $n!$ permutations are equally likely, the probability that a random permutation is a derangement is

$$\frac{D_n}{n!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n\frac{1}{n!},$$

which approaches $\dfrac{1}{e}$ as $n \to \infty$ (recall the series $e^{-1} = \displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$).

15 Introduction to Probability

15.1 The Hatcheck Problem Revisited

Recall the derangement counting result from the inclusion-exclusion principle. In the hatcheck problem, the total number of permutations of n hats is $n!$, and the number of derangements is

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right).$$

If the hats are returned randomly (all permutations equally likely), it is natural to say that the probability that a random permutation is a derangement is

$$p = \frac{D_n}{n!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!},$$

which approaches $1/e$ as $n \to \infty$. Probability theory makes this kind of reasoning precise.

15.2 Experiments, Sample Spaces, and Events

Definition. An *experiment* is a procedure that yields one of a given set of possible outcomes.

Definition. The *sample space* of the experiment is the set of all possible outcomes.

Definition. An *event* is a subset of the sample space.

Probability of an event (equally likely outcomes)

If S is a finite sample space of equally likely outcomes, and E is an event, i.e., a subset of S, then the probability of E is

$$p(E) = \frac{|E|}{|S|}.$$

So for equally likely outcomes, **probability = counting**: compute the size of the event and the size of the sample space.

15.3 Probability by Counting

Example 1. A bag contains 4 blue balls and 5 red balls. What is the probability that a ball chosen from the bag is blue?

Solution. There are $4+5 = 9$ balls in total, all equally likely to be chosen, so $|S| = 9$. The event "the chosen ball is blue" contains the 4 blue balls, so $|E| = 4$. Hence

$$p(E) = \frac{|E|}{|S|} = \frac{4}{9}.$$

Example 2. What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

Solution. By the product rule there are $6^2 = 36$ possible outcomes (each die shows one of 6 faces), all equally likely. The outcomes whose sum is 7 are

$$(1,6),\ (2,5),\ (3,4),\ (4,3),\ (5,2),\ (6,1),$$

six outcomes in total. Hence the probability of obtaining a sum of 7 is

$$p = \frac{6}{36} = \frac{1}{6}.$$

15.4 Poker and Lottery

A standard deck has 52 cards (13 ranks, 4 suits); a poker hand consists of 5 cards, so the total number of hands is $\binom{52}{5} = 2{,}598{,}960$.

Example. (Four of a kind) A four of a kind hand contains 4 cards of one rank and one card of another rank. What is its probability?

Solution. Choose the rank of the four cards: 13 ways. The four cards of that rank are then forced (all 4 suits). Choose the fifth card: any of the remaining 48 cards. So the number of four-of-a-kind hands is $13 \times 48 = 624$, and

$$p(\text{four of a kind}) = \frac{13 \times 48}{\binom{52}{5}} = \frac{624}{2{,}598{,}960} \approx 0.00024.$$

Example. (Full house) A full house contains 3 cards of one rank and 2 cards of another rank. What is its probability?

Solution. Choose the two ranks in order (first the rank of the triple, then the rank of the pair): $P(13,2) = 13 \cdot 12$ ways. Choose 3 of the 4 suits for the triple: $\binom{4}{3}$ ways. Choose 2 of the 4 suits for the pair: $\binom{4}{2}$ ways. Hence the number of full houses is $P(13,2)\binom{4}{3}\binom{4}{2} = 13 \cdot 12 \cdot 4 \cdot 6 = 3744$, and

$$p(\text{full house}) = \frac{P(13,2)\binom{4}{3}\binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2{,}598{,}960} \approx 0.0014.$$

Example. (Mark Six lottery) In the Mark Six lottery, a player chooses 6 numbers from 49; then 6 numbers are drawn plus 1 extra number. The probability that the player's 6 chosen numbers exactly match the 6 drawn numbers (first prize) is

$$\frac{1}{\binom{49}{6}} = \frac{1}{13{,}983{,}816},$$

since there are $\binom{49}{6}$ equally likely draws and only one of them matches the player's ticket exactly.

15.5 Sampling With and Without Replacement

Example. What is the probability that the numbers $11, 4, 17, 39$, and 23 are drawn in that order from a bin containing 50 balls labeled with the numbers $1, 2, \ldots, 50$ if (a) the ball selected is not returned to the bin before the next ball is selected (sampling without replacement), and (b) the ball selected is returned to the bin before the next ball is selected (sampling with replacement)?

Solution. There is exactly one way for the event to happen (draw exactly $11, 4, 17, 39, 23$ in that order), so the probability is 1 divided by the total number of possible ordered draws.

(a) Without replacement, the number of ordered draws of 5 balls from 50 is the number of permutations $P(50,5) = 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 = 254{,}251{,}200$. Hence

$$p = \frac{1}{P(50,5)} = \frac{1}{254{,}251{,}200}.$$

(b) With replacement, each of the 5 draws has 50 possible outcomes, so by the product rule there are $50^5 = 312{,}500{,}000$ ordered draws. Hence

$$p = \frac{1}{50^5} = \frac{1}{312{,}500{,}000}.$$

15.6 The Complement of an Event

Theorem. Let E be an event in a finite sample space S. The probability of the event $\overline{E}$, the complementary event of E, is given by

Complement rule

$$p(\overline{E}) = 1 - p(E).$$

Proof. Since E and $\overline{E}$ partition the sample space, $|\overline{E}| = |S| - |E|$. Therefore

$$p(\overline{E}) = \frac{|\overline{E}|}{|S|} = \frac{|S| - |E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E).$$

■

Example. A sequence of 10 bits is chosen randomly. What is the probability that at least one of these bits is 0?

Solution. Let E be the event that at least one of the 10 bits is 0. Then $\overline{E}$ is the event that all of the bits are 1s, i.e., the single outcome 1111111111. The size of the sample space S is $2^{10} = 1024$, so

$$p(\overline{E}) = \frac{1}{2^{10}} = \frac{1}{1024}, \qquad p(E) = 1 - p(\overline{E}) = 1 - \frac{1}{1024} = \frac{1023}{1024}.$$

Computing $p(E)$ directly would require counting strings with one 0, two 0s, ..., ten 0s — the complement rule avoids all that work.

15.7 The Union of Events

Theorem. Let E_1 and E_2 be events in the sample space S. Then

Union rule (addition rule)

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2).$$

Proof. For equally likely outcomes, this follows immediately from the inclusion-exclusion formula $|E_1 \cup E_2| = |E_1| + |E_2| - |E_1 \cap E_2|$ by dividing both sides by $|S|$. ■

Example. What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?

Solution. Let E_1 be the event that the integer is divisible by 2, and E_2 the event that it is divisible by 5. Then $E_1 \cup E_2$ is the event that the integer is divisible by 2 or 5, and $E_1 \cap E_2$ is the event that it is divisible by both 2 and 5, i.e., by 10. Counting multiples,

$$|E_1| = \left\lfloor \frac{100}{2} \right\rfloor = 50, \qquad |E_2| = \left\lfloor \frac{100}{5} \right\rfloor = 20, \qquad |E_1 \cap E_2| = \left\lfloor \frac{100}{10} \right\rfloor = 10.$$

It follows that

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2) = \frac{50}{100} + \frac{20}{100} - \frac{10}{100} = \frac{60}{100} = \frac{3}{5}.$$

Inclusion-exclusion principle for probability

Let $E_1, E_2, \ldots, E_n$ be events in a sample space S. Then

$$p\left(\bigcup_{i=1}^{n} E_i\right) = \sum_i p(E_i) - \sum_{i<j} p(E_i \cap E_j) + \sum_{i<j<k} p(E_i \cap E_j \cap E_k) - \cdots + (-1)^{n+1} p(E_1 \cap \cdots \cap E_n).$$

Additivity for disjoint events

If $E_1, E_2, \ldots$ is a sequence of pairwise disjoint events in a sample space S (i.e., $E_i \cap E_j = \emptyset$ for $i \neq j$), then

$$p\left(\bigcup_i E_i\right) = \sum_i p(E_i).$$

15.8 Probability Distributions

Definition. Let S be the sample space of an experiment with a finite number of outcomes. A *probability distribution* on S is characterized by a *probability mass function* (pmf) $p : S \to \mathbb{R}$ such that

(a) $0 \leq p(s) \leq 1$ for each $s \in S$,

(b) $\sum_{s \in S} p(s) = 1$,

where $p(s)$ is the probability of the outcome s.

Definition. (Uniform distribution) Suppose that S is a set with n elements. The *uniform distribution* assigns the probability $1/n$ to each element of S.

Example. For a fair die with 6 sides, we have $p(x) = \frac{1}{6}$ for all $x \in \{1, 2, 3, 4, 5, 6\}$. What is the probability that an odd number appears when we roll this die?

Solution. The event is $E = \{1, 3, 5\}$. Summing the probabilities of the outcomes in E,

$$p(E) = p(1) + p(3) + p(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}.$$

Example. (Non-uniform distribution) Suppose that a die is biased so that $p(2) = \frac{2}{7}$ and $p(x) = \frac{1}{7}$ for all other x ($x = 1, 3, 4, 5, 6$). What is the probability that an odd number appears when we roll this die?

Solution. First note this is a valid distribution: $\frac{2}{7} + 5 \cdot \frac{1}{7} = 1$. The odd outcomes are $1, 3, 5$, each with probability $\frac{1}{7}$, so

$$p(\text{odd}) = p(1) + p(3) + p(5) = \frac{1}{7} + \frac{1}{7} + \frac{1}{7} = \frac{3}{7}.$$

(For a fair die this probability was $\frac{1}{2}$; the bias toward 2 lowers it to $\frac{3}{7}$.)

Probability of an event under a general distribution

The probability of the event E is the sum of the probabilities of the outcomes in E:

$$p(E) = \sum_{s \in E} p(s).$$

Uniform distributions: probability = count ($|E|/|S|$). **General distributions:** probability = sum.

15.9 Independence

Example. Flip two coins. What is the probability that they both turn up heads? Each coin comes up heads with probability $1/2$, and the two coins do not affect each other; the sample space $\{HH, HT, TH, TT\}$ has 4 equally likely outcomes, so $p(HH) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$. The probability factors — this is the intuition behind independence.

Definition. Two events E and F are *independent* if and only if

$$p(E \cap F) = p(E)\, p(F).$$

How to check independence? Two unrelated events (e.g., two separate coins) are independent; otherwise, verify the definition $p(E \cap F) = p(E)p(F)$ directly.

Example. In a randomly generated bit string of length 4, let

- E: the string begins with a 1,
- F: the string contains an even number of 1s.

Are E and F independent?

Solution. All 16 bit strings are equally likely.

- $p(E) = \frac{8}{16} = \frac{1}{2}$ (the first bit is fixed to 1; the other 3 bits are free, giving $2^3 = 8$ strings).
- $p(F) = \frac{1}{2}$, since the number of strings with an even number of 1s is $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 1 + 6 + 1 = 8$, so $p(F) = \frac{8}{16} = \frac{1}{2}$.
- $E \cap F = \{1111, 1100, 1010, 1001\}$ (strings starting with 1 with 2 or 4 ones), so

$$p(E \cap F) = \frac{|E \cap F|}{16} = \frac{4}{16} = \frac{1}{4}.$$

Since $p(E \cap F) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = p(E)\, p(F)$, the events E and F **are independent**.

Example. Suppose a family has 3 children, each of which has equal probability to be a boy or a girl. Are the following two events independent?

- E: the family has children of both sexes,
- F: the family has at most one boy.

Solution. The sample space consists of $2^3 = 8$ equally likely outcomes (sequences of B/G in birth order).

- E excludes only BBB and GGG:

$$p(E) = \frac{|\{BBG, BGB, BGG, GBB, GBG, GGB\}|}{8} = \frac{6}{8} = \frac{3}{4}.$$

- F consists of outcomes with zero or one boy:

$$p(F) = \frac{|\{BGG, GBG, GGB, GGG\}|}{8} = \frac{4}{8} = \frac{1}{2}.$$

- $E \cap F$ = outcomes with both sexes and at most one boy = exactly one boy:

$$p(E \cap F) = \frac{|\{BGG, GBG, GGB\}|}{8} = \frac{3}{8}.$$

Check: $p(E)\,p(F) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8} = p(E \cap F)$. Hence E and F **are independent**.

15.10 Pairwise and Mutual Independence

Definition. The events $E_1, E_2, \ldots, E_n$ are *pairwise independent* if and only if

$$p(E_i \cap E_j) = p(E_i)\,p(E_j) \quad \text{for all pairs } i, j \text{ with } 1 \le i < j \le n.$$

Definition. The events $E_1, E_2, \ldots, E_n$ are *mutually independent* if

$$p(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_m}) = p(E_{i_1})\,p(E_{i_2}) \cdots p(E_{i_m})$$

whenever i_j, $j = 1, 2, \ldots, m$, are integers with $1 \le i_1 < i_2 < \cdots < i_m \le n$ and $m \ge 2$.

In particular, three events E_1, E_2, E_3 are mutually independent if all four conditions hold:

$$p(E_1 \cap E_2) = p(E_1)p(E_2), \quad p(E_1 \cap E_3) = p(E_1)p(E_3), \quad p(E_2 \cap E_3) = p(E_2)p(E_3),$$
$$p(E_1 \cap E_2 \cap E_3) = p(E_1)\,p(E_2)\,p(E_3).$$

Pairwise independence $\not\Rightarrow$ mutual independence

Mutual independence implies pairwise independence, but the reverse is not necessarily true.

Example. Flip a fair coin twice, and define

- E_1: the first flip is heads,
- E_2: the second flip is heads,
- E_3: exactly one of the first and second flips is heads.

The sample space is $\{HH, HT, TH, TT\}$ with probability $\frac{1}{4}$ each. Then

$$p(E_1) = p(E_2) = p(E_3) = \frac{1}{2}$$

($E_3 = \{HT, TH\}$). Check all pairwise intersections:

$$p(E_1 \cap E_2) = p(\{HH\}) = \frac{1}{4} = p(E_1)p(E_2),$$

$$p(E_1 \cap E_3) = p(\{HT\}) = \frac{1}{4} = p(E_1)p(E_3),$$
$$p(E_2 \cap E_3) = p(\{TH\}) = \frac{1}{4} = p(E_2)p(E_3).$$

Therefore E_1, E_2, E_3 are pairwise independent. However, $E_1 \cap E_2 \cap E_3 = \emptyset$ (if both flips are heads, it is impossible that exactly one of them is heads), so

$$p(E_1 \cap E_2 \cap E_3) = 0 \neq \frac{1}{8} = p(E_1)\,p(E_2)\,p(E_3).$$

Therefore E_1, E_2, E_3 are **not mutually independent**.

16 Conditional Probability and Bayes' Theorem

16.1 Conditional Probability

Definition. Let E and F be events with $p(F) > 0$. The *conditional probability of E given F*, denoted by $p(E \mid F)$, is defined as

Conditional probability

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)}.$$

Intuitively, $p(E \mid F)$ is the probability of E when we restrict the sample space to F: outcomes outside F no longer matter, and probabilities inside F are rescaled by dividing by $p(F)$.

Two immediate corollaries follow from the definition.

Corollary 1. (Product rule)

$$p(E \cap F) = p(F)\,p(E \mid F).$$

Corollary 2. (Chain rule) Applying the product rule repeatedly,

$$\begin{aligned} p(E_1 \cap E_2 \cap \cdots \cap E_n) &= p(E_1 \cap E_2 \cap \cdots \cap E_{n-1}) \cdot p(E_n \mid E_1 \cap E_2 \cap \cdots \cap E_{n-1}) = \cdots \\ &= p(E_1)\,p(E_2 \mid E_1)\,p(E_3 \mid E_1 \cap E_2) \cdots p(E_n \mid E_1 \cap E_2 \cap \cdots \cap E_{n-1}). \end{aligned}$$

Example. In a random bit string of length 4 (all 16 possible strings are equally likely), what is the probability that it contains at least two consecutive 0s, given that its first bit is a 0?

Solution. Let

- E: the event that the bit string contains at least two consecutive 0s,
- F: the event that the first bit is a 0.

We list the strings in $E \cap F$ (first bit 0, and containing 00 somewhere):

$$E \cap F = \{0000,\ 0001,\ 0010,\ 0011,\ 0100\}, \qquad p(E \cap F) = \frac{5}{16}.$$

Also, half of all strings start with 0, so $p(F) = \dfrac{8}{16} = \dfrac{1}{2}$. Therefore

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)} = \frac{5/16}{1/2} = \frac{5}{8}.$$

16.2 The Law of Total Probability

Theorem. (Law of Total Probability) Let F be an event with $p(F) > 0$. For any event E,

Law of total probability

$$p(E) = p(E \mid F)\, p(F) + p(E \mid \overline{F})\, p(\overline{F}).$$

Proof. Every outcome of E lies either in F or in $\overline{F}$, so

$$E = (E \cap F) \cup (E \cap \overline{F}).$$

The two sets $E \cap F$ and $E \cap \overline{F}$ are disjoint, so by additivity,

$$p(E) = p(E \cap F) + p(E \cap \overline{F}).$$

Applying the product rule $p(E \cap F) = p(E \mid F)p(F)$ and $p(E \cap \overline{F}) = p(E \mid \overline{F})p(\overline{F})$ to each term gives

$$p(E) = p(E \mid F)\, p(F) + p(E \mid \overline{F})\, p(\overline{F}).$$

■

Example. If a student knows 80% of the material in a course, what is the probability that she answers a question correctly on a well-balanced true-false test, assuming that she randomly guesses on any question for which she does not know the answer?

Solution. Let

- Q: the question is in the student's knowledge, with $p(Q) = 0.8$ and $p(\overline{Q}) = 0.2$;
- C: the student answers the question correctly.

If she knows the material she answers correctly for sure: $p(C \mid Q) = 1$. If not, she guesses randomly between true and false: $p(C \mid \overline{Q}) = 0.5$. By the law of total probability,

$$\begin{aligned} p(C) &= p(C \cap Q) + p(C \cap \overline{Q}) = p(Q)\, p(C \mid Q) + p(\overline{Q})\, p(C \mid \overline{Q}) \\ &= 0.8 \times 1 + 0.2 \times 0.5 = 0.8 + 0.1 = 0.9. \end{aligned}$$

So she answers correctly with probability 0.9.

16.3 The Birthday Problem

Question 1. What is the minimum number of people who need to be in a room so that there must be two people with the same birthday, assuming that there are 366 days in a year?

Answer. 367, by the pigeonhole principle: with 367 people and only 366 possible birthdays, two people must share a birthday.

Question 2. What is the minimum number of people who need to be in a room so that the probability that at least two of them have the same birthday is greater than 1/2? Assume 366 days in a year, that the birthdays of the people are independent, and that each birthday is equally likely.

Solution. Imagine the people entering the room one by one. Let

- D_n: the event that the first n people all have different birthdays;
- B_i: the birthday of the i-th person.

We want the smallest n such that $p(D_n) < 1/2$ (then $p(\text{at least one shared birthday}) = 1 - p(D_n) > 1/2$). Compute step by step using the chain rule:

- $p(D_1) = 1$ (one person cannot collide).
- The second person must avoid the first person's birthday (365 of 366 days):

$$p(D_2) = p(D_1) \cdot p(B_2 \neq B_1 \mid D_1) = 1 \times \frac{365}{366}.$$

- Given the first two birthdays differ, the third person must avoid both (364 of 366 days):

$$p(D_3) = p(D_2) \cdot p(B_3 \neq B_1 \wedge B_3 \neq B_2 \mid D_2) = 1 \times \frac{365}{366} \times \frac{364}{366}.$$

- In general, given D_{n-1} (the first $n-1$ birthdays all distinct), the n-th person must avoid $n-1$ birthdays:

$$p(D_n) = p(D_{n-1}) \cdot p\big(\forall i \in \{1, \ldots, n-1\},\ B_n \neq B_i \mid D_{n-1}\big) = 1 \times \frac{365}{366} \times \frac{364}{366} \times \cdots \times \frac{367-n}{366}.$$

Evaluating this product, it turns out that

$$p(D_{23}) \approx 0.494 < \frac{1}{2}.$$

Hence with only $n = 23$ people, the probability of at least one shared birthday is $1 - 0.494 = 0.506 > 1/2$. The answer 23 is surprisingly small — this is the famous birthday paradox.

16.4 Independence Revisited

Recall the definition of two events being independent:

$$p(E \cap F) = p(E)\, p(F).$$

Since by the product rule $p(E \cap F) = p(F)\, p(E \mid F)$ always holds (whether E and F are independent or not), the independence condition can be rewritten (when $p(F) > 0$) as

Independence via conditional probability

$$E, F \text{ independent} \iff p(E \mid F) = p(E) \iff p(F \mid E) = p(F).$$

This gives an alternative and more intuitive definition of independence: knowing that F occurred does not change the probability of E (and symmetrically).

Example. (Independence example revisited) In a randomly generated bit string of length 4: E = "it begins with a 1", F = "it contains an even number of 1s". Are E and F independent?

Solution. We compute $p(F)$ and $p(F \mid E)$ and compare.

- $p(F) = \frac{1}{2}$: among the 16 strings, $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 8$ have an even number of 1s.
- $p(F \mid E) = \dfrac{p(E \cap F)}{p(E)} = \dfrac{4/16}{1/2} = \dfrac{1}{2}$, since $E \cap F = \{1111, 1100, 1010, 1001\}$.

Because $p(F \mid E) = \frac{1}{2} = p(F)$, the events are independent. Intuitively, whether F happens or not only depends on the parity (even/odd) of the number of 1s just before the last bit together with the last bit, and the first bit alone does not bias this parity.

16.5 Bayes' Theorem

Motivating example. (Breathalyzer) Breathalyzers have an error rate of 5%:

- a sober driver is detected as "drunk" with probability 5%;
- a drunk driver is not detected with probability 5%.

Suppose 1 in 1000 drivers is driving drunk. A police officer stops a driver at random and the breathalyzer detects "drunk". What is the probability that the driver is really drunk?

A tempting answer is 95% — but this is **wrong**! It confuses $p(\text{drunk} \mid \text{detected})$ with $p(\text{detected} \mid \text{drunk})$. The correct answer, computed below, is only about 2%.

Theorem. (Bayes' Theorem) Suppose that E and F are events from a sample space S such that $p(E) \neq 0$ and $p(F) \neq 0$. Then

Bayes' theorem

$$p(F \mid E) = \frac{p(E \mid F)\, p(F)}{p(E)} = \frac{p(E \mid F)\, p(F)}{p(E \mid F)\, p(F) + p(E \mid \overline{F})\, p(\overline{F})}.$$

Proof. Just plug in the definition of conditional probability:

$$p(F \mid E) = \frac{p(F \cap E)}{p(E)} = \frac{p(E \mid F)\, p(F)}{p(E)},$$

where the numerator uses the product rule $p(E \cap F) = p(E \mid F) p(F)$. For the denominator, observe that $p(E \cap F) + p(E \cap \overline{F}) = p(E)$ (the two sets are disjoint and their union is E), i.e., by the law of total probability,

$$p(E) = p(E \mid F)\, p(F) + p(E \mid \overline{F})\, p(\overline{F}).$$

Substituting gives the second form. ■

Example. (Breathalyzer example explained) Let

- D: the driver is drunk,
- B: the driver is detected by the breathalyzer.

We know:

$$p(B \mid D) = 0.95 \text{ (drunk and detected)}, \quad p(B \mid \overline{D}) = 0.05 \text{ (sober but detected)},$$
$$p(D) = 0.001.$$

We want $p(D \mid B)$. Applying Bayes' theorem:

$$p(D \mid B) = \frac{p(B \mid D)\, p(D)}{p(B \mid D)\, p(D) + p(B \mid \overline{D})\, p(\overline{D})} = \frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.05 \times 0.999}$$
$$= \frac{0.00095}{0.00095 + 0.04995} = \frac{0.00095}{0.0509} \approx 0.02.$$

Although the test sounds accurate (95%), a detected driver is drunk only about 2% of the time, because drunk drivers are extremely rare: among 100,000 drivers, about 100 are drunk and 95 of them are detected, while 99,900 are sober and $0.05 \times 99{,}900 = 4995$ of them are (falsely) detected; true drunks are only $95/(95 + 4995) \approx 2\%$ of all detected drivers.

Example. (Two boxes) We have two boxes. Box 1 contains 2 green balls and 7 red balls. Box 2 contains 4 green balls and 3 red balls. Bob first picks one of the boxes at random, then selects a ball from that box at random. If he has a red ball, what is the probability that he picked box 1 at first?

Solution. Let

- E: Bob has chosen a red ball,
- F: Bob has chosen box 1.

We are given

$$p(E \mid F) = \frac{7}{9} \text{ (box 1)}, \qquad p(E \mid \overline{F}) = \frac{3}{7} \text{ (box 2)}, \qquad p(F) = \frac{1}{2}.$$

Applying Bayes' theorem,

$$\begin{aligned} p(F \mid E) &= \frac{p(E \mid F)\, p(F)}{p(E \mid F)\, p(F) + p(E \mid \overline{F})\, p(\overline{F})} = \frac{\frac{7}{9} \cdot \frac{1}{2}}{\frac{7}{9} \cdot \frac{1}{2} + \frac{3}{7} \cdot \frac{1}{2}} = \frac{\frac{7}{9}}{\frac{7}{9} + \frac{3}{7}} \\ &= \frac{\frac{49}{63}}{\frac{49}{63} + \frac{27}{63}} = \frac{49}{76} \approx 0.645. \end{aligned}$$

Example. (Disease testing) There is a test for a particular disease:

- the test's false negative rate is 1%, i.e., it gives a negative result with probability 1% when given to someone with the disease;
- the test's false positive rate is 0.5%, i.e., it gives a positive result with probability 0.5% when given to someone without the disease;
- on average, one person out of 100,000 has the disease.

Should someone who tests positive be worried?

Solution. Let D: the person has the disease; E: the person tests positive. We need to compute $p(D \mid E)$. The given data are

$$p(E \mid D) = 1 - 0.01 = 0.99, \qquad p(E \mid \overline{D}) = 0.005, \qquad p(D) = \frac{1}{100{,}000} = 0.00001.$$

By Bayes' theorem,

$$\begin{aligned} p(D \mid E) &= \frac{p(E \mid D)\, p(D)}{p(E \mid D)\, p(D) + p(E \mid \overline{D})\, p(\overline{D})} = \frac{0.99 \times 0.00001}{0.99 \times 0.00001 + 0.005 \times 0.99999} \\ &= \frac{0.0000099}{0.0000099 + 0.00499995} \approx \frac{0.0000099}{0.00501} \approx 0.00198 \approx 0.2\%. \end{aligned}$$

Even after testing positive, the person has the disease with probability only about 0.2% — the disease is so rare that false positives vastly outnumber true positives. One should take the result seriously but not panic.

What if the result is negative? Then

$$\begin{aligned} p(D \mid \overline{E}) &= \frac{p(\overline{E} \mid D)\, p(D)}{p(\overline{E} \mid D)\, p(D) + p(\overline{E} \mid \overline{D})\, p(\overline{D})} \\ &= \frac{0.01 \times 0.00001}{0.01 \times 0.00001 + 0.995 \times 0.99999} \approx \frac{10^{-7}}{0.995} \approx 10^{-7}. \end{aligned}$$

A negative result is extremely reassuring: the probability of having the disease drops from 10^{-5} to about 10^{-7}.

Example. (Bayesian spam filter) Given a set B of spam messages and a set G of non-spam ("good") messages, we want to compute the probability that a new email message is spam. Look at a particular word w, and count the number of messages in which it occurs in B and in G: call these $n_B(w)$ and $n_G(w)$.

- Estimated probability that a spam email contains w: $p(w \mid B) = n_B(w)/|B|$.
- Probability that a good email contains w: $p(w \mid G) = n_G(w)/|G|$.
- The prior probability that a new email is spam: $p(B) = \dfrac{|B|}{|B| + |G|}$.

Concretely: $|B| = 2000$ spam messages, $|G| = 1000$ good messages; the word "Rolex" occurs in 250 spam messages and 5 good messages. Then

$$p(w \mid B) = \frac{250}{2000} = 0.125, \qquad p(w \mid G) = \frac{5}{1000} = 0.005, \qquad p(B) = \frac{2000}{1000 + 2000} = \frac{2}{3}.$$

Suppose the new email contains "Rolex". The probability that it is spam is

$$p(B \mid w) = \frac{p(w \mid B)\, p(B)}{p(w \mid B)\, p(B) + p(w \mid G)\, p(G)} = \frac{0.125 \times \frac{2}{3}}{0.125 \times \frac{2}{3} + 0.005 \times \frac{1}{3}}$$
$$= \frac{0.25}{0.25 + 0.005} = \frac{0.25}{0.255} = \frac{50}{51} \approx 0.98.$$

So an email containing "Rolex" is spam with probability about 98%.

16.6 Generalized Bayes' Theorem

Theorem. (Generalized Bayes' Theorem) Suppose that E is an event from a sample space S and that $F_1, F_2, \ldots, F_n$ are mutually exclusive events such that $\bigcup_{i=1}^{n} F_i = S$ (i.e., $F_1, \ldots, F_n$ form a partition of S). Assume that $p(E) \neq 0$ and $p(F_i) \neq 0$ for $i = 1, 2, \ldots, n$. Then

Generalized Bayes' theorem

$$p(F_j \mid E) = \frac{p(E \mid F_j)\, p(F_j)}{\sum_{i=1}^{n} p(E \mid F_i)\, p(F_i)}.$$

Remark. This degenerates into the standard Bayes' theorem when $n = 2$ (with $F_1 = F$, $F_2 = \overline{F}$).

Proof. Similar to the two-event case. Since the F_i partition S,

$$E = \bigcup_{i=1}^{n} (E \cap F_i),$$

a union of pairwise disjoint sets, so $p(E) = \sum_{i=1}^{n} p(E \cap F_i) = \sum_{i=1}^{n} p(E \mid F_i) p(F_i)$. Then

$$p(F_j \mid E) = \frac{p(F_j \cap E)}{p(E)} = \frac{p(E \mid F_j)\, p(F_j)}{\sum_{i=1}^{n} p(E \mid F_i)\, p(F_i)}.$$

■

Example. (Market share and defect rates) The market share of a particular device is: Company A 80%, Company B 15%, Company C 5%. The defect rates of devices made by these companies are: A 4%, B 6%, C 9%.

(a) Given a random device from the market, what are the probabilities that it is made by each of the 3 companies?

(b) How do the results change if the random device is found to be defective?

Solution.

(a) Just the market shares: $p(A) = 0.8$, $p(B) = 0.15$, $p(C) = 0.05$.

(b) Let A, B, C be the events that the device is made by the three companies, and D the event that the device is defective. We are given

$$p(D \mid A) = 0.04, \qquad p(D \mid B) = 0.06, \qquad p(D \mid C) = 0.09.$$

First compute the overall probability of a defective device (the common denominator):

$$\begin{aligned} p(D) &= p(D \mid A)p(A) + p(D \mid B)p(B) + p(D \mid C)p(C) \\ &= 0.04 \times 0.8 + 0.06 \times 0.15 + 0.09 \times 0.05 = 0.032 + 0.009 + 0.0045 = 0.0455. \end{aligned}$$

Then by the generalized Bayes' theorem,

$$p(A \mid D) = \frac{p(D \mid A)p(A)}{p(D)} = \frac{0.032}{0.0455} \approx 0.7,$$

$$p(B \mid D) = \frac{p(D \mid B)p(B)}{p(D)} = \frac{0.009}{0.0455} \approx 0.2,$$

$$p(C \mid D) = \frac{p(D \mid C)p(C)}{p(D)} = \frac{0.0045}{0.0455} \approx 0.1.$$

So conditioned on being defective, the probability of coming from A drops from 0.8 to about 0.7, while C's share rises from 0.05 to about 0.1 (C has the worst defect rate).

16.7 The Monty Hall Problem

Setup. Behind one door is a prize; behind the others, goats. You pick a door, say door 1. The host (Monty) looks behind the 3 doors:

- if your pick is incorrect, the host opens the other door with a goat;
- if your pick is correct, the host randomly chooses another door and opens it.

Now he asks: do you want to change your mind?

Answer: Change!

Explanation 1. Your first pick is correct with probability 1/3. This does not change after the host opens a door (the host can always open a goat door, so this event gives no information about whether your pick was right). The third door then must have probability 2/3 to have the prize. So switching doubles your chance of winning: from 1/3 to 2/3.

Explanation 2. (Tree diagram) Consider all combinations of (prize door, your pick, Monty's choice). Prize location and your pick are uniform (1/3 each, independent). If the prize door equals your pick, Monty chooses between the two goat doors with probability 1/2 each, contributing leaves of probability $\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{18}$; if they differ, Monty's choice is forced (probability 1), contributing leaves of probability $\frac{1}{3} \cdot \frac{1}{3} \cdot 1 = \frac{2}{18}$. Summing the leaves: staying wins exactly on the leaves where your pick equals the prize door, with total probability $3 \times \frac{1}{18} \times 2 = \frac{1}{3}$ (staying wins with probability 1/3); switching wins on the remaining leaves, with total probability 2/3. So the winning probability is 1/3 if you stay and 2/3 if you switch.

Explanation 3. (Bayes' theorem) Let W: the door with the prize, M: the door Monty opens. We know

$$p(W = 1) = p(W = 2) = p(W = 3) = \frac{1}{3}.$$

Suppose we pick door 1 first. Then Monty opens door M according to:

$$p(M = 2 \mid W = 1) = \frac{1}{2}, \quad p(M = 3 \mid W = 1) = \frac{1}{2} \quad \text{(pick correct, Monty random)},$$

$p(M = 3 \mid W = 2) = 1, \qquad p(M = 2 \mid W = 3) = 1$ (Monty forced to a goat door), and of course $p(M = 2 \mid W = 2) = p(M = 3 \mid W = 3) = 0$ (Monty never opens the prize door). Suppose Monty opens door 2. Then by the generalized Bayes' theorem,

$$p(W = 1 \mid M = 2) = \frac{p(M = 2 \mid W = 1)\, p(W = 1)}{\sum_{j=1}^{3} p(M = 2 \mid W = j)\, p(W = j)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}}$$
$$= \frac{1/6}{1/2} = \frac{1}{3},$$

and $p(W = 3 \mid M = 2) = \frac{2}{3}$. Since $\frac{2}{3} > \frac{1}{3}$: **switch**.

Example. (Non-uniform priors) Now suppose the prize is placed non-uniformly:

$$p(W = 1) = 0.5, \qquad p(W = 2) = 0.3, \qquad p(W = 3) = 0.2.$$

Suppose we pick door 1 first (the most likely door), and consider both of Monty's possible moves.

Case 1: Monty opens door 2. The relevant conditional probabilities are

$$p(M = 2 \mid W = 1) = \tfrac{1}{2} \text{ (random choice)}, \quad p(M = 2 \mid W = 2) = 0 \text{ (prize door)},$$
$$p(M = 2 \mid W = 3) = 1 \text{ (forced)}.$$

Hence

$$p(W = 1 \mid M = 2) = \frac{p(M = 2 \mid W = 1)\, p(W = 1)}{\sum_{j=1}^{3} p(M = 2 \mid W = j)\, p(W = j)} = \frac{\frac{1}{2} \times 0.5}{\frac{1}{2} \times 0.5 + 0 \times 0.3 + 1 \times 0.2}$$
$$= \frac{0.25}{0.25 + 0.2} = \frac{0.25}{0.45} = \frac{5}{9},$$

and $p(W = 3 \mid M = 2) = \frac{0.2}{0.45} = \frac{4}{9}$. The probability that our initial pick contains the prize has increased (from 0.5 to 5/9) after door 2 is opened. So we should **not switch** to door 3 (keep door 1: $5/9 > 4/9$).

Case 2: Monty opens door 3. Now

$$p(M = 3 \mid W = 1) = \frac{1}{2}, \qquad p(M = 3 \mid W = 2) = 1, \qquad p(M = 3 \mid W = 3) = 0,$$

so

$$p(W = 1 \mid M = 3) = \frac{\frac{1}{2} \times 0.5}{\frac{1}{2} \times 0.5 + 1 \times 0.3 + 0 \times 0.2} = \frac{0.25}{0.25 + 0.3} = \frac{0.25}{0.55} = \frac{5}{11},$$

and $p(W = 2 \mid M = 3) = \frac{0.3}{0.55} = \frac{6}{11}$. The probability that our initial pick contains the prize has decreased (from 0.5 to 5/11) after door 3 is opened. So we should **switch to door** 2 ($6/11 > 5/11$).

Optimal strategy: choose door 1; keep door 1 if Monty opens door 2; switch to door 2 if Monty opens door 3. The overall winning probability of this strategy is

$$p(W = 1 \wedge M = 2) + p(W = 2 \wedge M = 3) = p(W = 1)\, p(M = 2 \mid W = 1)$$
$$+ p(W = 2)\, p(M = 3 \mid W = 2)$$
$$= 0.5 \cdot 0.5 + 0.3 \cdot 1 = 0.25 + 0.3 = 0.55.$$

Alternatively, using the posterior probabilities (note $p(M = 2) = 0.45$ and $p(M = 3) =$

0.55 from the denominators above),

$$p(\text{win}) = p(M=2)\,p(W=1 \mid M=2) + p(M=3)\,p(W=2 \mid M=3)$$
$$= 0.45 \times \frac{5}{9} + 0.55 \times \frac{6}{11} = 0.25 + 0.3 = 0.55.$$

Both computations agree: the strategy wins with probability 0.55, better than always staying (0.5). Is there a better strategy? See 2019 Spring Final Exam Q11.

17 Random Variables

17.1 Random Variables and Their Distributions

Random Variable

A **random variable** is a function from the sample space of an experiment to the set of real numbers, i.e., a random variable assigns a real number to each possible outcome.

Remark

The function itself is not random! The randomness comes from the random outcome of the experiment; once the outcome s is fixed, the value $X(s)$ is completely determined.

Example. Suppose that a coin is flipped three times. Let $X(t)$ be the random variable that equals the number of heads that appear when t is the outcome. Then $X(t)$ is the function where

$$X(HHH) = 3, \quad X(TTT) = 0,$$
$$X(HHT) = X(HTH) = X(THH) = 2, \quad X(TTH) = X(THT) = X(HTT) = 1.$$

Distribution of a Random Variable

The **distribution** of a random variable X on a sample space S is the set of pairs $(r, p(X=r))$ for all $r \in X(S)$, where $p(X=r)$ is the probability that X takes the value r, i.e.,

$$p(X=r) = \sum_{s \in S,\, X(s)=r} p(s).$$

Example. Continuing with the three-coin-flip example above (all 8 outcomes are equally likely, each with probability 1/8):

$$p(X=3) = \tfrac{1}{8}, \qquad p(X=2) = \tfrac{3}{8}, \qquad p(X=1) = \tfrac{3}{8}, \qquad p(X=0) = \tfrac{1}{8}.$$

17.2 Bernoulli Trials and the Binomial Distribution

Definition. Each performance of an experiment with two possible outcomes is called a **Bernoulli trial**. The two possible outcomes are generally referred to as success and failure.

Example. A coin is biased so that the probability of heads is 2/3. What is the probability that exactly four heads occur when the coin is flipped seven times?

Solution. There are $C(7,4)$ ways to choose which four of the seven flips are heads. Each such pattern has probability $(2/3)^4(1/3)^3$. Hence the answer is

$$C(7,4)\left(\tfrac{2}{3}\right)^4\left(\tfrac{1}{3}\right)^3.$$

Binomial Distribution

Theorem. Let X be the number of successes in n independent Bernoulli trials, with probability of success p and probability of failure $q = 1 - p$. Then

$$p(X = k) = b(k : n, p) = C(n, k)\, p^k q^{n-k}.$$

We say X follows the **binomial distribution** with parameters n and p, and $b(k : n, p)$ is the pmf (probability mass function).

Check. The probabilities sum to 1 by the binomial theorem:

$$\sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1.$$

Example. What is $\Pr(10 \le X \le 20)$?

Solution.

$$\Pr(10 \le X \le 20) = \sum_{k=10}^{20} \binom{n}{k} p^k q^{n-k},$$

which, unfortunately, has no closed form. For large n, and p not too close to 0 or 1, the **normal distribution** approximates the binomial distribution well; this is a special case of the central limit theorem. The Galton Board is a physical demonstration of this phenomenon.

Continuous Probability Distributions

A continuous distribution is defined over a domain of real numbers and is characterized by a **probability density function** (pdf). Example: the pdf of a normal (Gaussian) distribution is

$$f(x) = \frac{1}{\sqrt{2\sigma^2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Remark: $f(x)$ does not represent a probability! The probability of an interval is the integral: $\Pr(a \le X \le b) = \displaystyle\int_a^b f(x)\, dx$.

Example (Infinite sample space). Experiment: flip a fair coin until it turns up heads. Let the random variable X = the number of flips.

Solution. We have $p(X = x) = (1/2)^x$ for $x = 1, 2, 3, \ldots$. Verify that the probabilities sum to 1:

$$\sum_{x=1}^{\infty} p(X = x) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1.$$

The probability of the event that we need 3 or more flips is

$$p(X = 3) + p(X = 4) + \cdots = \frac{1}{8} + \frac{1}{16} + \cdots = \frac{1}{4}.$$

Geometric Distribution

Theorem. Let X be the number of independent Bernoulli trials until success, where the probability of success is p. Then

$$p(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

We say X follows a **geometric distribution** with parameter p.

Check. The probabilities sum to 1 (geometric series):

$$\sum_{i=0}^{\infty}(1-p)^i p = p \cdot \frac{1}{1-(1-p)} = 1.$$

Example. Experiment: roll 2 dice; the probability of seeing a sum of 6 is 5/36. Let the random variable X be the number of trials until getting a sum of 6. Then

$$p(X = 3) = \left(\tfrac{31}{36}\right)^2 \left(\tfrac{5}{36}\right).$$

17.3 Expected Value

Expected Value

The **expected value** (or expectation or mean) of a random variable X on a sample space S is

$$E(X) = \sum_{s \in S} p(s)\, X(s).$$

Example. Let X be the number that comes up when a fair die is rolled. What is the expected value of X?

Solution. The random variable X takes the values $1, 2, 3, 4, 5, 6$, each with probability $1/6$. It follows that

$$E(X) = \frac{1}{6}\cdot 1 + \frac{1}{6}\cdot 2 + \frac{1}{6}\cdot 3 + \frac{1}{6}\cdot 4 + \frac{1}{6}\cdot 5 + \frac{1}{6}\cdot 6 = \frac{21}{6} = \frac{7}{2}.$$

Expectation via the Distribution

Theorem. $E(X) = \sum_{r \in X(S)} p(X = r) \cdot r.$

This follows by grouping the outcomes: recall $p(X = r) = \sum_{s \in S, X(s) = r} p(s)$, so summing $p(s)X(s)$ over all s is the same as summing $r \cdot p(X = r)$ over all values r.

Example. Flip a fair coin twice; X is the number of heads. Compute $E(X)$ in both ways.

Solution. Using the definition $E(X) = \sum_{s \in S} p(s)X(s)$:

$$E(X) = \tfrac{1}{4}\cdot 2 + \tfrac{1}{4}\cdot 1 + \tfrac{1}{4}\cdot 1 + \tfrac{1}{4}\cdot 0 = 1$$

(corresponding to outcomes HH, TH, HT, TT with X-values $2, 1, 1, 0$). Using the theorem $E(X) = \sum_{r \in X(S)} p(X = r) \cdot r$, with distribution $p(X = 2) = 1/4$, $p(X = 1) = 1/2$, $p(X = 0) = 1/4$:

$$E(X) = \tfrac{1}{4} \cdot 2 + \tfrac{1}{2} \cdot 1 + \tfrac{1}{4} \cdot 0 = 1.$$

Example. Suppose that a coin is flipped three times. Let $X(t)$ be the random variable that equals the number of heads that appear when t is the outcome. We know that

$$p(X = 3) = \tfrac{1}{8}, \quad p(X = 2) = \tfrac{3}{8}, \quad p(X = 1) = \tfrac{3}{8}, \quad p(X = 0) = \tfrac{1}{8}.$$

So

$$E(X) = \tfrac{1}{8} \cdot 3 + \tfrac{3}{8} \cdot 2 + \tfrac{3}{8} \cdot 1 + \tfrac{1}{8} \cdot 0 = 1.5.$$

Example (Expectation of the binomial distribution). Recall that a random variable X with the binomial distribution with parameters n and p has

$$p(X = k) = \binom{n}{k} p^k q^{n-k}.$$

We can derive its expectation through a rather complicated sequence of derivations:

$$E(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{n-k}$$
$$= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j} = np(p+q)^{n-1} = np.$$

Hence $E(X) = np$. (A much simpler derivation using linearity of expectation is given below.)

17.4 Linearity of Expectations

Linearity of Expectations

Theorem. If X_i, $i = 1, 2, \ldots, n$ are random variables on S, and if a and b are real numbers, then

- $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$;
- $E(aX + b) = aE(X) + b$.

Proof. For two random variables X and Y on S, by the definition of expectation,

$$E(X + Y) = \sum_{s \in S} p(s)\big(X(s) + Y(s)\big) = \sum_{s \in S} p(s)X(s) + \sum_{s \in S} p(s)Y(s) = E(X) + E(Y).$$

The case of n random variables follows by repeated application (induction on n). For the second statement,

$$E(aX + b) = \sum_{s \in S} p(s)\big(aX(s) + b\big) = a \sum_{s \in S} p(s)X(s) + b \sum_{s \in S} p(s) = aE(X) + b,$$

since $\sum_{s \in S} p(s) = 1$. ■

Example (Binomial distribution revisited). X is the number of successes in n independent Bernoulli trials, with success probability p on each trial. Write

$$X = X_1 + X_2 + \cdots + X_n,$$

where $X_i = 1$ if the ith trial is successful and 0 otherwise. These X_i's are called **indicator random variables**.

Solution. For each i,

$$E(X_i) = p \cdot 1 + (1-p) \cdot 0 = p.$$

By linearity of expectation,

$$E(X) = E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n) = np.$$

Example (The Hatcheck Problem). A new employee started a job checking hats, but forgot to put the claim check numbers on the hats. So the n customers just receive a random hat from those remaining. What is the expected number of hats returned correctly?

Solution. Let X be the random variable that equals the number of people who receive the correct hat. We can write

$$X = X_1 + X_2 + \cdots + X_n,$$

where $X_i = 1$ if the ith person receives the correct hat and 0 otherwise. (Note that the X_i's are not independent — linearity of expectation does not require independence!) Person i receives his own hat with probability $1/n$, so

$$E(X_i) = \frac{1}{n}.$$

Therefore

$$E(X) = E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n) = n \cdot \frac{1}{n} = 1.$$

Example (The Hiring Problem). What is the expected number of hires made by the following algorithm?

```
Hire-Assistant(n):
    randomly permute all n candidates
    best <- 0
    for i <- 1 to n
        interview candidate i
        if candidate i is better than best then
            fire best
            hire candidate i
            best <- i
```

Solution. Let $X_i = 1$ if you hire candidate i and 0 otherwise. Let X = number of hires $= X_1 + \cdots + X_n$. Then

$$E(X_i) = p(X_i = 1) = \frac{1}{i},$$

because among the first i candidates, the best one has probability $1/i$ of being placed at the ith position. Therefore

$$E(X) = E(X_1) + \cdots + E(X_n) = 1 + \frac{1}{2} + \cdots + \frac{1}{n-1} + \frac{1}{n} = \Theta(\log n).$$

Example (Balls and Bins). Throw m balls into n bins randomly. How many bins will be empty in expectation?

Solution. Let $X_i = 1$ if bin i is empty, and 0 otherwise. Let X = number of empty bins $= X_1 + \cdots + X_n$. A fixed ball misses bin i with probability $1 - \frac{1}{n}$, and the m balls are thrown independently, so

$$E(X_i) = p(X_i = 1) = \left(1 - \frac{1}{n}\right)^m.$$

Therefore

$$E(X) = E(X_1) + \cdots + E(X_n) = n\left(1 - \frac{1}{n}\right)^m.$$

In particular, if throwing n balls into n bins, then $n\left(1 - \frac{1}{n}\right)^n \approx \frac{n}{e}$ for large n.

17.5 Average-Case Analysis of Algorithms

Definition. Let X be the random variable representing the running time (i.e., # of instructions) of an algorithm, when the input is drawn from a certain distribution (usually uniform). Then the **average-case running time** of the algorithm is $E(X)$.

Example (Linear Search). Consider the linear search procedure:

```
procedure linear search(x: integer, a1, a2, ..., an: distinct integers)
i := 1
while (i <= n and x != a_i)
    i := i + 1
if i <= n then location := i
else location := 0
return location
```

Assume that x is one of the a_i's with equal probability. For asymptotic analysis it is enough to count the number of iterations, denoted X.

Solution. If $x = a_i$, exactly i iterations are performed, and this happens with probability $1/n$. Hence

$$E(X) = \frac{1}{n} \cdot 1 + \frac{1}{n} \cdot 2 + \cdots + \frac{1}{n} \cdot n = \frac{n+1}{2} = \Theta(n).$$

Example (Insertion Sort). Consider insertion sort:

```
Insertion-Sort(A):
    for j <- 2 to n do
        key <- A[j]
        i <- j - 1
        while i >= 1 and A[i] > key do
            A[i + 1] <- A[i]
            i <- i - 1
        endwhile
        A[i + 1] <- key
    endfor
```

Assumption: the input is a random permutation. For asymptotic analysis, it is sufficient to count the total number of iterations of the inner loop, denoted X.

Solution. Let X_i be the number of iterations of the inner loop for inserting a_i. Then

$$E(X) = E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n).$$

The element a_i can be inserted into one of the first i locations with equal probability, because the first i elements of a random permutation form a random permutation of i elements, and the rank of the last element in a random permutation is one of $1, \ldots, i$ with equal probability. Therefore

$$E(X_i) = \frac{1}{i} \cdot 1 + \frac{1}{i} \cdot 2 + \cdots + \frac{1}{i} \cdot i = \frac{i+1}{2},$$

and

$$E(X) = \sum_{i=2}^{n} \frac{i+1}{2} = \frac{n^2 + 3n - 4}{4} = \Theta(n^2).$$

17.6 The Geometric Distribution and the Coupon Collector Problem

Recall the experiment: flip a fair coin until it turns up heads, and let $X =$ the number of flips. In the general case each trial succeeds ("heads") with probability p.

Definition. A random variable X has a **geometric distribution** with parameter $p \leq 1$ if

$$p(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Mean of the Geometric Distribution

Theorem. If X has a geometric distribution with parameter p, then

$$E(X) = \sum_{k=1}^{\infty} p(1-p)^{k-1}k = \cdots = \frac{1}{p}.$$

Proof. Write $q = 1 - p$. Using the derivative trick for power series, for $|q| < 1$,

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq}\left(\sum_{k=0}^{\infty} q^k\right) = \frac{d}{dq}\left(\frac{1}{1-q}\right) = \frac{1}{(1-q)^2} = \frac{1}{p^2}.$$

Hence

$$E(X) = \sum_{k=1}^{\infty} kpq^{k-1} = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

■

Variance of the Geometric Distribution

One can similarly compute $V(X) = \dfrac{1-p}{p^2}$, using $V(X) = E(X^2) - E(X)^2$ and differentiating the geometric series twice.

Example (The Coupon Collector Problem). Each box of cereal contains a coupon. There are n different types of coupons. Assuming a box contains each type of coupon equally likely, how many boxes do you need to open to have at least one coupon of each type?

Solution. Let $X_i =$ number of boxes you need to open between having i and $i+1$ distinct coupons. When you already hold i distinct coupons, each new box yields a new type with probability $p = \dfrac{n-i}{n}$, so X_i has a geometric distribution with parameter

$p = \dfrac{n-i}{n}$, and therefore

$$E(X_i) = \frac{n}{n-i}.$$

Let X = total number of boxes $= X_0 + X_1 + \cdots + X_{n-1}$. By linearity of expectation,

$$E(X) = \sum_{i=0}^{n-1} E(X_i) = \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{i=1}^{n} \frac{1}{i} = \Theta(n \log n).$$

17.7 Conditional Expectation and the Two Envelopes Paradox

The Two Envelopes Paradox. You are given two envelopes, with one containing twice as much money as the other. You pick one first. Then you are given the chance to switch to the other. Should you switch?

- Answer 1: The situation is symmetric. There is no point in switching.
- Answer 2: Suppose the first envelope has a dollars in it. The expected value in the other envelope is

$$\frac{1}{2} \cdot 2a + \frac{1}{2} \cdot \frac{a}{2} = \frac{5}{4}a,$$

so you should switch.

The two answers contradict each other — the paradox will be resolved after we introduce conditional expectation.

Conditional Expectation and the Law of Total Expectation

For a random variable X and an event A with $P(A) > 0$,

$$E(X \mid A) = \sum_x x \cdot P(X = x \mid A).$$

Law of total expectation. Suppose $F_1, F_2, \ldots, F_n$ are mutually exclusive events from a sample space S such that $\bigcup_{i=1}^{n} F_i = S$ and $P(F_i) > 0$ for $i = 1, 2, \ldots, n$. Then

$$E(X) = E(X \mid F_1)\, P(F_1) + \cdots + E(X \mid F_n)\, P(F_n).$$

Proof (Law of total expectation).

$$\begin{aligned}
\text{RHS} &= \sum_{i=1}^{n} E(X \mid F_i)\, P(F_i) = \sum_{i=1}^{n} \sum_x x\, P(X = x \mid F_i)\, P(F_i) \\
&= \sum_x \sum_i x\, P(X = x \mid F_i)\, P(F_i) = \sum_x x \sum_i P(X = x \mid F_i)\, P(F_i) \\
&= \sum_x x\, P(X = x) = E(X),
\end{aligned}$$

where we used the law of total probability: $\sum_i P(X = x \mid F_i) P(F_i) = P(X = x)$. ■

Example. Suppose we first flip a biased coin F that has probability $2/3$ of getting heads. Let (X, Y, Z) be

- a random permutation of $1, 2, 3$ if F is heads;
- the fixed permutation $(1, 2, 3)$ if F is tails.

Compute $E(X)$.

Solution. If F is heads, X is equally likely to be $1, 2$, or 3, so

$$E(X \mid F = \text{heads}) = \frac{1+2+3}{3} = 2.$$

If F is tails, $X = 1$ always, so

$$E(X \mid F = \text{tails}) = 1.$$

By the law of total expectation,

$$\begin{aligned} E(X) &= E(X \mid F = \text{heads})\, P(F = \text{heads}) + E(X \mid F = \text{tails})\, P(F = \text{tails}) \\ &= 2 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{5}{3}. \end{aligned}$$

Flaw in the Two-Envelopes Argument. Answer 2 is wrong because the sample space and prior distribution were never defined! Suppose the envelopes contain $\{1, 2\}$ or $\{2, 4\}$ with equal probability. Let X, Y be the amounts of money in the first and second envelope.

Solution. We compute the conditional expectations:

$$E(Y \mid X = 1) = 2, \qquad E(Y \mid X = 4) = 2$$

(if the first envelope contains 1, the other must contain 2; if it contains 4, the other must contain 2). For $X = 2$:

$$\begin{aligned} E(Y \mid X = 2) &= \sum_y y \cdot p(Y = y \mid X = 2) = 1 \cdot \frac{p(Y = 1, X = 2)}{p(X = 2)} + 4 \cdot \frac{p(Y = 4, X = 2)}{p(X = 2)} \\ &= \frac{1/4}{1/2} \cdot 1 + 4 \cdot \frac{1/4}{1/2} = \frac{5}{2}. \end{aligned}$$

(Note $p(X = 2) = 1/2$ since $X = 2$ occurs in both scenarios with total probability $1/4 + 1/4$.)

The expected gain from switching:

$$\begin{aligned} E(Y) &= p(X = 1)E(Y \mid X = 1) + p(X = 2)E(Y \mid X = 2) + p(X = 4)E(Y \mid X = 4) \\ &= \frac{1}{4} \cdot 2 + \frac{1}{2} \cdot \frac{5}{2} + \frac{1}{4} \cdot 2 = 2.25. \end{aligned}$$

The expected gain from keeping:

$$E(X) = p(X = 1) \cdot 1 + p(X = 2) \cdot 2 + p(X = 4) \cdot 4 = \frac{1}{4} \cdot 1 + \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 = 2.25.$$

So switching gains nothing: $E(Y) = E(X) = 2.25$. If we could decide after seeing the first envelope, then we would expect to get

$$\frac{1}{4} \cdot 2 + \frac{1}{2} \cdot \frac{5}{2} + \frac{1}{4} \cdot 4 = 2.75$$

(switch when $X = 1$ or 2, keep when $X = 4$). For any prior distribution, similar results hold.

17.8 Independent Random Variables

Definition. The random variables X and Y on a sample space S are **independent** if

$$p(X = r_1 \text{ and } Y = r_2) = p(X = r_1) \cdot p(Y = r_2) \qquad \text{for all } r_1, r_2.$$

Expectation of a Product of Independent Random Variables

Theorem. If X and Y are independent random variables on a sample space S, then

$$E(XY) = E(X)\,E(Y).$$

Proof. From the definition of expected value, and because X and Y are independent random variables, it follows that

$$\begin{aligned} E(XY) &= \sum_{s\in S} X(s)Y(s)\,p(s) = \sum_{r_1\in X(S),\, r_2\in Y(S)} r_1 r_2 \cdot p(X = r_1 \text{ and } Y = r_2) \\ &= \sum_{r_1\in X(S)} \sum_{r_2\in Y(S)} r_1 r_2\, p(X = r_1)\, p(Y = r_2) \\ &= \left(\sum_{r_1\in X(S)} r_1\, p(X = r_1)\right)\left(\sum_{r_2\in Y(S)} r_2\, p(Y = r_2)\right) = E(X)\,E(Y). \end{aligned}$$

Here the first equality groups outcomes by the values (r_1, r_2) taken by (X, Y), and the second uses independence $p(X = r_1, Y = r_2) = p(X = r_1)p(Y = r_2)$. ■

Note

Linearity of expectation holds no matter whether the random variables are independent or not; but the product rule $E(XY) = E(X)E(Y)$ requires independence.

Example. Let X and Y be random variables that count the number of heads and the number of tails when a fair coin is flipped twice. Show that X and Y are not independent.

Solution.

$$p(X = 2 \text{ and } Y = 2) = 0,$$

but

$$p(X = 2) = \frac{1}{4}, \qquad p(Y = 2) = \frac{1}{4},$$

so $p(X = 2 \text{ and } Y = 2) = 0 \neq \dfrac{1}{16} = p(X = 2)\,p(Y = 2)$, and hence X, Y are not independent.

Check the expectations: $XY = 1$ exactly when the outcome is HT or TH, so

$$E(XY) = 1 \cdot \frac{1}{2} = \frac{1}{2},$$

while $E(X) = E(Y) = 1$, hence $E(XY) \neq E(X)E(Y)$. However linearity still holds: $E(X + Y) = 2 = E(X) + E(Y)$ (indeed $X + Y = 2$ always).

17.9 Variance and Standard Deviation

Variance and Standard Deviation

Let X be a random variable on the sample space S. The **variance** of X, denoted by $V(X)$, is

$$V(X) = \sum_{s\in S} \big(X(s) - E(X)\big)^2 p(s) = E\Big[\big(X - E(X)\big)^2\Big].$$

The **standard deviation** of X, denoted by $\sigma(X)$, is $\sqrt{V(X)}$.

Example (Variance of a Bernoulli trial). $X = 1$ with probability p, and 0 otherwise. Then $E(X) = p$, and

$$V(X) = (1-p)^2 \cdot p + (0-p)^2 \cdot (1-p) = (1-p)p\big[(1-p)+p\big] = (1-p)p.$$

Computational Formula for Variance

Theorem. $V(X) = E(X^2) - E(X)^2$.

Proof. Expanding the square in the definition and using linearity of expectation,

$$\begin{aligned} V(X) &= \sum_{s\in S} \big(X(s) - E(X)\big)^2 p(s) \\ &= \sum_{s\in S} \big(X(s)^2 - 2X(s)E(X) + E(X)^2\big)p(s) \\ &= \sum_{s\in S} X(s)^2 p(s) - 2E(X)\sum_{s\in S} X(s)p(s) + E(X)^2 \sum_{s\in S} p(s) \\ &= E(X^2) - 2E(X)^2 + E(X)^2 = E(X^2) - E(X)^2. \end{aligned}$$

■

Corollary. $V(aX) = a^2 V(X)$ for any real number $a \geq 0$. (Indeed, $E(aX)^2 - E((aX)^2)$ gives $V(aX) = a^2E(X^2) - a^2E(X)^2 = a^2V(X)$.)

Example. What is the variance of a random variable X, where X is the number that comes up when a fair die is rolled?

Solution. First, $E(X) = \dfrac{1+2+3+4+5+6}{6} = \dfrac{7}{2}$.

Using the first definition:

$$V(X) = \frac{(1-3.5)^2 + (2-3.5)^2 + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2 + (6-3.5)^2}{6} = \frac{35}{12}.$$

Using the second definition:

$$E(X^2) = \frac{1^2+2^2+3^2+4^2+5^2+6^2}{6} = \frac{91}{6},$$

$$V(X) = E(X^2) - E(X)^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

Bienaymé's Formula

Theorem. If X and Y are two independent random variables on a sample space S, then

$$V(X+Y) = V(X) + V(Y).$$

Furthermore, if X_i, $i = 1, 2, \ldots, n$, are pairwise independent random variables on S, then

$$V(X_1 + \cdots + X_n) = V(X_1) + \cdots + V(X_n).$$

Proof of Bienaymé's Formula. Using the computational formula for variance and linearity of expectation,

$$\begin{aligned} V(X+Y) &= E\big((X+Y)^2\big) - E(X+Y)^2 \\ &= E(X^2+2XY+Y^2) - \big(E(X)+E(Y)\big)^2 \\ &= E(X^2)+2E(XY)+E(Y^2) - E(X)^2 - 2E(X)E(Y) - E(Y)^2. \end{aligned}$$

Because X and Y are independent, $E(XY) = E(X)E(Y)$, so the middle terms cancel:

$$V(X+Y) = \big(E(X^2) - E(X)^2\big) + \big(E(Y^2) - E(Y)^2\big) = V(X) + V(Y).$$

The proof of the case with n pairwise independent random variables can be constructed by generalizing the proof of the case for two random variables (expanding $E((\sum_i X_i)^2)$ and using pairwise independence to replace $E(X_iX_j)$ by $E(X_i)E(X_j)$ for $i \neq j$). ■

Example. Let X = the sum of the numbers on two dice. What is $V(X)$?

Solution. Write $X = X_1 + X_2$, where X_1 is the number on the first die and X_2 is the number on the second die. Since X_1, X_2 are independent and we already know $V(X_1) = V(X_2) = \dfrac{35}{12}$, Bienaymé's formula gives

$$V(X) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6}.$$

Example (Variance of the binomial distribution). Given that X follows the binomial distribution with parameters n and p, what is $V(X)$?

Solution. Write $X = X_1 + \cdots + X_n$, where $X_i = 1$ with probability p and 0 otherwise. The X_i's are independent Bernoulli trials, and we already know $V(X_i) = p(1-p)$. By Bienaymé's formula,

$$V(X) = np(1-p).$$

Example (Averaging independent measurements). Suppose we take a measurement by repeating the same experiment many times. Each trial returns a random measurement that is correct in expectation, but with some deviation. More precisely, let X_i be the measurement from the ith experiment; we have

$$E(X_i) = \mu, \qquad V(X_i) = \sigma^2.$$

If we repeat the experiment n times independently and report the average $Y = (X_1 + \cdots + X_n)/n$, what are $E(Y)$ and $V(Y)$?

Solution. By linearity of expectation,

$$E(Y) = \frac{E(X_1) + \cdots + E(X_n)}{n} = \frac{n\mu}{n} = \mu.$$

By Bienaymé's formula (the X_i's are independent) and the corollary $V(aX) = a^2V(X)$ with $a = 1/n$,

$$V(Y) = \frac{V(X_1) + \cdots + V(X_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

So averaging n independent measurements keeps the mean μ but shrinks the variance by a factor of n — the average is a more reliable estimate than any single measurement.

18 Graph Theory

18.1 Motivating Problems

The Seven Bridges of Königsberg. The city of Königsberg was divided by a river into four land areas (two banks A, D and two islands B, C) connected by seven bridges.

Q: Can you find a walk that crosses all seven bridges, each exactly once?

Reformulated as a graph problem: model each land area as a vertex and each bridge as an edge. Can you find a path in the graph that includes every edge exactly once? (Euler answered this negatively — see the Euler path theorem below.)

The Traveling Salesman Problem. **Q:** How to visit all places with the shortest total distance, and come back to the origin?

Reformulated as a graph problem: given a graph where edges have weights (lengths), how to find a cycle with minimum total weight that includes all vertices?

The World Wide Web. The web can be modeled as a web graph: each node is a web page, and each edge is a hyperlink from one page to another (a directed edge). For example: `cnn.com`, `cnnsi.com`, `novell.com`, `netscape.com`, `timewarner.com`, `hbo.com`, `sorpranos.com` link to one another.

18.2 Graph Terminology

Graph

A **graph** $G = (V, E)$ consists of a nonempty set V of vertices (or nodes) and a set E of edges. An edge is said to connect its two endpoints. The endpoints connected by an edge are called **adjacent** (or neighbors), and the edge is **incident** to its endpoints.

Example. A graph with vertex set $\{a, b, c, d\}$ may have four vertices and five edges.

Types of Edges and Graphs

- **Loop:** an edge connecting a vertex to itself.
- **Multiple edges:** edges connecting the same two vertices.
- **Simple graph:** a graph without loops and multiple edges.
- Edges can be directed or undirected: an **undirected edge** is $e = \{u, v\}$; a **directed edge** is $e = (u, v)$.

Degree

The **degree** of a vertex in an undirected graph is the number of edges incident to it, except that a loop at a vertex contributes two to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example. In an undirected graph G: $\deg(a) = 2$, $\deg(b) = \deg(c) = \deg(f) = 4$, $\deg(d) = 1$, $\deg(e) = 3$, $\deg(g) = 0$. In another graph H: $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, $\deg(d) = 5$.

The Handshaking Theorem

Theorem. If $G = (V, E)$ is an undirected graph with m edges, then

$$2m = \sum_{v \in V} \deg(v).$$

Proof. Each edge contributes two to the total degree of all vertices (one to each of its two endpoints; a loop contributes two to its single endpoint). Hence summing all degrees counts every edge exactly twice, giving $2m$. ■

Example. How many edges are there in a graph with 10 vertices, all of which have degree 6?

Solution. By the handshaking theorem, $2m = \sum_{v} \deg(v) = 10 \times 6 = 60$, so $m = 30$ edges.

Example. Show that at a party, the number of people who shake hands with an odd number of people must be even.

Solution. Model the party as an undirected graph: people are vertices, and a handshake is an edge. By the handshaking theorem, $\sum_{v} \deg(v) = 2m$ is even. Split the sum into odd-degree and even-degree vertices: the sum over even-degree vertices is even, so the sum over odd-degree vertices must also be even. A sum of odd numbers is even only if there is an even number of terms. Hence the number of people with an odd number of handshakes is even. ■

In-degree and Out-degree

The **in-degree** of a vertex v, denoted $\deg^-(v)$, is the number of edges which terminate at v. The **out-degree** of v, denoted $\deg^+(v)$, is the number of edges with v as their initial vertex. Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of the vertex.

Example. In a directed graph: $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 3$, $\deg^-(d) = 2$, $\deg^-(e) = 3$, $\deg^-(f) = 0$; $\deg^+(a) = 4$, $\deg^+(b) = 1$, $\deg^+(c) = 2$, $\deg^+(d) = 2$, $\deg^+(e) = 3$, $\deg^+(f) = 0$.

Degree Sum in Directed Graphs

Theorem. If $G = (V, E)$ is a directed graph with m edges, then

$$m = \sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v).$$

Proof. Each edge contributes one to the out-degree of its starting vertex and one to the in-degree of its terminal vertex. So summing all out-degrees counts every edge exactly once, and so does summing all in-degrees. ■

Special Graphs: K_n and C_n

- **Complete graph K_n:** a simple graph that contains exactly one edge between each

pair of distinct vertices. (K_n has $\binom{n}{2} = \frac{n(n-1)}{2}$ edges.)

- **Cycle** C_n (for $n \geq 3$): consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

Bipartite Graphs

A simple graph $G = (V, E)$ is **bipartite** if V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge connects a vertex in V_1 and a vertex in V_2. Equivalently, a bipartite graph is a graph where it is possible to color the vertices red or blue so that no two adjacent vertices are the same color.

Example. Show that C_3 is not bipartite.

Solution. C_3 has vertices v_1, v_2, v_3 with edges $\{v_1, v_2\}, \{v_2, v_3\}, \{v_3, v_1\}$. Try a 2-coloring: color v_1 red. Since v_2 is adjacent to v_1, v_2 must be blue. Since v_3 is adjacent to v_2, v_3 must be red. But v_3 is adjacent to v_1, and both are red — contradiction. Hence C_3 is not bipartite. (In general, a graph containing an odd cycle cannot be bipartite.) ■

Example. Of two displayed graphs G and H: G is bipartite (its vertices can be 2-colored with no adjacent pair sharing a color), while H is not bipartite (it contains a triangle, i.e., a C_3).

Complete Bipartite Graphs

A **complete bipartite graph** $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets V_1 of size m and V_2 of size n such that there is an edge from every vertex in V_1 to every vertex in V_2. ($K_{m,n}$ has mn edges.)

Definition. A **subgraph** of a graph $G = (V, E)$ is a graph (W, F), where $W \subseteq V$ and $F \subseteq E$, such that for any $e \in F$, both of its endpoints must be in W.

18.3 Representing Graphs

Definition. An **adjacency list** can be used to represent a graph with no multiple edges by specifying the vertices that are adjacent to each vertex of the graph. (Each vertex heads a list of its neighbors.)

Adjacency Matrix

Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Arbitrarily list the vertices of G as $v_1, v_2, \ldots, v_n$. The **adjacency matrix** $\mathbf{A}_G$ of G, with respect to this listing of vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j)th entry when v_i and v_j are adjacent, and 0 as its (i, j)th entry when they are not adjacent. In other words, if $\mathbf{A}_G = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Example. With the ordering of vertices a, b, c, d, write down the 4×4 adjacency matrix of a given simple graph. Properties of the adjacency matrix of a simple (undirected) graph:

- it is symmetric, i.e., $a_{ij} = a_{ji}$;
- each diagonal entry $a_{ii} = 0$ (no loops).

Adjacency List vs. Adjacency Matrix

Adjacency list is better for a sparse graph; adjacency matrix is better for a dense graph.

Adjacency matrices of non-simple graphs. A self-loop at the vertex v_i is represented by a 1 at the (i,i)th position of the matrix. When multiple edges connect the same pair of vertices v_i and v_j (or if multiple self-loops are present at the same vertex), the (i,j)th entry equals the number of edges connecting the pair of vertices. Adjacency matrices can also represent directed graphs — in that case the matrix may not be symmetric.

Incidence Matrix

Let $G = (V, E)$ be an undirected graph with vertices $v_1, v_2, \ldots, v_n$ and edges $e_1, e_2, \ldots, e_m$. The **incidence matrix** with respect to the ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Graph Isomorphism

Two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there is a bijection $f : V_1 \to V_2$ with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2, for all a and b in V_1. Such an f is called an **isomorphism**.

Example. Show that the graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic.

Solution. Use the bijection f with

$$f(u_1) = v_1, \qquad f(u_2) = v_4, \qquad f(u_3) = v_3, \qquad f(u_4) = v_2.$$

Then check adjacency preservation: u_1 is adjacent to u_2, u_3, u_4 in G, and $f(u_1) = v_1$ is adjacent to $f(u_2) = v_4$, $f(u_3) = v_3$, $f(u_4) = v_2$ in H; similarly for all other pairs. Since adjacency is preserved both ways, f is an isomorphism, so $G \cong H$. ■

Algorithms for Graph Isomorphism

The best algorithms known for determining whether two graphs are isomorphic have exponential worst-case time complexity (in the number of vertices of the graphs). However, there are algorithms with linear average-case time complexity.

18.4 Connectivity

Paths and Circuits

Let n be a nonnegative integer and G an undirected graph. A **path** of length n from u to v in G is a sequence of n edges $e_1, \ldots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \ldots, x_{n-1}, x_n = v$ of vertices such that e_i connects x_{i-1} and x_i, for $i = 1, \ldots, n$.

- For directed graphs, replace "e_i connects x_{i-1} and x_i" with "$e_i = (x_{i-1}, x_i)$".
- When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \ldots, x_n$.
- The path is a **circuit** (or a cycle) if $u = v$.
- A path or circuit is **simple** if it does not contain the same edge more than once.

Definition. In an undirected graph, two vertices u, v are **connected** if there is a path from u to v.

Example. Give a recursive definition of connectedness of two vertices without using the concept of a path.

Solution.

- Base step: for any edge $e = \{u, v\}$, u and v are connected;
- Recursive step: for any u, v, w, if u and v are connected, and there is an edge between v and w, then u and w are connected.

Connected Graph

An undirected graph is called **connected** if there is a path between every pair of vertices.

Example. Suppose in a wireless network of n mobile devices, each device is within communication range with at least $n/2$ other devices (assuming n is an even number). Show that all devices are connected.

Reformulated as a graph problem: let G be an undirected graph on n vertices where each node has degree $\geq n/2$. Show that G is connected.

Solution. Consider any two nodes u and v in G. There are two cases:

- If there is an edge $\{u, v\}$, then u and v are connected directly.
- If there is no direct edge between u and v, then they must have a common neighbor, say w, because there are $n-2$ nodes other than u and v, and u and v each have $\geq n/2$ neighbors among these nodes; since $\frac{n}{2} + \frac{n}{2} = n > n - 2$, the two neighbor sets must overlap (pigeonhole principle), so some w is adjacent to both. Thus there is a path u–w–v between u and v.

The above argument holds for any two nodes u, v, so the graph G is connected. ■

Think about it

If the threshold $n/2$ is changed to $n/2 - 1$, does the claim still hold? (Hint: consider two disjoint copies of $K_{n/2}$.)

Definition. A **connected component** of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G (i.e., a maximal connected subgraph).

Connectivity in Directed Graphs

- A directed graph G is **weakly connected** if the undirected graph obtained by ignoring the directions of the edges of G is connected.
- A directed graph $G = (V, E)$ is **strongly connected** if for any $u, v \in V$, there is a (directed) path from u to v.

Example. One displayed digraph is strongly connected (every vertex reaches every other vertex along directed paths); another is weakly connected but not strongly connected (the underlying undirected graph is connected, but some vertex cannot be reached from some other vertex following the directions).

Definition. The subgraphs of a directed graph G that are strongly connected but not contained in larger strongly connected subgraphs are called the **strongly connected components** of G.

18.5 Euler Paths and Circuits

Euler Paths and Circuits

An **Euler circuit** in a graph G is a simple circuit containing every edge of G. An **Euler path** in G is a simple path containing every edge of G.

(Recall: "simple" means no edge is used more than once. Euler paths/circuits use every edge exactly once; vertices may be revisited. This is exactly what the Königsberg bridge problem asks for, and is named after Leonard Euler (1707–1783).)

Example. Given several small graphs, which of them have Euler circuits? Which have Euler paths? (Apply the two theorems below: check connectedness and count odd-degree vertices.)

Euler Circuit Theorem

Theorem. An undirected connected graph has an Euler circuit if and only if all vertices have even degrees.

Proof. ($\Rightarrow$) The "only if" direction. An Euler circuit begins with a vertex a and continues with an edge incident with a, say $\{a, b\}$. The edge $\{a, b\}$ contributes one to $\deg(a)$. Each time the circuit passes through a vertex it contributes two to the vertex's degree (one edge in, one edge out — and the circuit uses each edge exactly once, so all these contributions are distinct edges). Finally, the circuit terminates at a, contributing one to $\deg(a)$. Therefore $\deg(a)$ is even (the initial edge plus the final edge plus pairs for each intermediate pass), and every other vertex also has even degree.

($\Leftarrow$) The "if" direction. We give an algorithm to find an Euler cycle when all degrees are even. First, consider the following simple algorithm:

```
u <- any vertex
while u has an edge not taken yet
    take that edge {u, v}
    u <- v
```

Observation: this algorithm always finds a cycle, because all degrees are even — whenever the walk enters a vertex $v \neq u$, it has used an odd number of edges incident with v, so at least one unused edge remains to leave v; the walk can only stop back at the starting vertex u.

Problem: this algorithm may not traverse all edges.

Fix: the subgraph consisting of all edges not yet traversed must still have even degrees at all vertices (removing a cycle removes an even number of edges at each vertex). So just repeat the algorithm on those remaining edges, and "splice" the new cycle into the old one at a common vertex:

```
c <- empty cycle
while there are still edges not taken yet
    u <- any vertex already seen (with an untaken edge)
    c' <- Find-Cycle(u)
    insert c' into c at u
```

Since the graph is connected, every untaken edge is reachable from some vertex already on the current cycle, so the loop eventually consumes all edges. The result is a simple circuit containing every edge — an Euler circuit. ■

Euler Path Theorem

Theorem. An undirected connected graph has an Euler path but not an Euler circuit if and only if there are exactly two vertices of odd degree.

Proof. (⇐) The "if" direction. Suppose u, v have odd degrees (and all other vertices have even degrees). Add an edge between u and v. In the new graph there are no odd-degree vertices, so by the Euler circuit theorem there is an Euler circuit. Removing the added edge from the Euler circuit turns it into a path, starting at u and terminating at v, that still contains every original edge exactly once — an Euler path.

(⇒) The "only if" direction. Suppose there is an Euler path from u to v and $u \neq v$. Add an edge between u and v in the graph as well as in the path. This turns the Euler path into a circuit containing all edges of the augmented graph. By the previous theorem, all degrees in the new graph are even. Then removing the added edge from the graph decreases the degrees of exactly u and v by one each, introducing exactly two vertices with odd degrees. ■

Back to Königsberg

In the Königsberg graph all four land areas have odd degree, so by the theorems above there is neither an Euler circuit nor an Euler path: no walk can cross every bridge exactly once.

18.6 The Chinese Postman Problem

Problem. Given a graph where the edges represent streets and vertices represent intersections, find the shortest tour so that the postman can traverse every edge at least once and return to the starting point. For each edge, its length is given.

Solution.

- Easy case: if all vertices have even degree, then just use the Euler circuit (every edge is traversed exactly once, which is clearly optimal).
- What if there are odd-degree vertices? Note that due to the handshaking theorem, there must be an even number of odd-degree vertices. Since an Euler circuit does not exist, some edges will have to be traversed more than once; we want to minimize the total length of such repeated edges.
- Medium case — two odd-degree vertices u and v: find the shortest path between them; add (duplicate) all edges on this path to the graph; now all vertices have even degrees (the duplicated path flips the parity of u, v and of every internal vertex twice), and we can find an Euler circuit.
- Hard case — more than 2 odd-degree vertices: for every pair of odd-degree vertices, find their shortest path; find the best pairing of these vertices such that the total length of the shortest paths between the paired vertices is minimized; add the shortest paths between the paired vertices to the graph and find the Euler circuit.

Example. On a weighted graph with two odd-degree vertices, compute the shortest path between them (e.g., by inspection), duplicate those edges, and output an Euler circuit of the augmented graph; its total weight equals the sum of all edge weights plus the length of that shortest path.

Further questions: How to find the shortest path? How to find the best pairing (matching)? These will be covered in COMP 3711.

18.7 Hamilton Paths and Circuits

Hamilton Paths and Circuits

Euler paths and circuits contain every edge exactly once. **Hamilton paths and circuits** contain every vertex exactly once (a Hamilton circuit returns to its starting vertex).

(Named after William Rowan Hamilton (1805–1865).)

No Efficient Algorithm Known

Determining whether a graph has a Hamilton path or circuit is **NP-hard** — unlike the Euler case, there is no simple degree characterization. On weighted graphs, finding a minimum-weight Hamilton circuit is exactly **the traveling salesman problem**.